

Nature-Based Solutions and Biomimicry for Urban Sustainability and Climate Change Mitigation and Adaptation: A Case Studies Analysis

Master's Thesis

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DECLARATION

I hereby declare that this thesis entitled “Nature-Based Solutions and Biomimicry for Urban Sustainability and Climate Change Mitigation and Adaptation: A Case Studies Analysis” was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references.

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ABSTRACT

This thesis reviews the role of nature-based and biomimetic solutions in improving the dimensions of urban sustainability and their effectiveness in addressing the growing pressures of climate change through mitigation and adaptation. The five case studies conducted on solutions in five cities from five continents suggest that the utilization of nature-based solutions in urban contexts creates benefits for all three dimensions of sustainability, albeit to different extents, despite the lack of policies that purposefully incorporate nature-based and biomimetic solutions in urban infrastructure. The study further revealed a lower level of usage of the solutions in climate change mitigation and adaptation, although there are some mitigation and adaptation co-benefits observed with some solutions. The economic dimension of development is undeniable, and from this perspective, there is still insufficient data that could be used to fully understand the implications and true value of nature-based and biomimetic solutions. This can also be said about quantitative data about the effects of the analyzed nature-based and biomimetic solutions in terms of air purification or energy efficiency. However, there is some data that implies these solutions support both energy efficiency, as well as bring environmental and health benefits to their users. Something that cities can and should do is develop policies that will promote the usage and monitoring of nature-based and biomimetic solutions in urban areas, in order to future-proof their infrastructure and to reap their social and economic benefits.

Keywords: nature-based solutions, biomimicry, urban infrastructure, climate change, sustainability

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INTRODUCTION

With global population growing and the population moving from the countryside into cities progressively over time, the topics revolving around the urban and the urbane are becoming more prominent among policy makers and withing political circles, as well as in academia. This does not come as surprise, as cities have always been places of more plentiful opportunities and higher quality of life, and this is still the case. With this increase in urban population, cities are also experiencing new and unprecedented challenges, which, when paired with global issues and uncertainties can seem daunting. However, research into urban challenges and pressures has been around for some time now, and it is today becoming increasingly intertwined with sustainability.

In order for cities to remain places that can sustain and receive all those interested in pursuing educational opportunities, career opportunities, and enjoying richer social and cultural life, their development must be sustainable. Without the aspect of sustainability, cities risk becoming dystopian realities ridden with crime, disease, and pollution, and this project looks at this exactly. This research looks at five cities around the world, Cape Town (South Africa), London (United Kingdom), Madrid (Spain), New York City (United States), and Seoul (South Korea), and explores the role that nature-based solutions and biomimicry play in improving urban sustainability.

The answers this thesis aims to provide possess a dual importance. First, there is the fact that this is still an interesting and growing field in academia. There are many facets to the topic that demand attention and further research, therefore this thesis has excellent thought experiment potential. Second, given the real-world cases that the project utilizes, everything that is a result of this research possesses real applicability and relevance. Challenges that cities are experiencing, and which are to varying extents addressed are at the root of this research.

In order to achieve its goals, this thesis will cover several sections. Following the introduction to the topic and the idea, the goals of the thesis will be introduced. Then, a literature review will be presented. Following the literature review treating several dozen academic sources, the thesis will present the methodology. In this section, the scientific method will be explained in more detail. Then, the case studies will be presented. In this section, which is the main body of the thesis, the content will be analyzed. From there, the case studies will be discussed and related to the research questions. The penultimate section will provide the reader with the takeaways and conclusions, as well as the recommendations for decision makers. The last section of the thesis are the references.

GOALS

The topic of nature-based solutions and biomimicry is a broad one, and research in this field can be taken in different directions. There are several questions that arose as appropriate for a project of this scope that the thesis aims to answer:

Research Questions	Thesis Sub-aims
What are the key factors that influence the effectiveness of nature-based solutions and biomimicry in improving urban sustainability in the selected cities?	Improving the understanding of nature-based and biomimicry solutions' performance in the context of urban sustainability.
What are the most important considerations when designing and implementing nature-based solutions and biomimicry projects in urban areas, such as site selection, materials, and maintenance?	Developing a better understanding of the factors of success for nature-based and biomimicry solutions' implementation.
How have the selected cities implemented nature-based solutions and biomimicry in their urban sustainability initiatives, and what are the challenges and opportunities faced in the implementation process?	Contributing to the development of local-specific knowledge about the implementation and outcomes of nature-based and biomimicry solutions in the selected urban areas.
What are the outcomes of nature-based solutions and biomimicry in improving urban sustainability in the selected cities, and how do these outcomes compare across the five cases?	
What can be learned from the selected cities about the implementation and effectiveness of nature-based solutions and biomimicry in improving urban sustainability, and how can these lessons be applied in other urban contexts?	Developing a better understanding of transposability of nature-based and biomimicry solutions across different geographies.
How can nature-based solutions and biomimicry be used across the five cities in the context of climate change mitigation and adaptation?	Developing a better understanding of how nature-based and biomimicry solutions can be integrated into climate change actions planes, namely mitigation and adaptation efforts.
How can the benefits of nature-based solutions and biomimicry be quantified and communicated to stakeholders, such as city officials, residents, and investors?	Contributing to the development of better understanding of optimal communication strategies for the benefits and co-benefits of nature-based and biomimicry solutions to different stakeholders.

Although the number of questions is fairly sizeable, all of them are integrated in a coherent whole, and the answers to these questions are all related and together shape a more accurate and relevant narrative about the topic. The thesis then, as can be seen from the questions, aims at explaining the relationship between urban sustainability and nature-based solutions and biomimicry. However, what sets it apart from previous research done in the field is that it goes beyond theoretical considerations and uses five global cases from four continents. In this sense, this thesis also aims to understand what can and cannot be replicated as a solution to common challenges, and also try and anticipate potential issues that might require the utilization of nature-based solutions and biomimicry in the future. Ultimately, the thesis also aims at building an understanding of the existing policy related to the topic, and the related communication of benefits of nature-based solutions and biomimicry to decision and policy makers.

LITERATURE REVIEW

This section will provide a comprehensive review of the relevant literature on nature-based solutions, biomimicry, and urban sustainability, with a focus on the role that the former play in improving the latter. Before commencing the review of the cities selected for this study, it is necessary to conduct a review of existing literature on nature-based solutions, biomimicry, and urban sustainability, as well as to integrate these three into a coherent whole. Since the purpose of the thesis is not codifying existing literature and the review is only a stage in the process of analyzing the five cities, the literature will also be probed for potential limitations, which can inform the forthcoming analysis. Additionally, the review will also serve as a theoretical and conceptual framework, encompassing the definitions of the terms, concepts, and ideas that the thesis is based on.

Nature-based Solutions

To commence the literature review, this thesis will look to defining the main terms – nature-based solutions (NBS), biomimicry, urban, urbane, and sustainability. Of course, there is other terminology that will be used in this work, however, these are the underlying elements which require to be well-defined. The definitions of nature-based solutions are plentiful, and different authors resorted to different approaches to explaining the concept.

Not to stray from the main goals of the literature review; for the purpose of this thesis, a specific interpretation of nature-based solutions will be built – one that will draw inspiration from academia and from practice, as well as balance between the environmental, social, and economic aspects. This will be done in order in order to better posit nature-based solutions and biomimicry, relative to urban sustainability and climate change adaptation and mitigation.

In more recent years, there are O'Sullivan, Mell, and Clement (2020) define nature-based solutions as an approach that relies on natural processes and ecosystems to address urban challenges, such as climate change, biodiversity loss, and social inequality. De Luca, Naumann, Davis, & Tondelli (2021) define nature-based solutions as solutions that are inspired and supported by nature, whose design is intended to address different environmental, social and economic challenges in a sustainable fashion. However, with the type of concept at hand and its usage in practice, it is necessary to also consider how it is defined by the leaders in the governance related to it. The International Union for Conservation of Nature (2021) defines nature-based solutions similarly Laforzezza and Sanesi (2018) as "actions to protect, sustainably manage, and restore natural or modified ecosystems, that address societal challenges effectively and adaptively, simultaneously providing human well-being and biodiversity benefits."

The European Commission (2021) defines nature-based solutions as "solutions that are inspired and supported by nature, which are cost-effective, simultaneously provide environmental, social and economic benefits and help build resilience." And the United Nations Environment Program (2021) defines nature-based solutions as "actions to protect, sustainably manage, and restore natural or modified ecosystems, that address societal challenges effectively and adaptively, simultaneously providing human well-being and biodiversity benefits."

Despite all six of these definitions being similar, all of them approach NBS from slightly different angles. Even more importantly, approaches do not constitute the only marked difference between the definitions; there are

also slightly different foci that can be observed among them. So, what do the definitions put into focus? Among these six definitions, all the authors and agencies seem to see NBS as tools or avenues to take toward a goal. NBS is something than be used or incorporated into an approach that will in turn bring about an improvement with respect to a faced challenge.

The most comprehensive definition is the one by the European Commission (2021). Unlike the other definitions, the European Commission touched upon the economic aspect, which, regardless of one's moral standpoints, plays an integral role in decision-making processes. It includes, in my opinion, all the qualities of nature-based solutions:

- Rooted in the natural world or biophilic design (Kellert, 2011),
- Provide benefits in different spheres
- Contribute to resilience building, and
- Economic or financial legitimacy.

This inclusion of the term cost-effective may come as striking, especially when contrasted with these other definitions. However, nature-based solutions are mostly discussed in terms of infrastructure, and regardless of whether it is infrastructure that is privately or publicly funded, the economic aspect can never be disregarded, as there are no agents who would want to invest more than is necessary. This is the reason why I will refer to this as to economic or financial legitimacy, as there is no scenario in which anyone will undertake any endeavor that is financially unfavorable, *ceteris paribus*. Moreover, as it will be presented in the subsequent sections, there are sometimes difficulties in gathering support for initiatives or techniques even when there are additional factors involved in the equation, such as climate change or overall environmental degradation.

Looking further at the literature on nature-based solutions it is interesting how these studies are conducted. A lot of the studies and articles done within the field of NBS or being tightly related to it utilize the case study method. Among others, Oral et al. (2020), Cohen-Shacham, Walters, Janzen, and Maginnis (2016), Frantzeskaki (2019), Keesstra et al. (2018), and Zwierchowska et al. (2019) have all written about nature-based solutions, and all of these authors have utilized case studies in some shape or form. This served as an indication that approaching nature-based solutions and the topic by developing case studies could be prudent and inspired the methodological approach for this thesis.

Bona, Silva-Afonso, Gomes, Matos, and Rodrigues (2022) and de Luca, Naumann, Davis, & Tondelli (2021) examine nature-based solutions in the European context and posit them in relation to sustainable development, and especially sustainable development of urban areas. Seddon et al. (2020), Cohen-Shacham, Walters, Janzen, and Maginnis (2016), Haase (2017), and Laforteza, Chen, Van Den Bosch, and Randrup (2018) all highlight the different potential of nature-based solutions in addressing societal challenges such as climate change, food security, and health, which is in line with the concept of sustainable development in the Brundtland Report (1987) as well as its more contemporary understandings. In this sense, nature-based solutions can be seen and understood as contributors toward the visions of more sustainable realities, where the environment and humanity need not necessarily be in conflict. This is also in line with Diamond (2005), who presents a broader perspective on sustainability and decay, analyzing how societies can fail or succeed in achieving long-term sustainability goals. There also other authors who touch upon the topic, such as Elmqvist et al. (2019), investigating the governance of the Earth system, emphasizing the need for better management of global sustainability challenges.

Ecosystem Services and Disservices

Within this wider framework of sustainable development and concepts that are of themselves grand, nature-based solutions and green infrastructure are integral components of ecosystem services. Building upon Fisher, Turner, & Morling (2009), ecosystem services can be understood as the benefits that humans enjoy on account of the natural world and healthy ecosystems. Gómez-Baggethun et al. (2013) and Kosanic and Petzold (2020) examine the social and economic values of ecosystem services in urban areas specifically, and argue that while urban ecosystems are often undervalued, they provide a wide range of benefits to human societies, including regulating services such as air and water quality, cultural services such as recreation and aesthetics, and provisioning services such as food and materials. All of this is also in line with the understandings of Potschin, Haines-Young, Fish, & Turner (2018) who provide a comprehensive overview of the concept of ecosystem services and its application in policy and decision-making. They discuss the different approaches to assessing and valuing ecosystem services, and highlight the importance of considering multiple perspectives and trade-offs when making decisions that affect ecosystem services, an approach that is especially relevant for discussions about nature-based solutions also.

Hoyle and Sant'Anna (2020), Wang and Banzhaf (2018), and Venkataramanan et al. (2019) all examine the infrastructure of biodiversity and ecosystem services, and they also stress the vital role of green infrastructure in sustaining ecosystem functions and services that are crucial for human well-being, just as they are for biodiversity. Furthermore, they argue that green infrastructure as an example of nature-based solutions, can provide multiple benefits for both humans and nature. This is again tied to the human-nature relationship, which is within the scope of sustainable development. Maes et al. (2012) consider the mapping of ecosystem services as a tool for policy support and decision-making in the European Union. The argument they make is that by virtue of understanding and mapping ecosystem services, the true value of nature can be integrated into decision-making processes, as well as promote more sustainable approaches to land use and management. Liu et al. (2007) consider this from a slightly different angle and introduce CHANS, or coupled human and natural systems, through which the interactions between the human and natural world can be understood better.

These three articles share a common focus on the importance of understanding and valuing ecosystem services and the role of green infrastructure in sustaining them. They also highlight another important detail, which is the necessity for interdisciplinary approaches in understanding the complexities of relationships between human and natural systems and promotion of sustainable development. Even more importantly in the context of nature-based solutions, they all emphasize the need for incorporating them into decision-making processes, and the consequential more sustainable land use and management practices.

Ernstson and Sörlin (2013) also consider ecosystem services in a global sense and the concept of value articulation. Globalization of ecosystem services presents both challenges and opportunities for the provision of nature's benefits to society. On the one hand, globalization can increase awareness of the interconnectedness of ecosystems and human societies, which may lead to more effective conservation and management practices. On the other hand, globalization can also create governance challenges, such as the need for new institutions and policy frameworks that can effectively manage the provision of ecosystem services across borders.

The diversity of ecosystem services is well-known, and some authors like Norgaard (2010) review not only them but also their antithesis – ecosystem disservices in urban areas. Ecosystem disservices are interesting because they come as the unintended consequences of exploitation or mismanagement of ecosystems. A

prominent example is soil degradation and erosion as results of deforestation. The name of the concept is one that catches attention, however, there is an injustice in it, as it is not the ecosystems that are at the root of these disservices, but rather the mismanagement of these ecosystems, which is of course the responsibility of their stewards.

Ecosystem disservices can also be connected to nature-based solutions and biomimicry, as they can bring about unintended consequences into the urban landscape, though these are objectively minimal. An example that immediately comes to mind are pollen allergies that can be caused by urban gardens. Although the wider intention is positive, increased pollen allergies can occur. In this sense, the works of Chow and Bakar (2019) are interesting, in their discussions about the potential of green roofs as nature-based solutions for sustainable urban development, as well as Nowak et al. (2018) examine the role of trees and shrubs in removing air pollution in urban areas. Furthermore, Pinto et al. (2022) highlight the benefits of urban green spaces for health, and they find that there is a correlation between the two, but paired with the concept of disservices, that health, for example, can be negatively affected by biophilic design. There is also a shared theme among these articles, which is the importance of ecosystem services for human well-being, especially in urban areas, where people are increasingly disconnected from nature.

Bowen and Lynch (2017), Kim and Song (2019), and Coutts and Hahn (2015) write on green infrastructure and land use planning, and find that it can provide a range of ecological benefits, including improving reducing stormwater runoff and providing habitat for wildlife. They also find that GI can provide important social benefits, such as improving public health and well-being, promoting social cohesion, and enhancing aesthetic value. Overall, their article highlights its potential as a nature-based approach to addressing environmental, social, and economic challenges in urban areas, and calls for greater attention and support for its integration into urban planning processes. This is an excellent example of the breadth of benefits of nature-based infrastructure also, as they go beyond the purely environmental which is often at the forefront, and also encompass the social dimension. Hautamäki (2021) conducted a study of NBS in urban planning in Finland and a wider Nordic context. The authors found that NBS can provide a holistic approach to urban planning and create a bridge between Finnish appreciation for nature and incorporating it into the urban environment.

Standard, traditional economic measures might not be able to best encapsulate the value of ecosystem services, especially non-market values like cultural and aesthetic benefits, or the psychological benefits of green spaces, all of which are often overlooked in economic assessments, particularly in the context of urban areas. For instance, a survey of Barcelona, Spain residents, found that what is mostly valued from nature-based solutions are social, cultural, recreational, and health-related benefits (Ramírez-Agudelo, Badia, Villares, and Roca, 2022), which implies all these categories or dimensions should also be considered when valuing and evaluating nature-based solutions.

Similarly, nature-based solutions can enhance human well-being and quality of life, as well as their capacity to provide economic benefits through reduced healthcare costs. Lee and Maheswaran (2011), Aerts, Honnay, Nieuwenhuysen. (2018), and Kondo, Fluehr, McKeon, & Branas (2018) and Vásquez's (2017) conducted research and found the importance of urban green spaces as critical components of healthy, sustainable cities, that simultaneously positively affect human health. While their research supports the potential health benefits of green spaces, more research might be needed to identify effective strategies to improve the quantity, quality, and accessibility of urban green spaces, particularly in disadvantaged neighborhoods.

Referring back to the very definition of nature-based solutions and its economic legitimacy facet, these two aspects work well with each other. Economic values would and should not be completely removed from

considerations about nature-based solutions, but should rather be complemented by socio-cultural values. In fact, I would argue that these non-economic values would only bolster the economic value of the investment in any particular case, as those who are the beneficiaries of the nature-based solution would be more satisfied, both with being included in the planning process, and with the final outcomes as they have co-designed them. These different approaches and foci reveal, if only in part, the beautiful complexity and intersections between the three dimensions of sustainability relating to nature-based solutions.

The types of benefits of nature-based solutions are plentiful and various. Hartmann, Slavíková, and McCarthy (2019), Huang et al. (2020), and Davis and Naumann (2017) outline their use for flood management. Herbert, Haya, Stephens, and Butsic (2022) and Regos, Pais, Campos, and Lecina-Diaz (2023) review solutions for landscape fire management by prescribed burning, fuel treatments, and landscape-scale approaches for reducing fire risk and restoring fire-adapted ecosystems. Hayes et al. (2022) focus on their effects on urban heat island mitigation, similar to Simperler, Ertl, and Matzinger (2020), and Lauwaet et al. (2022) or Turhan and Akkurt (2018) and Regos, Pais, Campos, and Lecina-Diaz (2023), who review the literature on urban heat islands. Gómez-Baggethun et al. (2013), Marando, Salvatori, Fusaro, and Manes (2016), Prigioniero, Zuzolo, Niinemets, and Guarino (2021), and Gopalakrishnan, Ziv, Hirabayashi, and Bakshi (2019) focus on the improvement of air quality by virtue of nature-based solutions, as well as reducing heat island effects, enhancing biodiversity, and promoting human well-being. Bottalico et al. (2016) and Nowak et al. (2018) consider the specific effects that urban forests have on addressing air pollution.

Oral et al. (2020) and Wild, Henneberry, and Gill (2017) reviewed the opportunities and limitations of nature-based solutions in water management and found that they can provide a cost-effective and sustainable approach for water management – with the potential to reduce water pollution, provide habitats for biodiversity, and enhance the aesthetic value of the landscape. Davis and Naumann (2017), Hartmann, Slavíková, and McCarthy (2019) and Huang et al. (2020) found that NBS-based measures such as floodplain restoration, green roofs, and permeable pavements are effective in reducing the impact of floods, but that their effectiveness varies with the context and specific characteristics of the site and recommend that a context-specific approach should be adopted in the design and implementation.

It is obvious that nature-based solutions can take different shapes and forms, and that their applications are also very broad. Considering this diversity in application and then also the benefits that different authors have identified, nature-based solutions have or at least can have a strong impact on urban sustainability. Helmrich et al. (2020) or Taylor Buck (2017) explored the relationship between biomimicry and green infrastructure, which by extension also applies to nature-based solutions. The authors argue that biomimicry can provide a design approach for green infrastructure that mimics the functional and structural properties of natural systems. This can enhance the effectiveness of green infrastructure in providing ecosystem services and improving urban sustainability and can serve as the link between nature-based solutions and biomimicry.

Biomimicry

At this point, the literature review will move toward biomimicry. Biomimicry is a concept under which nature informs decisions and offers solutions to various problems, and in this context especially with urban areas. Authors like Köktürk (2021) note that biomimicry possesses potential benefits of applying biomimicry in various fields, including engineering, architecture, and product design. Though this thesis focuses on the urban, architecture is not the only field of application. Indeed, engineering plays an instrumental role in achieving that cities and their infrastructure are providing the maximum of benefits to their residents. Product

design might seem less interesting than architecture for this context, but depending on the scale of the biomimetic project, product design can also be utilized and applied within the field of urban planning.

Benyus (1998) is an excellent starting point for reading on biomimicry. The author highlights the importance of looking to nature for creative solutions to human problems and argues that nature has already solved many of the challenges human societies face. The author also explores the ethics of biomimicry and the importance of preserving natural systems as the source of inspiration for human innovation. The author suggests the design of cities should follow the principles of natural ecosystems, where materials and energy are recycled and regenerated, and waste is minimized. The following examples of how can nature inspire sustainable urban design, including:

Vertical gardens and green roofs inspired by the way that plants use light, water, and nutrients in natural ecosystems to help with filtering air and water and reduce urban heat island effects	Hydroelectric power systems modeled after the way that salmon swim upstream
Green walls to mitigate noise and air pollution	Wave energy systems inspired by the way that kelp beds move in the ocean currents
Urban transportation systems inspired by the way that ants and other social insects coordinate their movements	Waste management systems inspired by the way that decomposers in natural ecosystems break down organic matter and recycle nutrients
Renewable energy systems inspired by the way that plants convert sunlight into energy through photosynthesis. - Wind turbines based on the shape and motion of humpback whale fins	Designing buildings that mimic the self-regulating, self-healing processes of natural systems. For example, buildings that can "breathe" like living organisms to regulate temperature and humidity
Energy storage systems based on the way that camels store water in their humps	Learning from the social organization and collective intelligence of natural systems to design more resilient, adaptive, and collaborative cities
Solar panels inspired by the structure and function of leaves	Using biomimetic materials and structures to reduce energy use and carbon emissions in urban areas. For example, using materials that mimic the insulating properties of polar bear fur or the self-cleaning properties of lotus leaves

Primrose (2020), on the other hand, writes about biomimetic materials, and defines them as inspired by natural systems and processes, and covers various topics, including structural and functional materials, nanomaterials, and biomimetic sensors and actuators, while also covering the roles of biomimetic materials in energy and sustainability, and other fields related to urban sustainability. Some biomimetic materials mentioned in the book include:

- Materials inspired by the structure of bones, such as bone cements and bone grafts
- Materials inspired by the structure of shells and exoskeletons, such as lightweight composites and protective coatings
- Materials inspired by the structure of plants, such as self-healing polymers and smart materials that respond to environmental stimuli

- Materials inspired by the structure of insects and animals, such as adhesives, sensors, and actuators that mimic the properties of natural systems

Biomimicry obviously has a variety of applications, and among of the more interesting applications of biomimicry that are the potential for public transportation improvement through biomimicry and built infrastructure with the ability of self-regeneration and self-regulate. Swarm intelligence, observed in ants can be used to optimize traffic flow, thus enabling for less traffic congestions (Nguyen and Jung, 2021). However, in order to have functioning transportation systems, the importance of integrating transportation systems with urban design to create more walkable and bikeable cities is also important. Self-healing materials such as concrete that can self-repair cracks and steel that can heal when exposed to heat are exquisite examples of self-regenerating materials. Benyus (1998) also discusses the use of smart materials in buildings, which can change their properties in response to external stimuli such as temperature or moisture. Furthermore, development of self-regulating building systems, which use feedback mechanisms to control the temperature, humidity, and lighting within buildings, drawing inspiration from termite mounds to regulate the temperature inside the building without the use of air conditioning.

Going more into the materials, Primrose (2020) gives examples of microcapsules filled with a healing agent embedded in a material or vascular systems. When a crack forms, the capsules rupture and release the healing agent, which reacts with the material to repair the damage. The use of vascular systems in materials, which are similar to the circulatory system in living organisms. When a crack forms, a fluid is released from the vascular system, which solidifies upon contact with air to repair the damage. The author also discusses the use of biomimetic materials for heat insulation in the context of building construction through "cool roofs," which are designed to reflect sunlight and reduce heat absorption. Inspired by the thermoregulatory properties of the Saharan silver ant's body a coating material is developed to reflect sunlight and radiate heat away from the surface, helping to cool buildings in hot climates. Other examples include the use of materials that mimic the structure of polar bear fur to provide insulation and the development of "smart windows" that adjust their transparency to regulate temperature and light levels based on environmental conditions.

Now, there is already an overview about the characteristics of biomimetics and biomimicry, but without a definition, as well as without a clarification about the potential difference between the two terms. Lurie-Luke (2014) understand biomimetics as a field of study that tries to uncover the underlying principles and mechanisms found in biological systems to and then adapt them to design new technologies, materials, and processes. It encompasses a wide range of disciplines, including biology, chemistry, materials science, engineering, and physics, and can be used to develop new products, optimize existing processes, and solve a variety of problems in different industries. Similarly, according to Sarikaya (1994) biomimetics is an interdisciplinary field that involves the study of natural systems and processes with the aim of imitating or adapting them for use in technology. It draws inspiration from biology, physics, chemistry, and materials science to design new materials, structures, and devices that can perform specific functions with greater efficiency and sustainability than existing technologies.

On the other hand, Benyus (1998) defines biomimicry as the practice of studying natural systems and processes to develop innovative solutions to human problems. It involves the application of biological principles, models, and systems to design products, technologies, and processes that are more efficient, sustainable, and resilient. Biomimicry draws on a wide range of disciplines, including biology, chemistry, materials science, engineering, and design, to create new approaches to innovation that are inspired by nature. Another definition comes from Pawlyn (2011) as the emulation of natural systems and processes for the purpose of solving human problems. It involves the study of biological models and systems, and the

application of their underlying principles to the design of products, technologies, and systems that are more sustainable, efficient, and resilient. Biomimicry draws on multiple fields, including biology, physics, chemistry, and engineering, to create innovative solutions that are inspired by the wisdom of nature.

The terms are sometimes used interchangeably in academic literature. For instance, “The promise of biomimicry is aligned with the “biomimetic promise”” (MacKinnon, Oomen, & Pedersen, 2020); “Biomimetics (which we here mean to be synonymous with ‘biomimesis’, ‘biomimicry’, ‘bionics’, ‘biognosis’, ‘biologically inspired design’ [...])” (Vincent, Bogatyreva, Bogatyrev, Bowyer, & Pahl, 2006); or “Biologically inspired design [...] is referred to as ‘biomimetics’. [...] Other words used include [...] biomimicry and biognosis.” (Bhushan, 2009).

Despite this usage, which is not entirely inaccurate, there are some subtle differences between the two concepts. Based on the definitions above, biomimicry is a design approach that involves emulating or copying natural systems, structures, and processes to create solutions for human problems, while biomimetics, on the other hand, is a broader field that encompasses biomimicry as well as other approaches to studying and utilizing biological systems. In other words, biomimicry is a specific application of biomimetics that involves using nature as a source of inspiration for design, while biomimetics encompasses a broader range of approaches to studying and utilizing biological systems.

The relevance of biomimicry, and by extension of biomimetics, for urban sustainability has already been researched and written about. Lepora, Verschure, and Prescott (2013)’s work notes the great promise of the field of biomimetics with significant potential for advancing sustainable innovation across multiple disciplines. They also, conversely, note that there are challenges to be addressed in the field, such as the need for interdisciplinary collaboration, the ethical implications of manipulating living organisms, and the need to develop new methods for translating biological principles into practical applications. Since this thesis addresses urban built environments, the manipulation of living organisms is not such a concern.

Verbrugghe, Rubinacci, and Khan (2023) also wrote about biomimetics and presented biomimetic design as a tool for creating sustainable buildings that draw inspiration from nature. Similarly, Jamei and Vrcelj (2021) and Beatley (2011) argue that incorporating nature's wisdom into urban design and planning can lead to more sustainable and resilient cities. Reed (2003) discusses how biomimicry is changing the way we think, live, and do business, highlighting the potential of biomimetic innovation to address global challenges. Sá and Viana (2023), Kanwal and Awan (2020), Alexandridis, Tzetzis, and Kyratsis (2016), and Ilieva et al. (2022) also argue that biomimicry possesses the potential to drive innovation in product design and sustainability.

Building upon Taylor Buck (2017) and Dicks et al. (2021) several potential benefits of applying biomimicry in urban design and planning, including improving the sustainability and resilience of urban systems, enhancing ecosystem services, fostering innovation and creativity, and enhancing human well-being can be identified. Biomimicry has the potential to help urban designers and planners create more efficient and effective urban systems, by learning from and emulating the strategies and processes of natural systems. However, there are also several challenges associated with applying biomimicry in urban contexts, including the need for interdisciplinary collaboration, access to and understanding of scientific knowledge, and the need for shift in the way urban design and planning are approached. This article can serve as a confirmation for the work of Uchiyama, Hayashi, and Kohsaka (2015), who explored the potential of biomimicry as a tool for sustainable urban design and argued that by emulating natural systems, urban design can become more sustainable, but did not provide substantial evidence for these claims.

These articles have a common theme in referring to resilience and sustainability, both relevant concepts for this thesis, and both also related to nature-based solutions. Sustainability and resilience are intrinsically connect, but even more importantly, one must take away the idea that sustainability cannot be achieved without a paradigm shift. Without redesigning the way societies think about all three pillars of sustainability, it is difficult they will ever be synchronized, and consequently that truly resilient systems can be built. McDonough and Braungart (2002) presented a framework for sustainable design based on the principles of "cradle to cradle," which might even provide an answer to wider sustainability considerations and questions.

Sustainability is also connected to innovation on a certain level, and innovative solutions are necessary to answer some challenges within the realm of sustainability. Alexandridis, Tzetzis, and Kyratsis (2016), for instance, highlight the potential of biomimicry in product innovation design, while Pawlyn, (2019), Oguntona and Aigbavboa (2017) and Rao (2014) reviewed the use of biomimicry principles in sustainable building design. They found that biomimicry can inform design in various areas, including energy efficiency, ventilation, lighting, and water management, among others. Now, biomimetics are also often put in the same context as innovation, and when considering the value of innovation and creativity, the link between sustainability and biomimetics becomes even more clear.

Based on the writings of Pohl and Nachtigall (2015), Dixit and Stefańska (2022), Ramzy (2015), Chayaamor-Heil (2023), Čolić-Damjanovic and Gadjanski (2016), and Gruber and Jeronimidis (2012), it is possible to identify several differences between biomimetic and traditional architecture. Here are some of the main differences:

Design process	Traditional architecture tends to rely on a linear design process, where the architect develops a concept and then creates a set of plans to realize that concept. In contrast, biomimetic architecture involves a more iterative design process, where the architect seeks inspiration from natural systems and uses biomimetic principles to guide the design process.
Materials	Traditional architecture often uses materials that are readily available and easy to work with, such as wood, brick, and concrete. In contrast, biomimetic architecture often looks to nature for inspiration in material selection, resulting in the use of novel materials and construction techniques.
Structural systems	Traditional architecture often uses a few standard structural systems, such as beams, columns, and trusses. In contrast, biomimetic architecture seeks inspiration from natural structures and systems, resulting in a wider range of structural systems that are more efficient and sustainable.
Energy efficiency	Biomimetic architecture often seeks to emulate the energy-efficient features of natural systems, such as passive heating and cooling, and natural ventilation. Traditional architecture may not prioritize energy efficiency to the same extent.
Aesthetics	Traditional architecture may prioritize aesthetic considerations over environmental concerns, resulting in buildings that are visually striking but may not be as environmentally sustainable. In contrast, biomimetic architecture seeks to balance aesthetic considerations with environmental concerns, resulting in buildings that are both visually appealing and environmentally sustainable.

Now, if nature-based solutions draw inspiration from nature and incorporate nature into the built environment and biomimicry is based on studying natural systems and using that knowledge to develop solutions to problems, there is clearly an overlap between the two. In a sense, all nature-based solutions are inherently products of or can be understood as biomimicry. Biomimicry, as a larger concept then, encompasses nature-based solutions as well.

Having covered several essential concepts related to the topic, the review will now look at literature on these concepts but in a more integrated manner. The report by Ten Brink et al. (2016) presented the potential of natural landscapes as sources of environmental and social benefits. They found that natural landscapes can provide numerous benefits such as clean water, air purification, biodiversity conservation, climate regulation, and mental health promotion.

This makes evaluating the effectiveness and effects of nature-based solutions and biomimetic projects on the urban environment more challenging, as these trade-offs need to be considered as well. Generally, this process is not simple due to the variety of factors that need to be considered in the evaluation process. Some assessment tools used to gauge urban sustainability are:

Connecting Nature Framework	The Connecting Nature Framework was developed to provide a comprehensive assessment of nature-based solutions in urban areas, taking into account a range of social, economic, and environmental factors (Collier et al., 2023).
LEED (Leadership in Energy and Environmental Design)	LEED, a green building rating system developed by the U.S. Green Building Council that assesses the sustainability of buildings across a range of categories, including water efficiency, energy and atmosphere, materials and resources, and indoor environmental quality (Li et al., 2021).
BREEAM (Building Research Establishment Environmental Assessment Method)	BREEAM, a sustainability assessment tool developed in the UK that assesses the environmental performance of buildings and other developments across a range of categories, including energy and water use, materials, pollution, and health and well-being (Felicioni, Lupišek, and Gaspari, 2023).
City Biodiversity Index	City Biodiversity Index, developed by the Convention on Biological Diversity to assess the biodiversity of cities and identify opportunities for integrating biodiversity into urban planning and development (Das, Shukla, Dhyani, 2022; Uchiyama, Hayashi, Kohsaka, 2015; Elmquist et al., 2019)
Global Reporting Initiative (GRI) Sustainability Reporting Standards	These standards provide a framework for organizations to report on their sustainability performance across a range of categories, including economic, environmental, and social performance (del Mar Alonso-Almeida, Llach, and Marimon, 2014).
Green and Blue Infrastructure (GBI) Assessment Framework	GBI Assessment Framework, developed by the European Commission's Directorate-General for Environment. The GBI framework provides a structured approach to assess the benefits and costs of green and blue infrastructure, including nature-based solutions, and to integrate these into policy and decision-making processes (European Commission, 2019).
Integrated Valuation of Ecosystem Services and Tradeoffs (InVEST) framework	InVEST framework is another assessment tool that can be used to evaluate the impacts of nature-based solutions on urban sustainability. InVEST is a spatially-explicit tool that can assess the trade-offs and synergies between different ecosystem services and help to identify the most effective nature-based solutions (Tallis and Polasky, 2009).

Overall, there are four main categories across which sustainability can be assessed, that are relevant to this thesis:

- Environmental sustainability: This category includes indicators related to the environmental impacts of urban development, such as air quality, water quality, and waste management.
- Social sustainability: This category includes indicators related to social well-being, such as access to education, healthcare, and cultural amenities, as well as social equity and inclusion.
- Economic sustainability: This category includes indicators related to economic well-being, such as employment, income, and economic diversity.
- Resilience: This category includes indicators related to the ability of urban areas to adapt and respond to environmental, social, and economic stresses and shocks.

Climate Change

O'Hogain and McCarton (2018), Cohen-Shacham, Walters, Janzen, and Maginnis (2016), and Maes and Jacobs (2017) all wrote about the potential of nature-based solutions to reduce urban temperatures, sequester carbon, and reduce the risk of flooding. They conclude that nature-based solutions have the potential to contribute to climate change mitigation and adaptation in urban areas, although their effectiveness depends on the specific context of each city. The terms climate change adaptation and mitigation have been popularized and defined by the United Nations Framework Convention on Climate Change (UNFCCC). According to UNFCCC (1992), climate change mitigation refers to "human intervention to reduce the sources or enhance the sinks of greenhouse gases," while climate change adaptation refers to "adjustment in natural or human systems in response to actual or expected climatic stimuli or their effects, which moderates harm or exploits beneficial opportunities."

In other words, climate change mitigation are steps taken to avert or slow down climate change, while climate change adaptation are steps taken to adjust the new realities wrought by climate change and prepare to endure these changes appropriately. Nature-based solutions and biomimicry can contribute to both mitigation and adaptation, and there are tools for planning both these processes.

An effective framework is that for climate change adaptation, developed by Major and Juhola (2021, p. 18). The wider work has already been referenced in some academic works, such as Krauss (2022), Lehmann et al. (2021), or Doust et al. (2021). The framework is comprised out of eight steps:

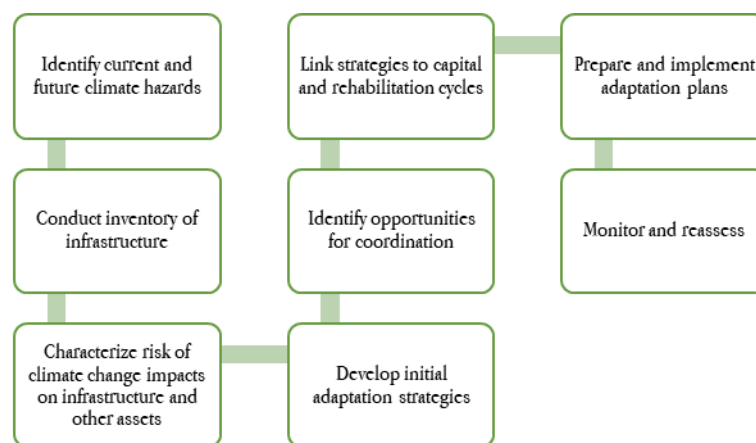


Figure 1. Framework for Urban Climate Change Adaptation. (Adapted from Major and Juhola, 2021. Visualization prepared by author.)

Aside from the many benefits of nature-based solutions and biomimicry, there are also challenges that are already discussed in academic literature. Some of these challenges, and perhaps shortcomings of nature-based solutions are discussed by Seddon et al. (2020), Croeser et al. (2021), and Nelson, Bledsoe, Ferreira, and Nibbelink (2020), and Bona, Silva-Afonso, Gomes, Matos, and Rodrigues (2022):

- Issues of participation and equity – a common concern in urban planning whereby the process of selection and implementation of certain solutions without the participation of those who the solutions is considered. Then, even if the ability to participate in decision making processes is given, one must also consider whether everyone has equal access to this platform. Ultimately, this is a concern even when the solutions are implemented, as it raises the question of who can participate in their usage also arises.
- Economic valuation and cost-effectiveness – this is something that arose as a result of this research as well, and that is that it is difficult to assess the economic benefits and value of nature-based solutions, without concrete information about the exact value of the projects themselves, and when data is not collected about exact economic effects of the solutions. This, however, does not go to say that there are no economic benefits, but that more quantitative data would be desirable in order to make more broad conclusions.
- Scalar and topical mismatches – finding the right type of solution, with the correct type of vegetation, and with the correct desired outcomes can be challenging, especially in cities with no prior experience in implementing such solutions. A concern arises especially if the, for instance, the selected type of vegetation becomes invasive or cannot thrive in the environment it is introduced to.
- Integration of natural and built infrastructure – this concern relates to both practicality and aesthetics, as nature-based solutions can and should not be something independent from existing infrastructure. In this sense, it is important to appropriately place the solutions, so that they deliver the desired benefits, as well as fit in well with the surrounding.
- Governance – the concerns arise with respect to who and how designs, manages, and decides on the solutions, as well as the policy surrounding nature-based solutions.
- Lack of monitoring – whereby, similar to the economic valuation category, it is challenging to draw objective conclusions without continued monitoring of the solutions' implementation.

The topic at hand is quite broad and encompasses a variety of inter-connected topics. This literature reviewed some of the most important academic sources and mapped out the conceptualization and understanding of all main facets of the topic. In addition to academic literature, the review also included several international organization sources when defining certain terms, as some of the come from practice, and not academia. Therefore, when defining terms, it was important that they are understood the way it was intended by those who developed them.

METHODOLOGY

This study will use a comparative case study analysis to evaluate the effectiveness and implementation of nature-based solutions and biomimicry in improving urban sustainability in five cities: Cape Town, South Africa; Madrid, Spain; London, United Kingdom; New York City, United States; and Seoul, South Korea.

Data collection methods

Data collection methods will include both secondary and primary sources. Secondary sources of data will be obtained from relevant literature, government reports, academic journals, and online databases. This will provide an understanding of the current state of knowledge and existing literature on the use of nature-based solutions and biomimicry in improving urban sustainability.

Primary sources of data will include data collected through structured observations and document analysis. This will provide information on the current policies, initiatives, and projects related to nature-based solutions and biomimicry in each of the five cities.

Data collection methods for this study will include several different strategies to gather information about the effectiveness and implementation of nature-based solutions and biomimicry in improving urban sustainability in the selected cities. These methods include:

1. **Literature Review:** A comprehensive review of the existing literature and research studies on the topic will be conducted. This will involve a systematic search of academic databases, such as SCOPUS, Web of Science, JSTOR, Google Scholar, and others, to identify relevant articles, reports, and studies. The aim is to gain an understanding of the current state of knowledge on the use of nature-based solutions and biomimicry in urban sustainability, and to identify the key challenges and opportunities in this field.
2. **Document Analysis:** Official documents, such as city plans, sustainability reports, and policies, will be analyzed in each of the selected cities. This will provide valuable information on the current strategies and approaches to urban sustainability, and how nature-based solutions and biomimicry are being used in these cities.
3. **Direct Observation through site visits:** Direct observation of selected nature-based solutions and biomimicry initiatives in each of the cities will be conducted. This will provide an opportunity to understand the implementation of these solutions in practice, and the challenges and opportunities that exist in their implementation. Site visits to selected initiatives, such as green roofs, urban parks, and other nature-based solutions, will be conducted in some of the cities. This will allow for a deeper understanding of the design, construction, and management of these solutions, and their impact on urban sustainability.
4. **Secondary Data Analysis:** Secondary data, such as census data, air quality data, and climate data, will be analyzed in each of the cities, where available. This will provide an understanding of the current state of the urban environment, and the impact of nature-based solutions and biomimicry on urban sustainability. Statistical, historical data, was obtained from national statistical bureaus, and the cities' open data platforms.

By using a combination of these methods, this study will provide a comprehensive understanding of the effectiveness and implementation of nature-based solutions and biomimicry in improving urban sustainability in the selected cities, and provide valuable insights for potential further research in this field.

Data Analysis Methods

The collected data will be analyzed using a combination of qualitative content analysis and comparative analysis. The content analysis will be used to identify themes and patterns in the implementation and outcomes of nature-based solutions and biomimicry in the five cities. The comparative analysis will be used to compare and contrast the findings from the five case studies and to identify commonalities and differences in the implementation and effectiveness of nature-based solutions and biomimicry in the cities. This will involve identifying and categorizing themes, patterns, and relationships in the data.

Sampling Strategy

A purposive sampling strategy was used for the selection of the five cities for the comparative case study analysis. The five cities were chosen based on their reputation as global cities, their commitment to sustainability, and their track record in implementing nature-based solutions and biomimicry. The cities, with respect to their country scale, are also comparable population- and size-wise.

Cape Town, South Africa: known for its unique and diverse environment, including Table Mountain, one of the world's most iconic natural landmarks. The city also faces major environmental challenges, including water scarcity and degradation of natural ecosystems, making it an ideal case study for evaluating the effectiveness of nature-based solutions and biomimicry in addressing environmental challenges in a developing city context.

Madrid, Spain: the capital of Spain and one of Europe's largest cities, Madrid has a rich history and a diverse cultural heritage. The city has also been proactive in implementing sustainability initiatives, including nature-based solutions and biomimicry, making it an interesting case study for evaluating the impact of these approaches on urban sustainability.

London, United Kingdom: one of the largest and most influential cities in Europe, London has a long history of urban innovation and sustainability leadership. The city has also been proactive in implementing nature-based solutions and biomimicry to address a range of environmental and sustainability challenges, making it a valuable case study for evaluating the impact of these approaches on urban sustainability.

New York City, United States: one of the largest and most influential cities in the world, New York City represents a unique blend of cultural, economic, and technological elements. The city has a history of innovation and a commitment to sustainability, making it an interesting case study for evaluating the role of nature-based solutions and biomimicry in improving urban sustainability in a large and complex urban environment.

Seoul, South Korea: the capital and largest city in South Korea, Seoul represents a unique blend of traditional and modern cultural, economic, and technological elements. The city has a history of implementing innovative urban sustainability solutions and is known for its ambitious sustainability goals, making it an ideal case study for evaluating the role of nature-based solutions and biomimicry in urban sustainability.

By selecting these five cities, the thesis will provide a diverse and representative sample of the global experience with nature-based solutions and biomimicry in urban sustainability, thus purporting meaningful conclusions and recommendations for future research and practice.

Limitations

This study has some limitations related to data collection and analysis. Even though one would expect an abundance of information being generated in and about cities every year, there seems not to be a lot of data that can be analyzed in order to assess urban sustainability. London and New York City had some data available about air pollution, population, and unemployment (enabling an insight into the three dimensions of sustainability), but the other three cities did not have any official data that could be used. This is an unfortunate fact because the idea about the challenges and the situations in the cities is based on qualitative data and publications. There was data available on national levels, however, data on sub-national levels is either not available or not published. National statistical agencies/bureaus were consulted together with the city's open data platforms in both English and local languages, but the historical data could not be found.

Furthermore, sustainability is very often influenced by a more complex set of factors. These factors sometimes play a negligent role, though sometimes they also have a more prominent role in setting the premises for a city to be sustainable. Despite these limitations, this study will provide valuable insights into the effectiveness and implementation of nature-based solutions and biomimicry in improving urban sustainability in the selected cities, and contribute to the existing literature in this field.

CASE STUDIES ANALYSIS

This section will present the five selected case studies. First, each city's profile will be presented, noting all the most relevant information about the city, its current sustainability- and climate change-related challenges. Then the case study will provide detailed descriptions of selected nature-based solutions and biomimicry projects, followed by some regulatory or legislative initiatives in each city.

New York

New York City is the largest city in the on the United States' East Coast, and with 8,467,513 residents in the city itself it is the most populous city in the United States of America (U.S. Census Bureau, Population Division, 2022). New York City has five boroughs – Manhattan, Brooklyn, Queens, the Bronx, and Staten Island; the Bronx being the only borough that is not an island. The reason New York City has been selected to be featured in this study is, among others, its importance for the entirety of the United States, its size, and the fact that as a coastal city it is especially vulnerable to climate change. New York is also one of the most places in the United States, with a strong immigration influence, beautiful architecture, it is the seat of two major universities – Columbia University and New York University (NYU), and the seat of the United Nations Headquarters.

The Hudson river separates the City from the state of New Jersey. The East River separates Manhattan and Bronx from Long Island, while the Harlem River separates most of Manhattan from the Bronx. Both the East River and the Harlem River are tidal straits, meaning they are not actual rivers in nature, but rather connect water bodies in between two land masses. Lastly, the Bronx River, though not separating any boroughs, is the only entirely freshwater river in the city (Berger, 2010). This leaves the city with four rivers, all emptying into the Atlantic Ocean. The mouth of the Hudson River is also the location of New York Harbor, which is one of world's largest natural harbors (Lopate, 2008). New York City's climate is humid subtropical (Cfa) as per the Köppen climatic classification. Winters are normally cold and the summers are hot, whereas rainfall is stable and evenly distributed throughout the seasons (Weather Atlas, n.d.).

In Figure 2. below, one can see that the population of New York is generally growing, with a significant change in the growth rate around the middle of the observed period. However, toward the end of the observed period, one can see that the trend is changing, and that the population is slowly decreasing. It is important to note that the graph shows only the population of the five city boroughs and not the metropolitan area population.

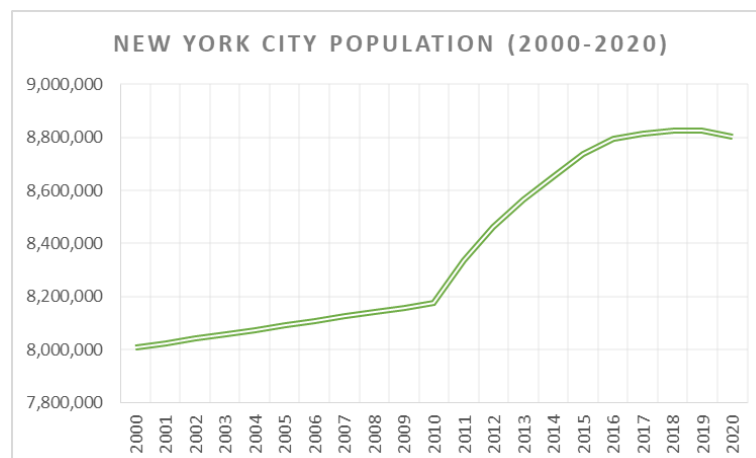


Figure 2. New York City Population (2000-2020). (Data source: DEP, 2022. Data visualization prepared by author.)

Given that water scarcity is among issues often tied to future climate change scenarios, it is interesting to observe the current water usage patterns of the city’s residents, in Figure 3. below The daily per capita water consumption in New York City is 111 gallons (approximately 420 liters) from 2020 is somewhat lower than the U.S. average of 156 gallons, but it is still several larger than the averages in India or Mali, at 38 and 3 gallons respectively (CDC, 2020).

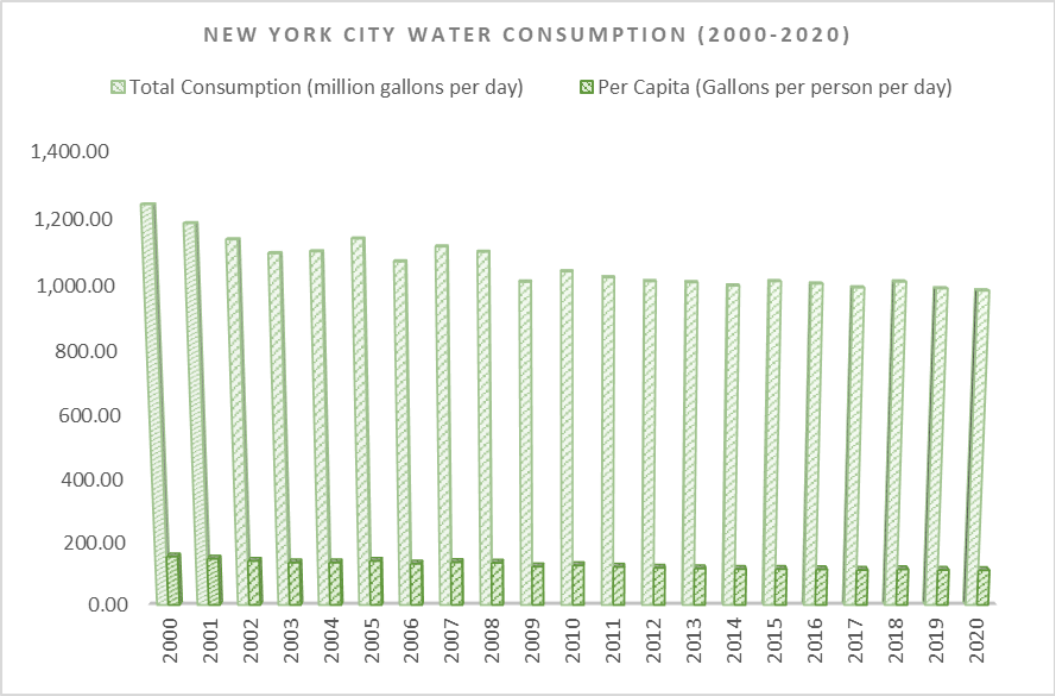


Figure 3. New York City Water Consumption. (Data source: DEP, 2022. Data visualization prepared by author.)

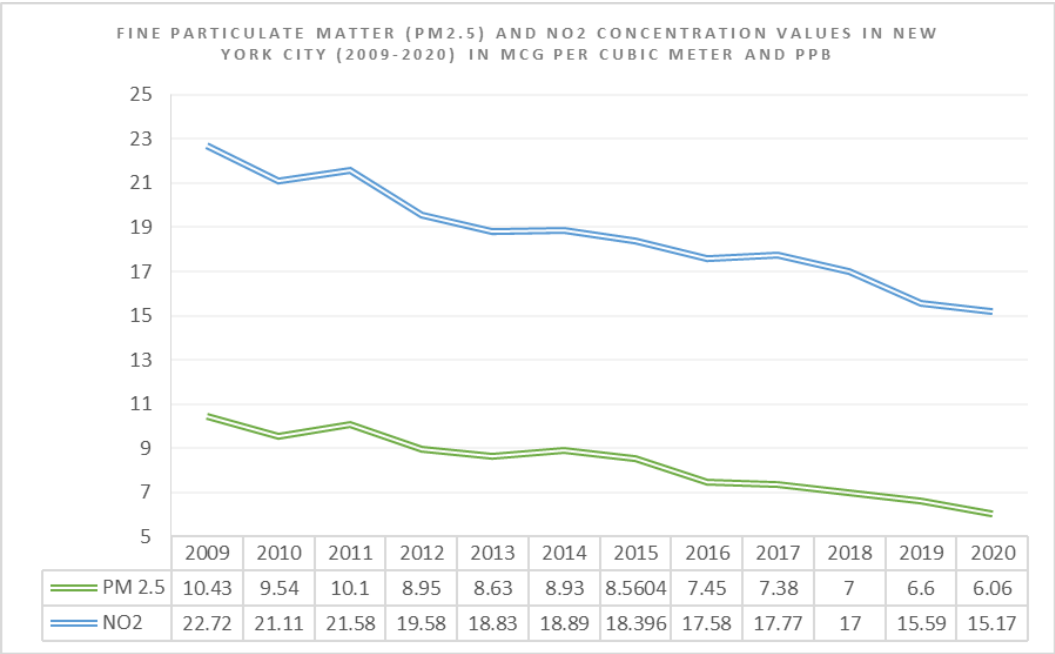


Figure 4. New York City Air Pollutants PM2.5 and NO2 values (2009-2020). (Data source: DOHMH, 2022. Data visualization prepared by author.)

In Figure 4. above, one can see two major pollutants' values – PM2.5 and NO2, between 2009-2020. Both pollutants' air concentrations are decreasing, despite the increase in population in the city. A similar trend can be observed in Figure 5. below, for SO2, the concentration of which has also been decreasing.

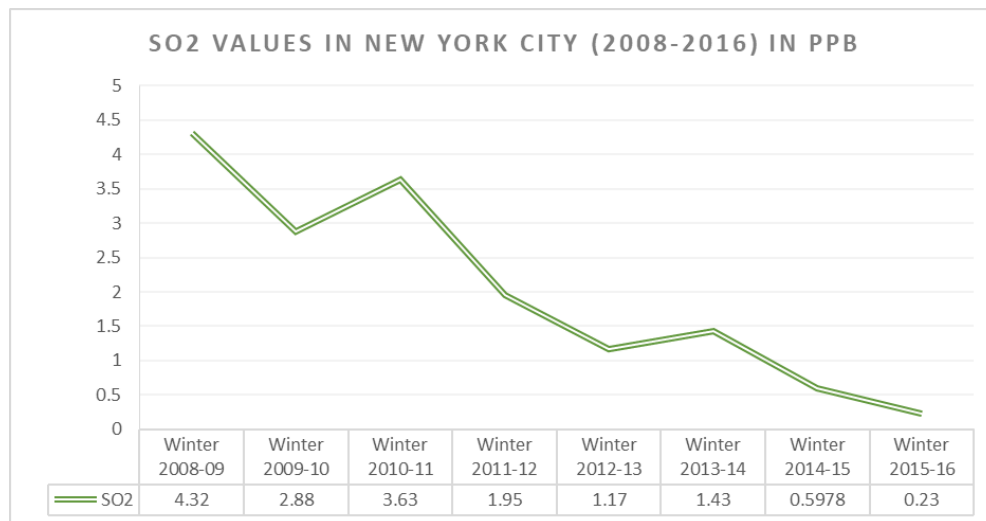


Figure 5. New York City Air Pollutant SO2 values (2008-2016). (Data source: DOHMH, 2022. Data visualization prepared by author.)

Despite these generally positive trends, which are also demonstrated in the trends of health effects of PM2.5, there are still health concerns caused by this pollutant. For instance, between 2015-2017, over 78 per every 100,000 children ended up at the Emergency Department due to PM2.5-related Asthma issues, as presented in Figure 6. below.

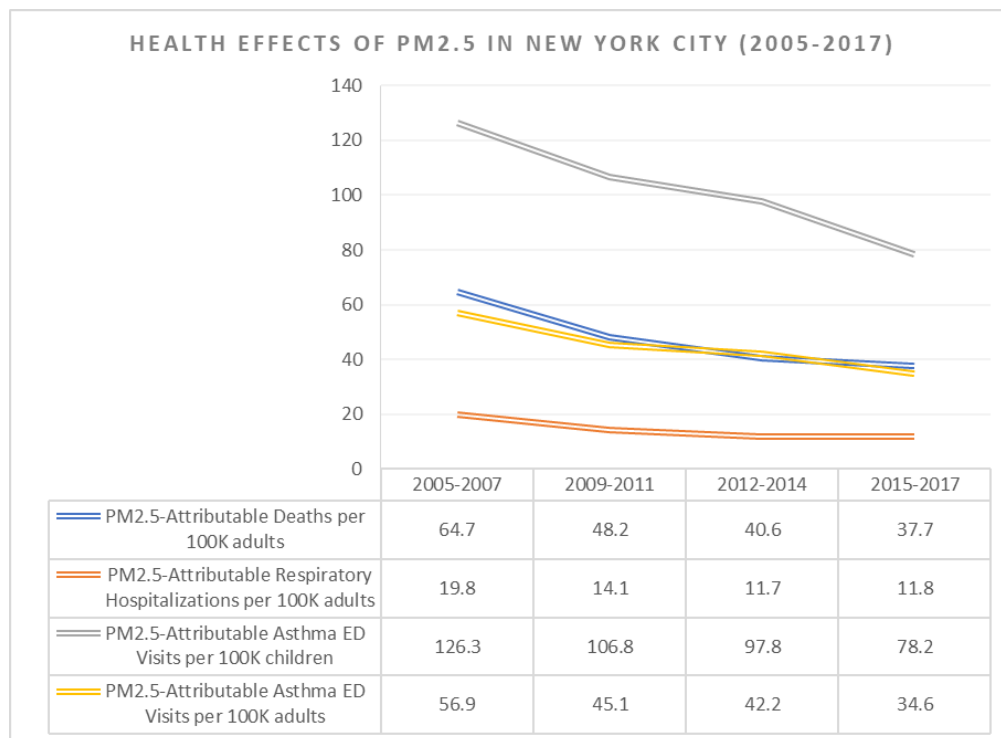


Figure 6. Adverse Health Effects of PM2.5 in New York City (2005-2017). (Data source: DOHMH, 2022. Data visualization prepared by author.)

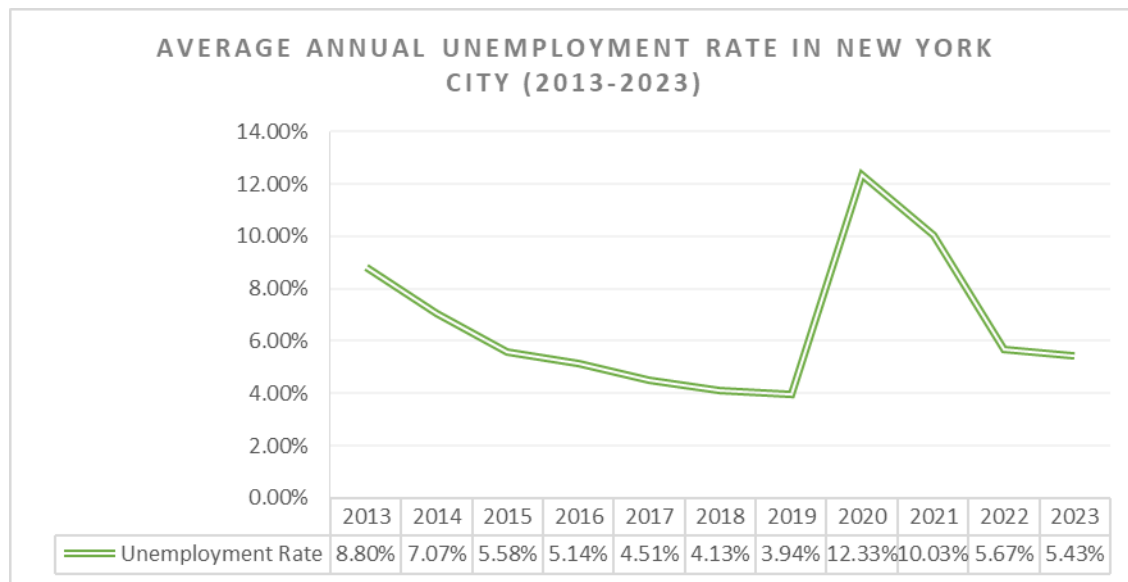


Figure 7. New York City Unemployment Rate. (Data Source: BLS, 2023. Data visualization prepared by author.)

In Figure 7. above, one can also see the city’s unemployment rate, which is now showing a decreasing tendency. What is positive is that the city seems to slowly be recovering from the pandemic employment-wise, as unemployment is almost back to its pre-pandemic levels.

In the last years, New York has experienced several major sustainability challenges, as well as climate change effects, including (Bernard, 2022; Mayor's Office of Sustainability, n.d.):

- Floods – results of climate change and general extreme weather events
- Availability of clean and healthy air, water, and food
- Energy sustainability – greenhouse gases emissions and clean energy

These challenges are only some of the issues that the city is facing, and also some of the ones that the deployment of nature-based and biomimetic solutions can contribute to solving.

Sustainability and Climate Change Policies

New York City has a fully integrated strategy for its future – OneNYC 2050 by New York City's Mayor's Office (2019). This strategy comprehensively addresses all areas of sustainability and how the city relates to them, while simultaneously also presenting the ways in which it plans to deal with climate change mitigation and adaptation. The strategy is conceptualized so that envisions the city in 2050, through several thematic areas. However, the strategy is not organized in a way that facets of sustainability are separated, but all three facets of sustainability are applied through the visions in an integrated manner.

Out of the focus areas from the OneNYC 2050 strategy, these are the ones where nature-based and biomimicry solutions can be used – inclusive economy, thriving neighborhoods, healthy lives, livable climate, and modern infrastructure, despite the term being used only several times in the entire document. Expectedly, the solutions are mentioned under the Livable Climate’s Initiative 21 that addresses mitigation of physical risks of climate change. Nature-based solutions, such as restored wetlands and forests, are seen as an ailment to hazard mitigation, like coastal floods or erosion.

However, although the strategy primarily mentions nature-based solutions only briefly, the application and application potential of nature-based and biomimicry solutions in the strategy is broad. The strategy plans on restoring and conserving the city's natural environment by improving over 250 acres of forested areas annually, planting over 5,000 trees in neighborhoods with highest exposure to heat-related risks, and these have been milestones only until the end of 2021. Aside from these instances, the city has the potential to use the solutions to deal with noise pollution, upgrading existing infrastructure, and improving parks and green recreational areas. Most importantly, perhaps, the city can use nature-based solutions and biomimicry to address flooding hazards. Some of the solutions already used in the city can be seen below.

Nature-based and Biomimetic Projects

Table 1. below presents the basic information about the reviewed biomimetic and nature-based solutions analyzed in New York City. Each solution's description is paired with features that make it nature-based and/or biomimetic, as well as the cost of the projected. Information about the costs could not be found for all the reviewed projects.

Table 1. Solutions from New York City. (Prepared by author)

New York	Project Name	Brief Description	Relevant Nature-based and Biomimetic Features	Cost
1	The Living Breakwaters Project	Underwater breakwaters aim to protect the local community from future climate change hazards and build social and environmental resilience	Permeable concrete, oyster-tecture, wetlands restoration, rain gardens, and bioswales	\$107M (SCAPE, 2023)
2	The Vessel	Visitor attraction and art installation resembling a beehive	Hexagonal / honeycomb / beehive structure	\$200M (Warren, 2020)
3	The Bank of America Tower	Building praised for a variety of its sustainability features aiming to reduce energy dependency	Green roof, rainwater harvesting system, co-generation natural gas plant, air filtration system, high-performance curtain wall system	\$1B (Di Justo, 2005)
4	The Brooklyn Grange Rooftop Farm	Urban agriculture project aiming at both fostering local food production and reducing the city's reliance on food import	Green roof, rainwater harvesting system, modular planting beds, composting	Over \$1.38M (NYC DEP, 2019)
5	The Newtown Creek Wastewater Treatment Plant	Green roof aiming to increase biodiversity and manage stormwater effectively	Green roof	No data
6	The Cooper Union Building	Private college building with an array of sustainability features aiming at both improving biodiversity and energy efficiency	Green roof, rainwater harvesting system, geothermal heating and cooling system, natural ventilation	\$166M (Carlson, 2011)

7	The Barclays Center	Sports and recreation venue focused on sustainability and aiming at improving biodiversity and managing stormwater	Green roof	No data
8	The Solaire	Residential building and one of the city's first green buildings with sustainable features that help it save energy and preserve the environment	Green roof, solar panels, on-site renewable energy generation	\$17.25M (NRDC, n.d.)
6	The High Line	Elevated park and recreational area and an example of repurposing grey infrastructure, also aiming at enabling easier access to green spaces in a heavily built environment	Park	\$242.3M (NYC Mayor's Office, 2012)

Most of the solutions implemented in New York are green roofs and have a component that harvests rainwater. Outliers are three solutions – breakwaters, attraction, and park. The solution that is purely biomimetic in design and does not include any nature-based components is the only one that does not contribute to greening an area. All the other solutions contribute to making the area greener or more biodiverse.

The High Line

The High Line is an elevated park in the borough of Manhattan that incorporates green infrastructure and native plant species in an attempt to improve biodiversity and provide a recreational space for residents and visitors. The High Line is a linear park developed as a repurposing project on a former elevated railway. The park opened in 2009 and stretches over 1.45 miles (2.33 kilometers) from Gansevoort Street to 34th Street (Friends of the High Line, n.d.). It is designed as a public space, offering views of the city skyline and the Hudson River, and one can access it for free.

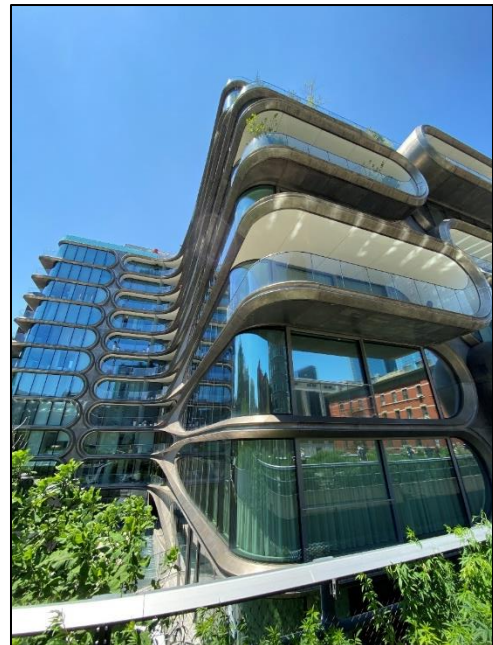


Figure 8. The High Line. (Aziz Hreljić, 2022)

One of its most striking features is its incorporation of green infrastructure and over 500 species of plants, including trees, shrubs, and grasses, most of which are native to the area (see Figure 8.) These plants provide habitat for birds and insects and help to improve air quality and reduce the urban heat island effects. Moreover, the park also features some biomimetic properties, as its design is based on the natural processes that occurred on the abandoned railway tracks before the park was built. The planting design was inspired by the self-seeded grasses and wildflowers that had grown on the tracks over time, aiding the creation of an aesthetically pleasing and ecologically functional landscape, providing habitat for pollinators and other wildlife. Beyond the inspiration, the park also incorporates elements that mimic the natural hydrological cycle, where rainwater is collected in planters and infiltrates the soil, reducing stormwater runoff and improving soil health (Larson, 2019).

In addition to its ecological benefits, The High Line also possesses recreational value, as it features several seating areas and gathering spaces, as well as spaces for art exhibitions and performances. The park is surrounded by residential units of more recent build (see Figure 9.)

Figure 9. Residential Complex Next to the High Line. (Aziz Hreljić, 2022)



The Solaire

The Solaire is a residential building located in Battery Park (See Figure 10., second building from the right), New York City, which was completed in 2003. It was among the pioneer green buildings in the City, and its sustainable features have since served as an inspiration for other green building projects (EPA, 2020). These sustainable features include a green roof (see Figures 11.-13.), solar panels, and a highly efficient heating and cooling system. The green roof serves multiple functions, including reducing the building's energy consumption by providing insulation and reducing the urban heat island effect by absorbing and reflecting sunlight (New York City Department of Parks & Recreation, 2011). This type of roofing also serves as a habitat for birds and insects, improving local biodiversity.



Figure 10. The Solaire. (The Solaire, n.d.a)



Figure 11. The Solaire's Green Roof. (The Solaire, n.d.b)

The building's solar panels generate electricity for the building's common areas and reduce its reliance on the power grid. The highly efficient heating and cooling system uses a heat recovery system to capture and reuse waste heat generated by the building's mechanical systems, further reducing energy consumption (EPA, 2020). In addition to these features, the Solaire was also designed with a focus on indoor air quality and resource conservation. The building's ventilation system includes high-efficiency filters that remove pollutants from the air, and the building's plumbing fixtures are low-flow to conserve water (New York City Department of Parks & Recreation, 2011).

The Rockefeller Brothers Fund (2013) note that the building incorporates a range of energy-saving features, including energy-efficient lighting, high-performance windows, and a building-wide energy management system, which together have resulted in a 35% reduction in energy use compared to a conventional building of similar size. The same publication notes that the Solaire has contributed to urban sustainability is through its use of on-site renewable energy. The building incorporates a 5,000 square foot rooftop solar array, which generates approximately 5% of the building's annual electricity needs. This combination of reduction in energy use and on-site renewable energy generation has resulted in a corresponding reduction in greenhouse gas emissions, helping to mitigate the impacts of climate change, as well as reduced the building's reliance on grid electricity, which is often generated from non-renewable sources.



Figure 12. The Solaire's Green Roof. (The Solaire, n.d.c)



Figure 13. The Solaire's Green Roof. (The Solaire, n.d.d)

The Barclays Center

The Barclays Center is a sports and entertainment venue and arena located in Brooklyn, New York, and its green roof is one of the largest in the city (see Figure 14.), spanning approximately 130,000 square feet or 12,077 square meters (Gotham Greens, n.d.). The green roof is planted with a mix of **sedum, grasses, and other hardy and drought-tolerant plants**, and it helps reduce the building's energy use in several ways.

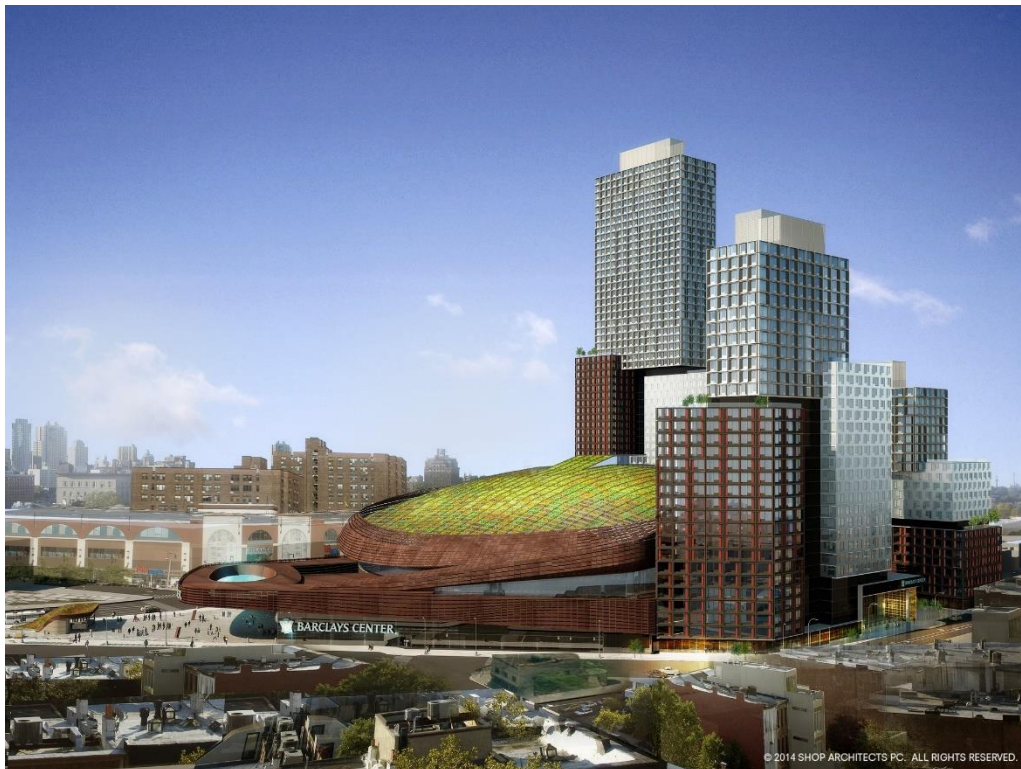


Figure 14. The Barclays Center Green Roof. (SHoP Architects, n.d.)

First, the vegetation absorbs and deflects solar radiation, reducing the amount of heat that penetrates the building and lowering the need for air conditioning. Further, the green roof serves as an insulator, reducing heat loss during winters and further lowers the need for heating. Finally, the green roof contributes to the

reduction of stormwater runoff by absorbing and filtering rainwater, therefore offsetting the pressure on the building's stormwater management system and helps mitigate the impacts of urban flooding. Of course, in addition to all these energy-saving and stormwater management benefits, the green roof also provides ecological benefits as a habitat for bird and insect species, as well improving air quality by absorbing pollutants (Barclays Center, n.d.). Beyond all practical benefits, the roof possesses an aesthetic appeal, thus positively impacting the surroundings.

The Cooper Union Building

The Cooper Union Building, located in Manhattan's East Village is a sustainable architecture project designed by Morphosis Architects (see Figure 15.) The building houses The Cooper Union for the Advancement of Science and Art, a private college established in 1859. The building incorporates several sustainable features, such as a green roof, a rainwater harvesting system, and a geothermal heating and cooling system. The rainwater harvesting system collects rainwater from the roof and uses it for irrigation and toilet flushing. The geothermal heating and cooling system regulates the temperature of the building by using the temperature of the earth, thus reducing its energy consumption. Furthermore, the design integrates a range of passive solar strategies, including shading devices and natural ventilation, which further reduce energy use. The building facade comprises a series of angled panels that reflect natural light into interior spaces (see Figure 14.), reducing the need for artificial lighting, whilst the interior incorporates energy-efficient systems such as LED lighting and motion sensors to further reduce energy consumption (Cooper Union, n.d.; Morphosis Architects, n.d.). The building has achieved LEED Platinum certification from the U.S. Green Building Council (USGBC), which recognizes buildings that meet high standards of sustainability and environmental performance (U.S. Green Building Council, 2021).

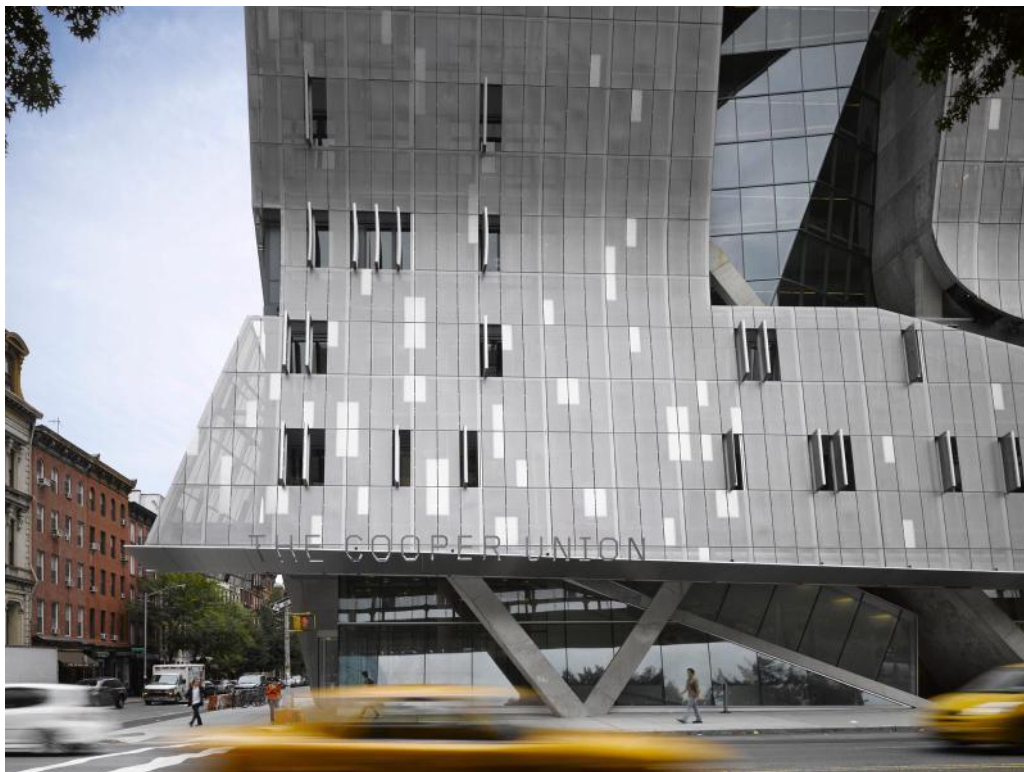


Figure 15. The Cooper Union Building. (Roland Halbe, n.d.a)

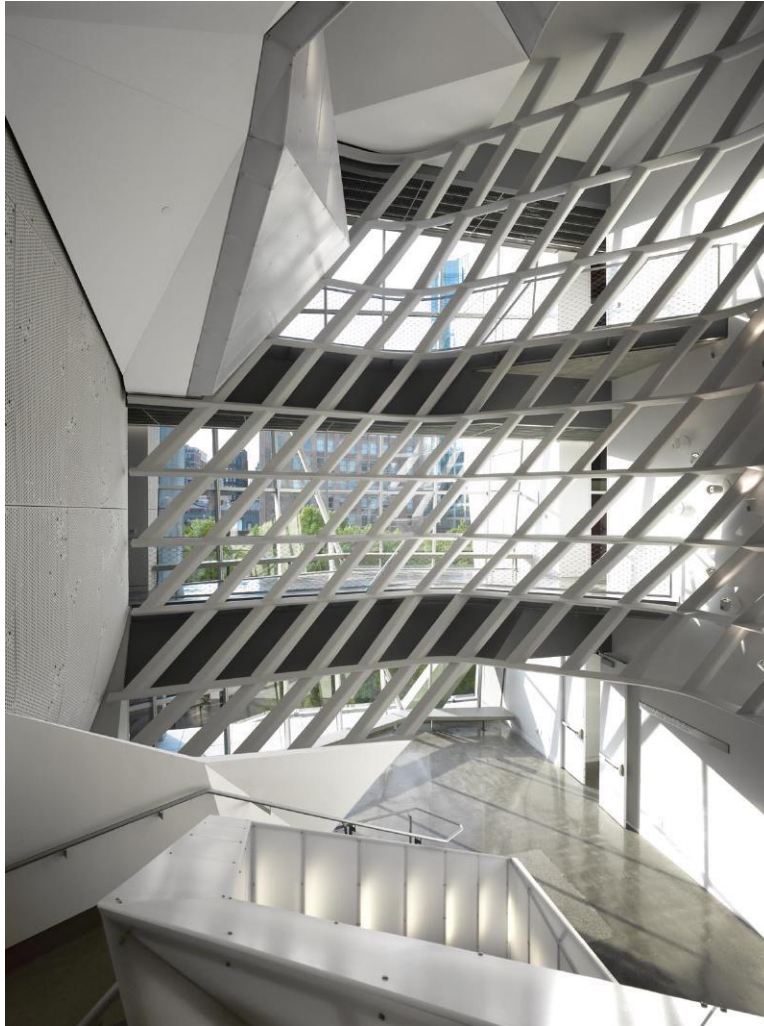


Figure 16. Interior of The Cooper Union Building. (Roland Halbe, n.d.b)

The Newtown Creek Wastewater Treatment Plant

The Newtown Creek Wastewater Treatment Plant in Brooklyn is a sustainable infrastructure project designed to minimize its environmental impact. One of the most notable features of the plant is its green roof, that reduces energy use by providing insulation to the building, reducing the need for heating and cooling. In addition, the green roof also helps to manage stormwater runoff, reducing the amount of water that flows into nearby waterways during rain events, while the ecological benefit is it being a habitat for insects and birds. It features a variety of plant species, including sedums and grasses, which require minimal maintenance. The use of native plant species helps to support local biodiversity and provide habitat for pollinators such as bees and butterflies (NYCDEP, 2019). This, as all other green roofs, also provides additional benefits, as reducing the urban heat island effect, improving air quality, and reducing noise pollution.



Figure 17. The Newtown Creek Wastewater Treatment Plant's Green Roof. (Dorkys Ramos, n.d.a)



Figure 18. The Newtown Creek Wastewater Treatment Plant's Green Roof. (Dorkys Ramos, n.d.b)

The Brooklyn Grange Rooftop Farm

The Brooklyn Grange Rooftop Farm is an innovative project that combines urban agriculture with sustainable design in an attempt to address several environmental challenges. The farm is located on the roof of a commercial building in Brooklyn and is one of the largest rooftop farms in the city, covering an area of more than two acres. The farm utilizes several nature-based solutions to improve its environmental performance, including green roofs, rainwater harvesting, and composting. The green roofs used in the farm are designed to absorb rainwater, reduce stormwater runoff, and provide insulation, which helps to reduce the building's energy use. The farm also features rainwater harvesting systems that capture rainwater from the roofs and store it in tanks for later use in irrigation, which aids the conservation of water and reducing the demand for municipal water supplies, which is important especially during droughts (Brooklyn Grange, n.d.).

In addition to being a nature-based solution, biomimicry is also incorporated in the farm's design. The farm uses a modular design for its planting beds, which allows for efficient use of space and resources. The modular planting beds are inspired by the growth patterns of natural ecosystems, which tend to maximize resource efficiency (Lundholm, 2006).

The Bank of America Tower

The Bank of America Tower in Midtown Manhattan possesses numerous sustainable features and systems, and according to Cook Fox Architects (n.d.), the tower's design integrates several green infrastructure strategies, including a high-performance curtain wall system that maximizes natural light and reduces solar heat gain. The building also features an air filtration system that improves indoor air quality and reduces energy consumption (see Figure 21. in Appendix 2 for a schematic about the building's heating-cooling and air circulation systems). Moreover, the tower's green roof covers more than 56,000 square feet and provides insulation, reduces stormwater runoff, and serves as an outdoor space for building occupants. The green roof also features a rainwater harvesting system that collects and stores rainwater for irrigation purposes. The tower also includes a co-generation, natural gas, plant that generates electricity and provides heating and cooling to the building, reducing its reliance on the city's electrical grid. Looking at the ground floor of the building one can see that the space was designed with openness in mind, and it provides access to certain amenities as well (see Figure 20. in Appendix 2 and Figure 22.)

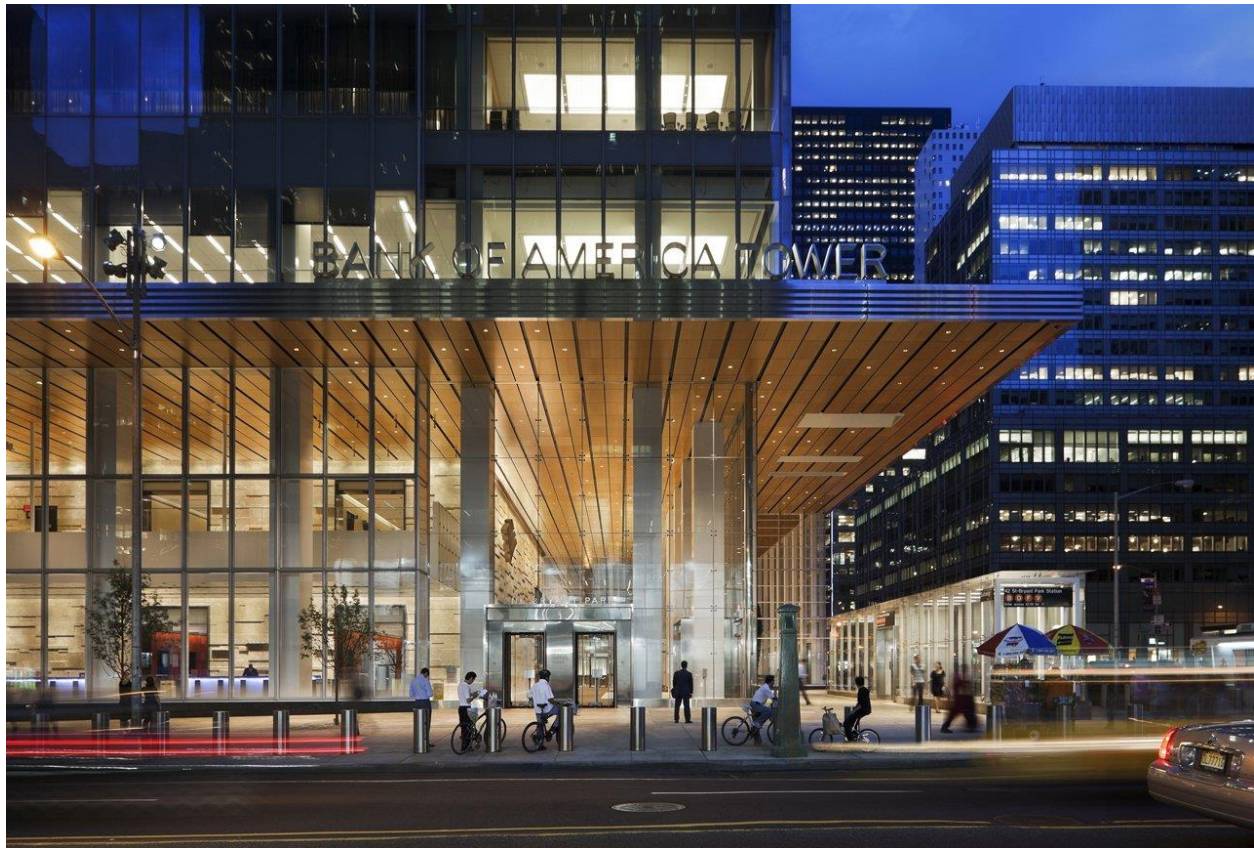


Figure 19. One Bryant Park. (David Sundberg, n.d.)

The tower's energy-efficient systems have helped it achieve significant energy savings, with the 4.6 megawatt cogeneration plant accounting for 70% of the building's annual energy supply. Although the tower might not possess overtly noticeable biomimetic properties, the building is rooted in biophilic design and is based on materials' natural properties to make the building sustainable. In fact, the propensity toward light is somewhat typical of biophilic design, as plenty of natural light possess both pleasing qualities, as well as positive sustainability connotations (Vinnitskaya, 2012). The Bank of America Tower has also been awarded the LEED Platinum certification, due to its sustainable properties (MKA, n.d.).

The Vessel

The Vessel is a public art installation located in the Hudson Yards development on the west side of Manhattan. Designed by Thomas Heatherwick, this structure is made up from 154 flights of stairs and almost 2,500 individual steps, with 50 feet of width at its base, expanding to 150 feet at the apex, makes for an interesting sight (see Figures 23. and 24.). The architect notes that he has drawn inspiration from ancient Indian stepwells (Heatherwick Studio, n.d.).



Figure 20. The Vessel Exterior. (Aziz Hreljić, 2022)

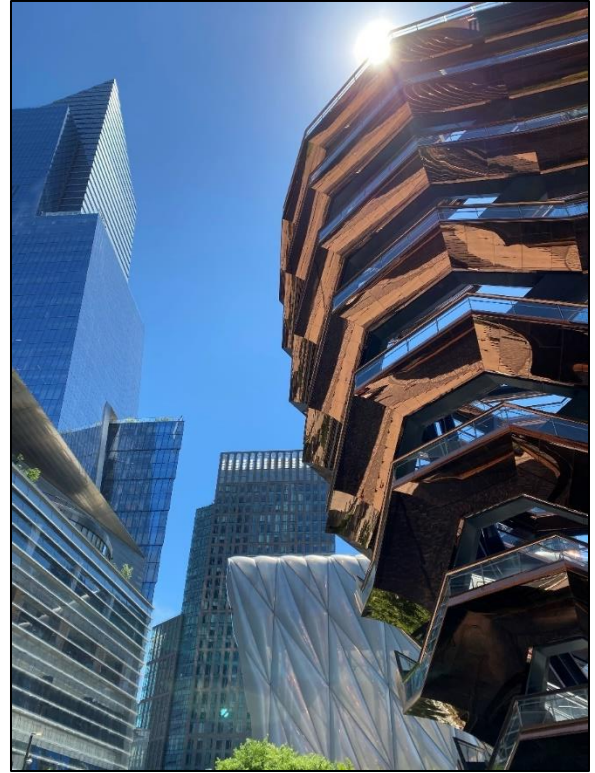


Figure 21. The Vessel and Its Surroundings. (Aziz Hreljić, 2022)

Figure 22. The Vessel Interior. (Aziz Hreljić, 2022)



However, what is prominent about the structure is that it bares striking resemblance to a beehive, and that the “cells” are also similar to honeycomb (see Figure 25. below and Figures 26. and 27. in Appendix 2), both of which would imply biomimicry. Now, although the structure brings no practical ecological or environmental benefits, it was purposefully selected for its social and economic sustainability implications, as a tourist attraction and a sight that blends in with the surroundings marvelously.

The Living Breakwaters Project

The Living Breakwaters Project in Staten Island deploys nature-based solutions and biomimicry to reduce the impact of storm surges and erosion on the coastline. The project, which is currently under construction, features a series of breakwaters that mimic natural forms and functions, providing habitats for marine life and creating calmer waters (see Figures 28. and 30. in Appendix 2.) that can help mitigate wave energy during storms (New York State Governor's Office of Storm Recovery, n.d.a).

The 2,400 feet of breakwaters are designed to reduce the impact of storm surges and erosion by absorbing and dissipating the energy of waves, reducing the risk of flooding and erosion on the coastline. The breakwaters are also designed to create habitats for marine life, with the use of textured concrete that encourages the growth of algae and other organisms. They are made of large, permeable concrete structures that are designed to create a range of habitats for marine life, including fish, shellfish, and seaweed (see Figure 29. in Appendix 2.) The design of the breakwaters is based on the concept of "oyster-tecture," which draws inspiration from the role of oyster reefs in protecting coastlines and creating diverse marine habitats. The project also aims at building social resilience through educational programs and a coalition of local stakeholders called the Citizens' Advisory Committee (Scape, n.d.b).

Moreover, the project includes a range of other nature-based solutions to enhance the resilience of the coastline. These solutions include the restoration of wetlands, the creation of oyster reefs, and the installation of green infrastructure such as rain gardens and bioswales, which help absorb stormwater and reduce runoff (New York State Governor's Office of Storm Recovery, n.d.b).

London

London is the capital of and the largest city in the United Kingdom. With 8,796,628 residents in 2021 (ONS MYE, 2022) distributed in the city's five sub-regions, it is a bustling metropolis worth researching. The City of London is located in the southeastern part of England and the United Kingdom, and it lies on the River Thames that empties into the English Channel. The city is the seat of the British Government, many renowned universities, such as the London School of Economics, and well as many historic landmarks. Considering its relevance, and even more so its geography, it is an interesting city to research in the contexts of sustainability and climate change. The city's climate, according to the Köppen-Geiger classification is a temperate oceanic climate (Cfb). This climate is characterized by cool winters, warm summers, and precipitation being consistently present all year round (Peel & Finlayson, 2015). The average temperature during winter is around 6.7° C (44F) and during summer the average is around 19° C (66F), according to the Weatheronline (2023).

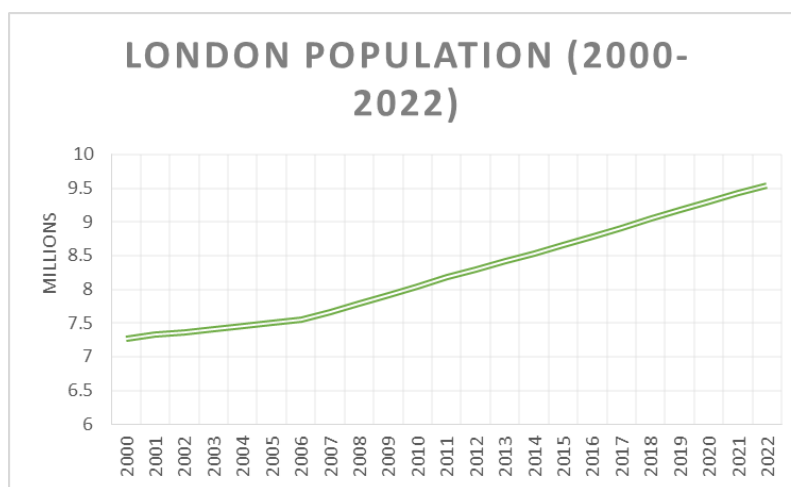


Figure 23. London Population (2000-2022) in Millions. (Source: Macrotrends, n.d.; Graph prepared by author.)

In Figure 31. above, one can see that the population of the City of London is growing over time. Below, in Figure 32., one can see that despite this growth, the unemployment rate is steadily decreasing.

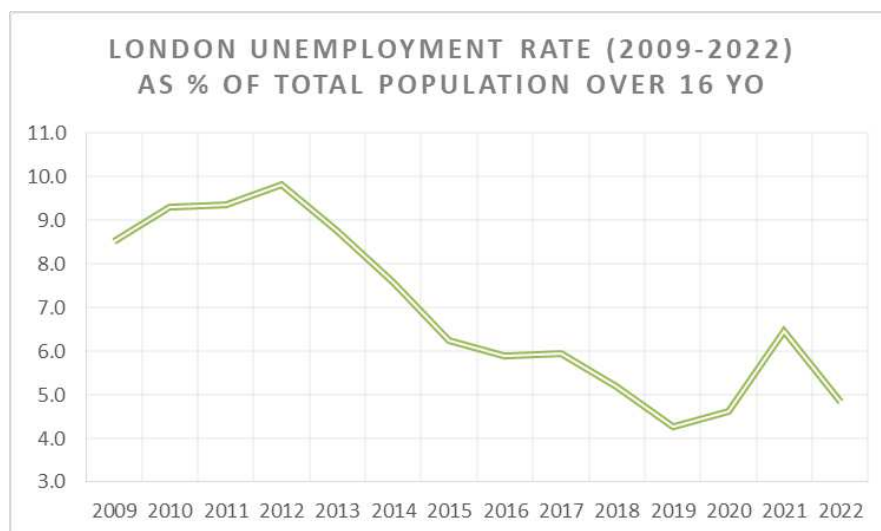


Figure 24. London Unemployment Rate for Population 16 and Older. (Source: ONS, 2023; Graph prepared by author.)

In figure 33. below, one can see four major air pollutant levels – NO₂, PM₁₀, PM_{2.5}, and SO₂, between 2008 and 2019. The fluctuations that can be observed can be attributed to the time of year, so for PM₁₀ and PM_{2.5}, spikes happen during winter, for instance.

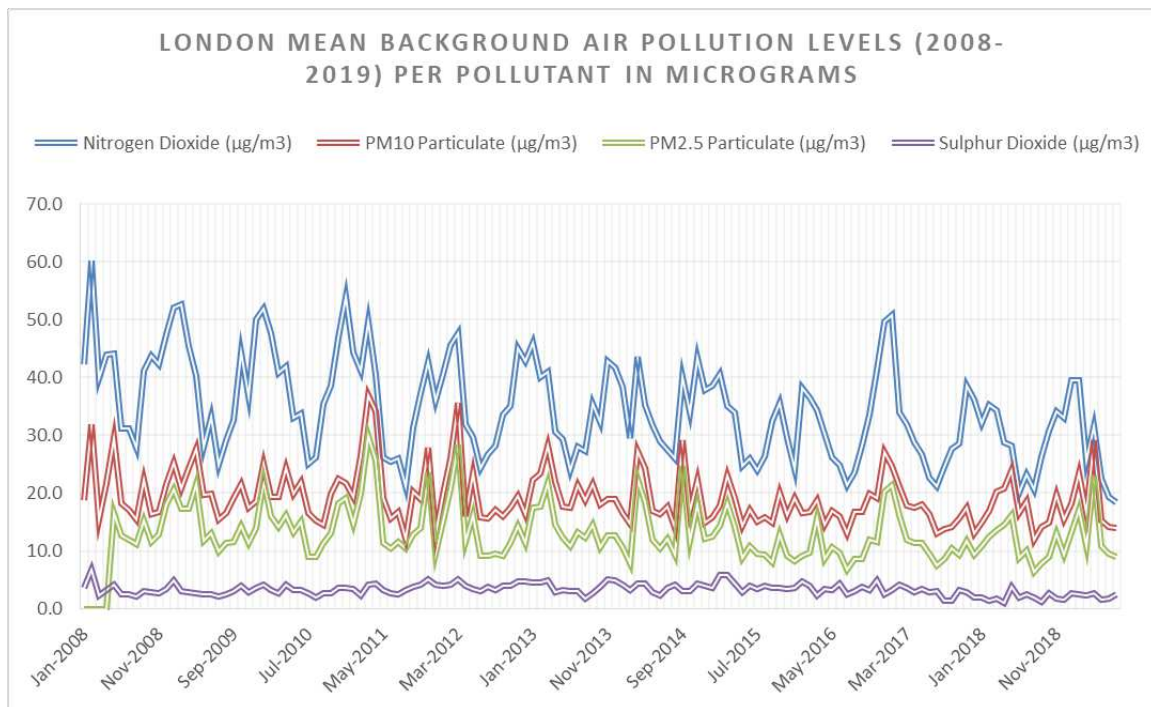


Figure 25. London Air Pollution Between 2008-2019, per Pollutant. (Source: King's College London, 2019; Data visualization prepared by author.)

London, as most other cities, faces many challenges looking into the future, including (Edwards, 2020; London Assembly, n.d.):

- Energy - green sources, home usage and equipment
- Transportation - greenhouse gases emissions and overall functioning of the city
- Extreme weather – flooding and heat waves
- Air quality
- Noise pollution

These are only some of the city's challenges, which the city aims to resolve by proactive legislation, and with practical action.

Sustainability and Climate Change Policies

The City of London has two main policy documents that relate to sustainability and climate change. Sustainability is, like in the other cities also considered mostly from the environmental perspective. The documents in question are the London Environment Strategy and the Climate Action Strategy 2020-2027. The city also possesses a resilience strategy, but since this thesis does not nominally consider resilience, this document will not be analyzed.

The London Environment Strategy by the Greater London Authority (2018) is a strategy that responds to the city's environmental sustainability challenges. This is the city's first integrated environmental sustainability strategy, and it aims to turn London into a zero carbon city by the year 2050 and make other improvements to make the city more livable. The strategy addresses several thematic areas, including air quality, green infrastructure, climate change mitigation and adaptation, and noise pollution.

Nature-based solutions are ingrained in the strategy by virtue of the strategy's chapter on green infrastructure. The main idea of this part of the strategy is to promote the idea of turning half of the city's area green, and this infrastructure is used to provide benefits to the citizens. The strategy considers the quality of green spaces and their various benefits and co-benefits, ranging from biodiversity preservation, over their recreational value, and all the way to their relationship with climate change.

Within this aspect of the strategy, nature-based solutions already effect all three facets of sustainability. The city wants to manage its green areas as nature-based solutions, thus creatin environmental benefits. Further, the strategy notes that green infrastructure also possesses economic value, such that public parks provide approximately GBP5 billion annually which means that every GBP invested in them is returned 27-fold (Mayor of London, 2017). The strategy also refers to a study (Mayor of London, 2015) according to which eight million London trees provide GBP133 billion in economic benefits through combating pollution. Access to these green areas is also considered from a social perspective, and the city mapped out the areas with lower access. The city understands trees and woodlands, rivers and wetlands, gardens, green roofs, and sites of importance for nature conservation as green infrastructure, all of which can be or can be managed as nature-based infrastructure.

The city also intends to further the use of nature-based solutions in water management and flood risk management. This relates primarily to the improvement of water quality in the city's rivers, and through both sustainable drainage systems and river restoration. Both these approaches provide more than one benefit and address biodiversity concerns, habitat restoration amenity benefits, and water quality and quantity. The city plans to utilize natural systems' capacity to regulate water quantities, therefore reducing the risks of flooding.

The Climate Action Strategy by the City of London Corporation (2020) is a significantly more succinct document focusing on the actions that the city needs to take in response to climate change. This is not particularly surprising as the environmental strategy also includes climate change. The action plan includes both mitigation and adaptation efforts, and is designed so that it tries to turn risks into opportunities and find cooperation linkages with other national and international strategies and plans.

Although the plan does not mention nature-based solutions or biomimicry by name, some of the actions it proposes are in fact such solutions. For instance, the city plans to use climate resilient landscaping, adapt buildings to better cope with higher temperatures and prioritize water conservation, as well as use urban greening absorb rainwater and developing flood defense improvements. Some of these can be considered nature-based solutions as they are, whilst others can use these solutions to improve the response they offer. Below are some examples of nature-based and biomimicry infrastructure that the city already has.

Nature-based and biomimicry projects

Table 2. below presents the basic information about the reviewed biomimetic and nature-based solutions analyzed in London. Each solution's description is paired with features that make it nature-based and/or biomimetic, as well as the cost of the projected. Information about the costs could not be found for all the reviewed projects.

Table 2. Solutions from London. (Prepared by author)

London	Project Name	Brief Description	Relevant Nature-based and Biomimetic Features	Cost
1	The Bloomberg European Headquarters	Sustainable urban development utilizing biomimetic principles and appropriate materials to achieve energy efficiency and a harmonious appearance	Material efficiency, natural ventilation system	\$1.26B (Wainwright, 2017)
2	30 St Mary Axe (The Gherkin)	Sustainable high-rise relying on biomimetic principles to improve its energy efficiency	Natural ventilation system (building's lungs), glazed façade	\$174M (BBC News, 2014)
3	Camley Street Natural Park	Natural park or reserve in a heavily built urban setting preserving biodiversity and providing educational experiences to visitors	Natural park/reserve	No data (refer to The King's Cross Central Development)
4	Bagley Walk	Elevated park built upon repurposing old grey infrastructure that was not being used anymore and aiming at improving local biodiversity	Park	No data (refer to The King's Cross Central Development)
5	Gasholder Park	Small park in a heavily built setting providing not only access to a green space but also preserving a piece of historic infrastructure	Park	No data (refer to The King's Cross Central Development)
6	Regent's Canal	Redevelopment of the river canal making it both more resilient to climate change and attractive to visitors, as well as improving local biodiversity	Gardens, moorings, green walls	No data (refer to The King's Cross Central Development)

7	Jellicoe Gardens	Public square and garden drawing inspiration from ancient Persian gardens	Garden	No data (refer to The King's Cross Central Development)
8	The King's Cross Central Development	Redevelopment of an entire area of the city that has been neglected and unused, using principles of nature-based and biomimetic solutions, aiming at increasing the areas biodiversity, making it more attractive to visitors, and creating new educational and economic activity opportunities	Parks, canal, gardens, rain gardens, swales, moorings, green walls	\$3.77B (Wainwright, 2018)
9	The Sky Garden	Indoor rooftop garden and viewpoint with various plants aiming at greening grey built architecture and providing access to green spaces	Indoor roof garden	No data
10	The Crystal	Sustainable building inspired by crystals' light refraction properties and focused on energy conservation and water resilience	Rainwater harvesting and processing/purification system, six types of high insulation glass	\$48.7M (Thomas, 2012)

Most of the solutions in London are related to the provision of additional green spaces in the form of parks, green roofs, or natural parks. However, there are some outliers, where the solutions utilize specific materials or design features, that gives them their nature-based or biomimetic properties. In these cases, a tendency toward energy efficiency is also observed.

The Crystal

The Crystal is a sustainable building located in East London that incorporates a range of green infrastructure and energy-efficient systems to reduce its environmental impact (see Figure 35. in Appendix 3.) The project's 6,300 square meters in area houses exhibition areas, conference facilities, as well as an innovation center (see Figure 34. for the building's exterior). This immense space's design achieved the highest international sustainability credentials for a building – BREEAM Outstanding and LEED Platinum. Its design showcases a progressive water agenda, where rainwater is being harvested and purified to drinking-quality levels, and the building is 90% water self-sufficient. (The Crystal / Wilkinson Eyre Architects, 2012).

Figure 26. The Crystal Exterior. (WilkinsonEyre, n.d.)



The all-glass building, features six different types of glass of high insulation, each possessing different levels of transparency, in order to moderate solar gain. With this in mind, the backward-leaning facets oriented toward the sun are made of reflective glass, while inner faces angled towards the ground use transparent glass (The Crystal. WilkinsonEyre, n.d.). The building's namesake is what inspired the design, and it is actually the building's link to biomimicry. Crystals, present in nature are coveted for their light refraction properties, and the Crystal emulates this and uses it to control internal temperature and lighting (see Figure 36. in Appendix 3.)

The Sky Garden

The Sky Garden is a public garden on top of the Fenchurch Building (the Walkie-Talkie) in London and it is a representative example of a nature-based solution amidst grey infrastructure (see Figure 37.) The highest garden in London was designed by the architectural practice Gillespies, and it is freely accessible to visitors

(with a pre-booking) to enjoy the 360° views of the city's skyline, the lush greenery, and an open-air terrace (What is Sky Garden - Sky Garden – London, n.d.).

The garden stretches across three floors, where each terrace is modeled after a specific landscape theme. The lead landscape architects, Gillespies cooperated with Growth Industry as the planting specialists, Biotecture as the greenwall design specialists, and Willerby Landscapes as the landscape contractor on this project. One of the aims for the architects was to convey a high quality de-cluttered setting for the Fenchurch building, which resulted in an accessible and welcoming open space at street level (Sky Garden at 20 Fenchurch Street Opens - News – Gillespies, n.d.).

Figure 27. The Sky Garden. (Sky Garden, n.d.)



Although it might seem strange, the inspiration for the project was drawn from ancient forests. However, due to the finite nature of the space where the glass vaulted roof creates a particular microclimate, the garden was conceptualized as a series of micro-landscapes. The landscape design framework was rooted in the ‘Evolution of plants’, thanks to which the garden is split into three zones: shade tolerant forest, a transition zone and flowering plants. The terraces progress from colorful plants on the lower level, over a transition from ancient cycads to the forest, all the way to the top terrace that is a shaded groove of trees of fern and fig (The Sky Garden at 20 Fenchurch Street - Projects – Gillespies, n.d.). The garden is a proper example of type 3 nature-based solutions, where the built environment is made green.

The King's Cross Central Development

King's Cross is an example of a redevelopment of an area, where an otherwise decaying and desolate landscape with derelict and disused buildings, railway sidings, warehouses and contaminated and the burdens of crime, unemployment, and a poor quality environment was transformed into a vibrant and popular hub (see Figure 38.) An underused industrial zone was reimagined and revamped remodeled with new infrastructure, including nature-based solutions. The planning process for the project began at the start of the 21st century, and it took several years of participatory planning for the masterplan to be created. This plan was developed by Allies and Morrison, Porphyrios Associates, and Townshend Landscape Architects, and the permissions from 2006 was granted for approximately 50 new buildings, 20 new streets, 10 new major public spaces, the restoration and refurbishment of 20 historic buildings and structures, as well as up to 2,000 homes. Between then and now, a university, several hospitality establishments, and companies such as Google and Facebook have laid roots here. The development also includes new streets, squares, parks, and gardens and made this a popular site for residents and visitors (About the redevelopment of the King's Cross area of London. (n.d.). Several nature-based solutions have been selected from this development: Jellicoe Gardens, Gasholder Park, Bagley Walk, and Camley Street Natural Park.



Figure 28. Section of the King's Cross Redeveloped Area. (King's Cross Central Limited Partnership, n.d.a)

Jellicoe Gardens is designed by the architect Tom Stuart-Smith and the project draws inspiration from the Persian Bagh-e Fin – one of, if not the most iconic of all Persian gardens, both in design and layout. The focal point of the garden is a tall, open pavilion surrounded by oriental plane trees, which is an homage to the lofty columned canopies which served as the entrances to traditional Persian garden palaces. The pavilion is also decorated with a narrow thin channel, which serves as a water basin (see Figure 40.) The garden features small paths with scattered benches, providing resting spots for visitors (see Figure 39.), as well as an opportunity to take in the scenery. The planting features meadow-like properties, with somewhat wild swathes of color and texture from highly scented and nectar-rich vegetation, that attracts bird and insect species year-round (Jellicoe Gardens at King's Cross. (n.d.).



Figure 29. Jellicoe Gardens. (King's Cross Central Limited Partnership, n.d.b)



Figure 30. Jellicoe Gardens Basin. (King's Cross Central Limited Partnership, n.d.c)

Regent's Canal is nine-mile (15 kilometer) waterway running from the River Thames through the King's Cross. It is home to green space (see Figure 41.) amidst one of the city's most densely populated areas, and features a towpath for walking or cycling, parks, gardens, bridges, moorings, reclaimed green walls, as well as canal side hospitality establishments (see Figure 42.) In addition to all the services, it also doubles as an important habitat for birdlife, insects, amphibians, and vegetation (Regent's Canal at King's Cross, n.d.).



Figure 31. Restored biodiversity at the Regent's Canal. (King's Cross Central Limited Partnership, n.d.d)



Figure 32. Establishments Along the Regent's Canal. (King's Cross Central Limited Partnership, n.d.e)

Along the Canal, one can find the Gasholder Park, a small park the view of which is dominated by a cast iron frame, that used to be on the canal's opposite bank and held 1.1 million cubic feet of gas. With its volume, Gasholder No. 8 is the largest of gasholders in King's Cross. The lengthy, two-year, process of relocating the iron structure ended with it being placed among sculpted canopy and a lush circular lawn. Bell Phillips Architects designed the park, while the planting and lighting are designed by Dan Pearson Studio and Speirs + Major, respectively (Gasholder Park - the new park at King's Cross, n.d.). The park is a space open to the public and it is frequented by families, students, and office workers in need of respite from the city (see Figures 43. and 44.)



Figure 33. Gasholder Park. (King's Cross Central Limited Partnership, n.d.f)

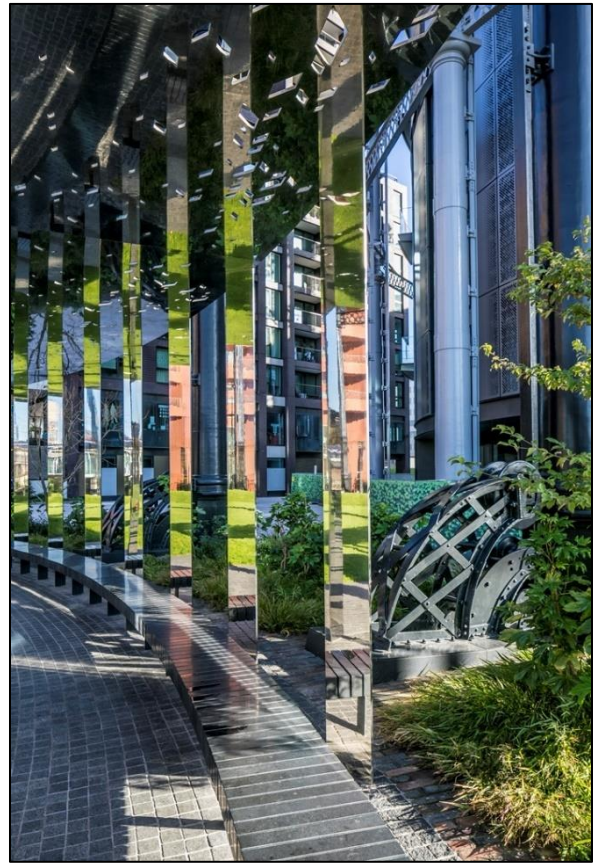


Figure 34. Gasholder Park. (King's Cross Central Limited Partnership, n.d.g)

Bagley Walk is an elevated park designed by Dan Pearson, built on a former railway viaduct also attached to Regent's Canal and connecting Somers Town Bridge and Granary Square. It features lush greenery and forms a triptych view together with Camley Street Natural Park and Coal Drops Yard. The park is yet another example of repurposing in the King's Cross development, where railway tracks used to unload coal carts have now become a lively mélange of Victorian coal infrastructure and boutiques and restaurants (see Figure 46.) The greenery selected for the park was chosen with biodiversity in mind and includes a variety of ornamental and edible plants, such as licorice, strawberries, and fig, that both celebrate the seasons and serve as a habitat for fauna (see Figure 45.) Low hedges emulate the rhythm of the brick arches that support the viaduct, and present seats that are both viewing areas and resting spots (Bagley Walk King's Cross, n.d.).



Figure 35. Bagley Walk Vegetation. (King's Cross Central Limited Partnership, n.d.h)



Figure 36. Bagley Walk Establishments. (King's Cross Central Limited Partnership, n.d.i)

Camley Street Natural Park brings two acres of wild green space to the Regent's Canal, right in the heart of the city and in close proximity to one of the busiest rail interchanges in the United Kingdom. This park, or rather reserve, provides a habitat for vegetation and wildlife, such as mallards, coots, swans, moorhens, kingfishers, herons, amphibians, insects, and fungi. The park is run by the London Wildlife Trust, and hosts a variety of activities for children and adults (see Figure 48.), including family drop-in days for pond-dipping, mini beast hunting, or simple strolls along the nature trail (see Figure 47.), through woodland, grassland, and wetland habitats. Aside from all of the above, the park serves an educational purpose with its visitor center where one can learn about the area and conservation work (Camley Street Natural Park, King's Cross, n.d.).



Figure 37. Camley Street Natural Park. (King's Cross Central Limited Partnership, n.d.k)



Figure 38. Floating Platform at Camley Street Natural Park. (King's Cross Central Limited Partnership, n.d.j)

30 St Mary Axe (The Gherkin)

30 St Mary Axe is mixed use, 41 floors tall, circular plan project, with over 46,000 square meters of office spaces, shops, and hospitality establishments, and a club room adorning its apex and offering a 360° panoramic view of London. The building is designed by Foster + Partners and it embodies radial geometry – it widens as it rises, and then grows more narrow toward the summit, achieving both a slender and elegant profile and minimizing the use of precious space at street level (see Figure 49. below and Figure 50. in Appendix 3.) However, the background for this design decision is not purely practical and aesthetic and has

wind and light elements in mind. It reduces wind deflection which creates the conditions for the building's natural ventilation system, while the diagonally braced structure made it possible to have a glazed façade and to eliminate columns from the floor space, opening the building up to plenty of natural light.



Figure 39. The Gherkin. (Foster + Partners, n.d.a)

When all of this is paired with spaces between floors that are envisioned as the building's lungs and distribute fresh air provided through opening façade panels, the building's reliance on air conditioning is reduced, and energy use is halved compared to traditionally air-conditioned office towers (30 St Mary Axe | Foster + Partners, n.d.). The building is colloquially referred to as the Gherkin, because of its resemblance to the vegetable, but this resemblance is not the reason for attributing the building biomimicry attributes. This shape found in nature is where biomimetics come into play, as that is how the building's natural ventilation system is enabled and building maximizes solar gain.

The Bloomberg European Headquarters

Another Foster + Partners project is the Bloomberg European Headquarters. The project takes up an entire city block, spanning 3.2 acres and is comprised of two buildings connected by bridges covering a pedestrian arcade (see Figure 51.), three public plazas, and hospitality establishments at ground level. The façade is given its definition by a structural sandstone frame and large-scale bronze fins that cover the outer glazing. These fins do not provide only shading and control of solar exposure, but also create a visual hierarchy and a natural ventilation system and enable solar gain. In addition to fins which draw inspiration from nature, the building also wisely uses aluminum petal panels (see Figure 52.) the cooling and acoustic properties of which converted the office ceiling into an integrated energy-efficient system. The project conforms to the standards of sustainable development, with a BREEAM Outstanding rating, as well as the highest ever design-stage score awarded to a major office complex (Bloomberg | Foster + Partners, n.d.).



Figure 40. Bloomberg European HQ Exterior. (Bloomberg.com, 2017)



Figure 41. Bloomberg European HQ External Panels. (Foster + Partners, n.d.b)

Madrid

Madrid is the capital, and with its 3,305,408 inhabitants (Instituto Nacional de Estadística, 2022), the largest city in Spain. The city is divided into 21 municipal districts, with Madrid Central being the actual City of Madrid, and the rest belonging to its metropolitan region. The city has a long history and beautiful classical architecture. The city, apart from being the seat of the Spanish government and monarchy, also hosts several universities, such as the Autonomous University of Madrid and the Madrid Polytechnic University, as well as the World Tourism Organization.

The city's climate enjoys a Mediterranean climate, or Csa according to the Köppen climate classification (Kottek et al., 2006), and it is characterized by hot summers and mild winters. In fact, enjoying this climate might not be entirely accurate, as the city is not a coastal city, which makes the summer difficult to bare. Summer average temperatures are 24° C (75F), whereas the city has an average temperature of 7° C (44F) during winter (Agencia Estatal de Meteorología., n.d.). The city, although not coastal, lies on the River Manzanares.

The city is faced primarily with issues related to air pollution and greenhouse gas emissions (Buces, 2023) and stronger heatwaves (McMurty, 2022). These are of course not the only problems the city faces, as it is common knowledge that the temperatures in the city have been rising in the last years, which can be attributed to climate change.

Sustainability and Climate Change Policies

The City of Madrid has a strategy that integrates environmental sustainability and action against climate change – Madrid 360 – Roadmap Towards Climate Neutrality by 2050. The strategy was developed by the city and its main aim is to harmonize the city's actions with European Union's air quality regulations, by aligning municipal policies with European and national policies. The strategy, in a sustainable scenario, plans for the city to reduce local greenhouse gas emissions by 65% and achieve carbon neutrality by 2050. Despite not possessing a singular sustainability strategy for sustainability, this strategy also underlines its correlation to socio-economic processes and refers to economic and social data when envisioning the future. However, although the dimensions are referred to, the strategy does not explore them, and therefore, the impacts of nature-based and biomimicry solutions on the social or economic spheres of the city are not assessed. (City of Madrid, 2021a).

The strategy mentions nature-based solutions in two instances, both in relation to climate change adaptation, but it does not refer to biomimicry explicitly. First, they are mentioned under Strategic objective: A cooler city. The strategy posits that nature-based solutions should be used to contribute to a cooler city, and in response to higher temperatures and heat waves, underlining their capacity to improve the incorporation of green and blue infrastructure into urban environments. Under this strategic objective, the city aims to increase green areas and tree canopy, replace impermeable with permeable and natural surfaces, promote the usage of materials with high light reflection properties, and develop urban green infrastructure.

Interestingly enough, nature-based, as well as biomimetic, solutions can contribute to all four strategic objectives, even though the strategic contextualizes them mostly as a tool for improving the quantity of green surfaces. Light reflection is a common biomimetic concept and it is used to help with infrastructural thermal regulation, decreasing dependance on artificial air conditioning, as well as control solar gain. Greening areas

by tree planting is a nature-based approach to creating protection from solar exposure, as the strategy acknowledges as well as permeable pavement.

Second, nature-based solutions are mentioned under Strategic objective: Transforming rainfall management in Madrid. The strategy notes nature-based solutions' potential to reduce peak flows when rainwater enters the drainage system and consequently minimize the release of polluted water into natural water sources. This application of solutions is actually already connected to the previous objective, although here, purification potential is also explored.

In addition to this strategy, the city also has a Localization Strategy for the Sustainable Development Goals of the 2030 Agenda in the City of Madrid. This strategy gives some more insight into the way the city approaches all dimensions of sustainability, and it is a way for the city to take a better lead on furthering the agenda, by aligning municipal policies with the overall framework of the agenda. The strategy has 104 local targets, developed for 16 out of the 17 SDGs, as well as a monitoring and evaluation system (City of Madrid, 2021b).

The strategy does not mention nature-based or biomimetic solutions at all. Despite not being mentioned in the strategy, there are several targets that the city has set for itself, where they might be useful. The first is related to Goal 3 – Good Health and Wellbeing, and city target 3.1. Under this target the city wants to have 30 healthy routes, something that nature-based solutions can help design and implement. Moving forward to Goal 6 – Clean Water and Sanitation, and city targets 6.1 and 6.3. The city can use nature-based solutions, to assist in wastewater treatment processes, as well as to increase riparian ecosystem biodiversity. Perhaps somewhat unexpectedly, the solutions can also be connected to Goal 8 – Decent Work and Economic Growth, and city targets 8.11 and 8.13. Under these, the city wants to become more sustainable and accessible as a tourist destination and become more recognizable as a filming location. With their effects on overall pleasantness and aesthetics of areas where they are implemented, nature-based and biomimetic solutions can contribute to both these goals.

The net goal where the solutions are applicable is Goal 11 – Sustainable Cities and Communities, and city targets 11.5, 11.11, 11.12, and 11.13. In its aim to increase the available cycling infrastructure, the city can resort to nature-based solutions in its design and implementation. As nature-based solutions can help with air purification, using them to increase the coverage of green areas in the city can help meet both these targets. Moreover, in relation to the design of the city's sustainable drainage system, nature-based solutions can be used, as was already mentioned with respect to Goal 6.

Under Goal 12 – Responsible Consumption and Production, nature-based solutions can be related to target 12.7. Here, the city aims to increase the number of urban gardens and their accessibility. These are, often, typical nature-based solutions in and of themselves. Under Goal 15 - Life on Land, the city has target 15.1. The city intends to increase the urban green areas and tree canopy, which are yet again nature-based solutions, even though not mentioned as such. The areas where the solutions can be applied to this Madrid's strategy are fairly numerous, and progress is already being made toward some of these targets, as can be seen in the examples of projects.

Nature-based and biomimicry projects

Table 3. below presents the basic information about the reviewed biomimetic and nature-based solutions analyzed in Madrid. Each solution's description is paired with features that make it nature-based and/or biomimetic, as well as the cost of the projected. Information about the costs could not be found for the majority of the reviewed projects.

Table 3. Solutions from Madrid. (Prepared by author)

Madrid	Project Name	Brief Description	Relevant Nature-based and Biomimetic Features	Cost
1	3D printed bridge in Alcobendas	World's first large-scale 3D printed bridge	Parametric design, micro-reinforced concrete, sustainable use and recycling of materials	No data
2	The Vertical Garden of Torre de Cristal	Indoor vertical garden designed to make the space greener and purify air	Vertical garden, hydroponics	No reliable data
3	The Vertical Garden of Caixa Forum Museum	Outdoor, publicly accessible vertical garden improving the visual appeal of the plaza and makes the space greener	Vertical garden, polyurethane base and non-biodegradable felt-type blanket, automated irrigation and fertilization system	No reliable data
4	Avenida de Portugal	Public park that includes nature both in planting and the design, with cherry trees in extended bloom	Park	No data
5	Salon de Pinos	Linear green space and children's park built with sustainable natural materials	Park	No data
6	The Madrid Río Park Redevelopment	Redevelopment project focused on restoring the city's access to the river and creating an easily accessible green space	Parks, river's hydraulic architectural restoration	\$6.62B (Archello, n.d.)

The solutions observed in Madrid display a fine balance between the provision of green spaces in different ways – by restoring natural systems through parks and the creation of new vertical gardens. In addition, something that is generally different is the fact that 3D printing is utilized to create a new structure, which is in line with some biomimetic principles and overall sustainability practices.

The Madrid Rio Park

The first nature-based solution in Madrid is the **Madrid Rio** project. This multi-component development dates back to 2005 when the need to design the area reclaimed from a section of the M30 motorway arose. The reclaimed area is approximately 200 acres in area, and the team comprised of Burgos & Garrido, Porras La Casta, Rubio & Álvarez-Sala, and West 8 that was selected to breathe new life into the area decided to resolve the urban challenge solely by means of landscape architecture, and they were the winners in the international competition. The redevelopment was conceived so that the area is divided into smaller parts, each with its own subproject. There are many completed subprojects, but the most notable ones include: the Salón de Pinos, Avenida de Portugal, Huerta de la Partida, Jardines de Puente de Segovia, Jardines de Puente de Toledo, Jardines de la Virgen del Puerto and the Parque de la Arganzuela (Minner, 2011).



Figure 42. Madrid Rio Park Redevelopment. (Jeroen Musch, City of Madrid, n.d.)

The motorway section that was buried had run along the Manzanares River and because of it the river was rendered inaccessible, and city's connection to the river was fading. The large public space that then appeared was not only a restorative landscaping project, but a way to reconnect the city with its river and incorporate it into the project. The entire redevelopment features new sports facilities, communal and artistic spaces, an urban beach, playgrounds, squares, boulevards, parks, bridges, and restored river's hydraulic architectural heritage (MADRID RÍO — Burgos & Garrido, 2021).

Salon de Pinos is one of the subprojects in the Madrid RIO redevelopment by West 8 and it is a nature-based solution with biomimicry elements. It features a grove of black locust trees various playground equipment developed for children of all age groups, as well as children with disabled children (Magic Forest in Salon de Pinos, Madrid RIO - West 8, 2019). Its design is that of a linear green space, and it found its place almost entirely on top of the motorway tunnel. What the designers also had in mind in selecting the vegetation for this green space are the mountains on the outskirts of Madrid. The pine trees that can survive in harsh environments are featured in various planting forms and with different sculpting to create a botanical monument of sorts, both natural and sculptural. The park was carefully planned with sustainable materials and drawing inspiration from bull's horns for the design of tree support (Minner, 2011).



Figure 43. Salon de Pinos. (West 8, n.d.)

Avenida de Portugal is a boulevard garden created once the road was relocated to a tunnel and parking was provided underground. The key design elements are cherry trees and Portuguese paving. The elements are selected because the road extending from the boulevard leads toward Lisbon in Portugal, and one part of that road passes through a valley known for its cherry blossoms amidst the otherwise desolate and inhospitable climate of the Estremadura (Minner, 2011). This pairing of design elements and the pleasing aesthetics of prolonged cherry blossoming, due to diverse types of cherry trees planted, has made this a popular public space. The designers also decided to keep the blossoms year-round, as the paving also features cherry blossoms.



Figure 44. Avenida de Portugal. (Ayuntamiento de Madrid, n.d.)

The Vertical Garden of the Caixa Forum Museum

The Caixa Forum Museum is a well-known modern art gallery in a former power station and features a beautiful green wall. The museum is in the center of Paseo del Prado, often considered Madrid's cultural district, across the Prado Museum and botanical gardens and the Reina Sofia Museum (see Figures 56.)



Figure 45. Vertical Garden of Caixa Forum Museum. (Patrick Blanc, n.d.a)

The garden, created by the famous botanist Patrick Blanc, contains between 15,000 and 17,000 plants from almost 300 species selected by Blanc himself with Madrid's climate in mind, including: *Dianthus deltoides*, *Lonicera nitida*, *Yucca filamentosa*, *Cistus purpureus*, *Cedrus deodara*, *Pilosella aurantiaca*, *Arenaria montana*, *Bergenia cordifolia*, *Cornus sanguinea*, *Lonicera pileata*, *Sedum alpestre*, *Campanula takesimana*, *Garrya elliptica*, and different *Begonias*. The public can access the wall from the nearby plaza, and feel and explore the plants on it. The wall is designed in typical Blanc style, with a base sheet of polyurethane, followed by plastic mesh and non-biodegradable felt-type blanket with pouches to pot the plants. This method has plants root well, and it is estimated that the total vertical garden weight is 30 kg/m². Due to the wall not being the museum's property, the set up is carefully put together so that moisture would not damage it. There is not direct contact between the base polyurethane sheet and the wall, and this gap houses a network of pipes and emitters used for the automated irrigation and fertilization system (Caixa Forum Museum Vertical Garden - Greenroofs.com, 2022).

The Vertical Garden of the Torre de Cristal

Another vertical garden by Patrick Blanc can be found in the Crystal Tower (Torre de Cristal), and fittingly, Spain's tallest building is adorned with the tallest vertical garden in Europe at 30 meters tall. This garden is also hydroponic, i.e. plants do not require substrate or soil to grow and water is used instead (Messa, 2022). This garden, though the project is much grander, features fewer, less than 50 plant species and over 24,000 plants, distributed over 600 m² (Carranco, 2008). There is another notable difference between this and Blanc's other garden in Madrid, and it is that the project in Torre de Cristal is entirely indoors (see Figure 57.)



Figure 46. Vertical Garden at the Torre de Cristal. (Patrick Blanc, n.d.b)

3D printed bridge in Alcobendas, Madrid

The 12 meters long and 1.75 meters wide 3D printed footbridge (see Figure 58.) in the Castilla-La Mancha park of Madrid's Alcobendas is an engineering feat of the Institute for Advanced Architecture of Catalonia (IAAC) and ACCIONA, and it is the first of its kind in the world. The bridge is made of micro-reinforced concrete and its relevance lies in the fact that it is the first large-scale 3D printing project and because the technology had not been used in civil engineering up to that point. It was developed using parametric design and it represents complex natural forms, enabling the optimization of material usage and minimizing waste through recycling raw materials throughout the manufacturing process. The design and the project also display enviable freedom of form and structural performance as a result of computational design and generative algorithms (see Figure 59. in Appendix 4.), making it a true next generation approach to construction (3D printed bridge – IAAC, 2021).



Figure 47. 3D Printed Bridge in Alcobendas. (Nicolas Janberg, n.d.)

Seoul

Seoul is the capital, and with a population of 9,664,666 people (SMG, 2023a) the largest city, of South Korea. The city is located in the northwestern part of the country, and its area is approximately 0.61% of the country's entire area. Administratively, it is divided into 25 autonomous districts, while the Han River divides the city into the northern and southern parts (SMG, 2023b). The city has a humid tropical or Dwa climate, characterized by monsoons and great yearly variation and precipitation, according to the Köppen classification (Weatherbase, n.d.) with average temperatures in winter averaging 1° C (30F) and summer temperatures averaging 21 ° C (76F) (I: Korea Meteorological Administration, 2021). The city is experiencing certain sustainability issues, such as managing the Han River ecosystem (Kim et al., 2018), but then also those that come with climate change.

Sustainability and Climate Change Policies

The city of Seoul, or the Seoul Metropolitan Government, displays an integrated approach to environmental and climate change policy through the theme of environment and energy, which although touching upon topics related to the social and economy facets of sustainable do not address them in detail. However, this information is available only on the Metropolitan Government's website, and the city does not appear to have any publicly available policy documents that relate to sustainability, or perhaps does not have any such documents published online in the English language. What the city does have are the city's Sustainable Development Goals for 2030, in which all aspects of sustainability are addressed. Similarly, some information on what the city does in relation to climate change can be found on official sites, but there is one document from 2015 that the Metropolitan Government published - Promise of Seoul: Taking Action Against Climate Change, that serves as the city's climate change action plan.

Kim et al. (2018) Seoul Sustainable Development Goals or "17 Ways to Change Seoul" prepared for the City of Seoul contextualizes the city's development within the Agenda 2030 agenda by developing 96 detailed targets across the 17 overarching goals. This strategy comes after the city first made a decision to establish local SDGs in early 2016 and several months of planning across various committees and subcommittees over the following months. The strategy focused on applying the 17 goals' dimensions to the city of Seoul both from the international level and national goals in South Korea.

The strategy does explicitly mention nature-based solutions or biomimicry, but there are several instances where I believe they are implied. Under SDG 12 – Cities/Spaces, the city's 12.2 target is to achieve sustainable use and management of natural resources. Nature-based solutions can be seen as a tool to achieving this target, as they often rely on sustainable resource use as, while one of their main benefits is sustainable management of natural wealth. Under SDG 15 – Terrestrial Ecology/Resources, the city has several targets that actually rely on nature-based solutions. Target 15.1 where the city wants to sustainably use urban ecosystem services and manage urban ecosystems can again be achieved through nature-based solutions. Target 15.2 under which the city explains its forestation plans is also a measure in alignment with nature-based solutions. Target 15.4 can be achieved in the same way, as the city wants to create small-scale biotopes in an effort to improve biodiversity. Under SDG 9 – Industrialization/Innovation, the city could make use of both biomimicry and nature-based solutions to achieve its targets. Target 9.1 to expand environmentally friendly infrastructure that is available to all societal strata is an idea rooted in nature-based solutions, and can be achieved by applying biomimetic principles. Targets 9.3 can be seen as the city's future-proofing efforts, and can be

connected to biomimicry, as its principles are generally related to technology and require significant research and development.

Seoul Metropolitan Government's (2015) policy on climate change can be found in The Promise of Seoul, which is the city's comprehensive and integrated strategy to address both adaptation and mitigation, as well as all various fields that are affected by climate change, including air quality, water, ecology, and urban planning. The policy is goal oriented and suggests different visions and actions for the goals to be achieved, and is a way for the city to recommit itself to the role it has in the active fight against climate change.

The entire strategy makes no mention of nature-based or biomimicry solutions as responses to climate change in any context. The city makes 11 promises, and among them to reduce its CO₂ emissions, reduce greenhouse gas emissions by water management measures, transform into a climate change-resilient and eco-city with a wealth of biodiversity, and engage in capacity building in response and prevention of climate disasters. The strategy addresses 10 thematic areas, including air quality, water, ecology, health, and urban planning. These pledges and areas can be connected to biomimicry and nature-based solutions.

The city intends to make itself safe against fine particles (PM) and this can be addressed by utilizing green walls and vertical gardens, as well as forestation or greening projects. When it come to water, the city plans on collecting and utilizing rainwater, which can be aided by rain and rooftop gardens, as well as green surfaces in general. By the appropriate selection of plant species and sedum, rain absorption can be increased, while simultaneously decreasing the demand for other water sources.

When the strategy addresses ecology, it aims at lowering average temperatures by extending green spaces, as well turn the city green and biodiverse. Lower average temperatures, and their effects can be addressed with biomimicry solutions, both through cooling systems and layout planning. An increase in green areas can, while bolstering biodiversity also helps with heat absorption and therefore makes higher temperatures more bearable. In relation to health, the city wants to reduce vulnerable groups' exposure to heat waves and reduce citizens' overall exposure to air pollution. An increase in green areas can help with both these aims. In fact, by addressing some of the issues mentioned above, the health benefits can be considered co-benefits of these nature-based solutions.

Lastly, the strategy's section on urban planning aims at lowering the city's energy consumption through strategic urban planning. Aside from the solutions contributing to lower energy usage in completed infrastructural projects, the city would also benefit from lower energy usage during the implementation phase, as the resources used are mostly renewable.

Neither of these policies seem to deploy nature-based and biomimetic solutions to address their challenges. Some of the areas above point to how these could be included to improve the plans, but the list is, of course, non-exhaustive. Additionally, some of the solutions already implemented in the city are addressing the city's challenges. This claim is also supported by the writings of Song & Müller (2022) who found that when urban development can be made climate friendly through climate change and urban regeneration policy integration, and cite several other authors on the usage of nature-based solutions as a measure of upgrading existing buildings.

The city has also made a plan to achieve carbon neutrality by the year 2050 through cutting 35 million tons of carbon emissions by several means (SMG, 2022). This plan also does not include nature-based or biomimetic solutions, but there are several instances in which they can be inferred or included to improve the city's plan.

The plan includes five main target areas, and nature-based and biomimetic solutions can be used in at least four – pursuing energy efficiency in one million buildings, removing concrete and asphalt in favor of green areas, climate-proofing the city, and engaging citizens in combating climate change. Some of these impacts and co-benefits can already be seen in some of the implemented solutions below.

Nature-based and biomimicry projects

Table 4. below presents the basic information about the reviewed biomimetic and nature-based solutions analyzed in Seoul. Each solution's description is paired with features that make it nature-based and/or biomimetic, as well as the cost of the projected. Information about the costs could not be found for the majority of the reviewed projects.

Table 4. Solutions from Seoul. (Prepared by author)

Seoul	Project Name	Brief Description	Relevant Nature-based and Biomimetic Features	Cost
1	Seoul National University Green Roof	Green roof with both small-scale agricultural and decorative purpose aiding the building in lowering the roof temperature and managing rainwater	Green roof	No data
2	Seoul City Hall	Sustainable building as a seat of the local government with several sustainable elements which enable it to be energy efficient and green	Green wall – vertical garden, geothermal energy	No data
3	Seoul Energy Dream Center	Zero-energy building dedicated to energy sustainability and efficiency in style and performance	Rainwater harvesting system, thermal storage, geothermal energy	\$22M (Kim, 2013)
4	Garden of the Sun in Jonggak Subway Station	Underground green space created utilizing technology to transfer light under ground aiming at greening idle grey infrastructure	Underground garden	No data
5	The Galleria in Gwanggyo	Retail store with biomimetic elements to its design and a green roof accessible to the public	Green roof	No data

6	The Weaves	Revitalization project aiming to restore a local river to its natural state and repurpose existing grey infrastructure, as well as improve biodiversity and create additional social and economic activity opportunities	Riverbank restoration, purification streams water retention pools, multi-level park	No data
7	Seoullo 7017	Elevated linear park created as a project to repurpose existing grey infrastructure and enable easier access to green spaces	Park, urban tree nursery	\$50.5M (Hartman, 2018)
8	Dongdaemun Design Plaza and Park	Futuristic building with nature-based elements enabling more access to green spaces	Park, green roof, 3D digital construction services	\$451M (Cha, 2014)
9	Cheonggyecheon Stream Restoration	Restoration and repurposing project where grey architecture has been turned into a green area in the center of the urban area aiming to promote biodiversity, access to green spaces, and address sustainability issues	Green space, spawning grounds for fish, flood protection walls, reestablished waterway connections	No reliable data

In Seoul, there are several different approaches to the observed solutions. Something that was fair to expect is that a significant number of them aims at the provision of green spaces, which one does very creatively, by bringing it underground. Then, one solution utilizes 3D printing, similar to what was seen in Madrid, however, this is done on a different scale. Two projects are related to water and restoring aquatic ecosystems. One more outlier is a solution dedicated entirely to energy sustainability, both in build and purpose.

Cheonggyecheon Stream Restoration

The first set of nature-based solutions in this part of the review is the one from the Cheonggyecheon Stream Restoration Project. This restoration project was intended to revive a stream in downtown Seoul by removing an elevated freeway (Urban Nature Atlas, 2023) that had covered it for decades, in order to give back the area a green public space with all the accompanying ecological, social, and economic benefits (see Figures 60. and 61.), as well as indicate a shift in South Korean approaches to planning and sustainability (Commission for Architecture and the Built Environment, n.d.).



Figure 48. Elevated Freeway Prior to the 2003 Start of Construction on the Stream Restoration Project. (SMG, n.d.a)



Figure 49. Restored Cheonggyecheon Stream After Construction Was Complete in 2005. (SMG, n.d.b)

The entire restoration project was huge, but it is not relevant for this research so the focus will be put on nature-based solutions related to the project. The project was initiated in 2002, and the master plan dating to 2004 included measures to restore the area's natural environment, and the entire process was rooted in participatory planning, involving a research group, the local government, and citizens, through the so-called triangular implementation system. The restored Cheonggyecheon opened up to visitors in late 2005, which is a fairly short period, especially considering flooding tests that had to be conducted during the rainy season. The development included approximately 40 acres of new green space open to the public with a number of benefits – space for pedestrian and cultural activities downtown, habitats for flora and fauna, and lower temperatures amidst the heavily built environment. After the project was completed and the revival of green areas, the difference in land prices in immediate vicinity to the Cheonggyecheon Stream compared to those that are more than half a kilometer away had doubled from 15% to 30% (The World Bank, n.d.).



Figure 50. Cheonggyecheon River Natural Stone Walkways. (Alexander Robinson, n.d.a)

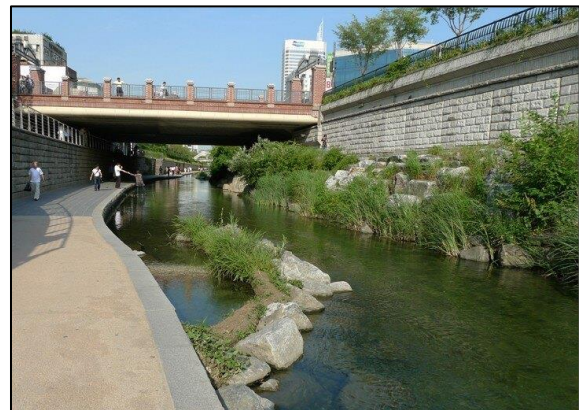


Figure 51. Cheonggyecheon River Slower Water Pools. (Alexander Robinson, n.d.b)

Landscape Performance Series (2020) cataloged all the sustainability features of the new stream area, and here are the nature-based solutions that are a part of the development:

- Over 3.5 miles (5.5 kilometers) of a Green corridor for pedestrians, bicyclists, and wildlife;
- Reestablished connections between waterways – Cheonggyecheon and Jungraechon streams leading into the Han River, as well as an ecological conservation area in the wetlands at their meeting point;
- Consistent flow of water, at 40 centimeters (15.5 inches) in the stream thanks to subway pump stations and the Han River;

- 29 locations with native willow swamps, shallows and marshes, serving as a habitat for fish, amphibian, insect, and bird species;
- Spawning grounds for fish at confluence of Cheonggyecheon and Jungnangcheon;
- Flood protection walls;
- Walkways of natural stone help regulate water speeds and levels (see Figure 62. and 63.)

Dongdaemun Design Plaza and Park

The Dongdaemun Design Plaza and Park is an exciting and futuristic project designed by Zaha Hadid and it represents a green oasis for relaxation in the urban and fast-paced Dongdaemun district of Seoul. The design of this project is a pioneer in many ways, such as it being the first public project in South Korea utilizing 3D digital construction services like Building Information Modelling – BIM (Zaha Hadid Architects, n.d.). Although it seems like a futuristic building and nothing more, the reason this architectural feat has been included in the review is because of the nature-based solutions incorporated in its design. The building is accompanied by a park, which is a nature-based solution in itself, but its roof is also green, thus providing both elevated and on-ground vegetation, thus balancing the design that stands out from its surroundings (see Figure 64.)



Figure 52. Dongdaemun Design Plaza Complex. (Edmond Leong, n.d.)

Seoullo 7017 Skygarden

Seoullo 7017 is a plant village in the form of a 983-meters long linear park at the elevation of 16 meters created by MVRDV, and it is located atop a former highway running through Seoul's inner city. The name Seoullo is Korean for "toward Seoul" and 7017 is a not to the initial construction year of the overpass 1970 and its current function as a park from 2017. The park brings together 228 species from 0 plant families in 24,000 plantings arranged according to the Korean alphabet and divided into gardens with their own composition, character, scents, and identity, including: maple, cherry, rhododendron, as well as various conifers and shrubs. Aside from bringing biodiversity to the city now, it is envisioned that the overpass will become an "urban nursery" and help grow trees for surrounding districts extending the positive influence to its surroundings (see Figures 65.-67.) The park also uses blue lights at night, as they are most beneficial to nature, and provides a pleasing contrast to the bright lights of the city (MVRDV, n.d.).



Figure 53. Seoullo 7017. (MVRDV, n.d.a)



Figure 54. Scale of the Seoullo 7017 Skygarden. (MVRDV, n.d.b)



Figure 55. Traffic Below Seoullo 7017. (MVRDV, n.d.c)

The Weaves

MVRDV is currently working on another project in Seoul – the Weaves, which is an attempt at returning the Tancheon River to its natural state by enabling it to naturally flow and replacing the grey built infrastructure with green riverbanks and restore a part of the Han River’s waterfront (see Figure 68.) The central idea of this revitalization project is to create the conditions for a healthy aquatic ecosystem, that will support thriving flora and fauna with purification streams water retention pools, and islands. Since this project is also a response to the government wanting to demolish parts of highways around the site, the architects suggested an alternative approach of transforming grey infrastructure into elements of the paths that will shape the multi-level public park (see Figure 69.) with amphitheaters, plazas, and viewpoints that will accompany the revitalization project. The interconnected, winding paths are where the project got its name, and the inspiration for them was found in silk threads and Jamsil neighborhood’s history of silk production (MVRDV, n.d.). The project is expected to be completed in 2024 (Baldwin, 2019).



Figure 56. The Weaves Concept. (MVRDV, n.d.d)



Figure 57. The Weaves Park Computer Visualization. (MVRDV, n.d.e)

The Galleria in Gwanggyo

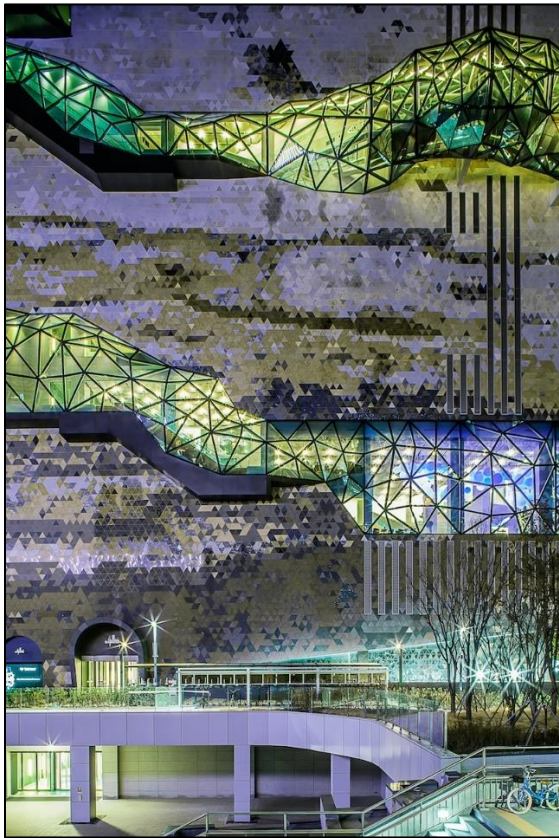


Figure 58. The Path to the Galleria's Roof. (OMA, n.d.)

The Galleria's retail store in Gwanggyo, which is technically outside of Seoul proper in the town of Gwanggyo but still in the same province and relatively close to Seoul, is included in this research due to its biomimetic features. The building features a mosaic stone façade, which connects it to the nature of Suwon Gwanggyo Lake Park, while the sculpting helps it blend in with the surrounding building at the intersection. Aside from being nature-inspired, this building also features a rooftop garden which easily accessible not only to the store's visitors, but also to the public via a route starting on the sidewalk outside. The path to the garden is framed in glass and appears as though excavated from the stone,



Figure 59. The Galleria in Gwanggyo. (OMA, n.d.)

Garden of the Sun in Jonggak Subway Station

The Garden of the Sun project was designed by the architect James Ramsey (Raad Studio), who is currently working on the Lowline project in New York City, and it is essentially an underground idle space converted into an underground garden through innovative natural light control technology. This kind of technology entails collecting sunlight via remote condensers and passing it through specialized lenses to underground spaces with minimal loss of its properties. Light collected in eight condensers is then retracted underground through transparent pillars of the subway station and used to create a livable environment for 37 different plants, including citrus and kumquat trees (see Figure 72.) Even though weather conditions above ground change, the entire system is automated to switch to LED lighting in case there is no solar gain to be collected. The name of the project was decided by the public through a civic contest, with 1,139 citizens participating and approximately 2,750 having been proposed (Park, 2021; SMG, 2018).



Figure 60. Garden of the Sun in Jonggak Subway Station. (Seoul City Hall, n.d.)

Seoul Energy Dream Center

Seoul Energy Dream Center, occupying 3,500 m² of floor space in World Cup Park is an exemplar of renewable energy in both form and function – a zero energy building, meaning it produces as much energy as it produces within a year. The building consists of three floors and its shape's extension resembles that of a trapezoid or an inverted pyramid, widening from the ground upwards. This shape provides both protection from the elements and solar exposure, as well as a structural role in stabilizing the construction. When it comes to energy sustainability the building possesses a strong foundation in passive energy measures, such as thermal storage granted by the massive ceilings and ample natural lighting from the building's shape. When this is coupled with earth probes that serve as thermoregulators and ventilation system with two-step heat recovery, the consumption of energy for heating and cooling is decreased by 70% compared to the standard in South Korea. Energy demand is met by virtue of geothermal energy and grid-connected photovoltaic roof panels (Fraunhofer ISE, n.d.). The building also features a rainwater harvesting system that recycles rainwater and uses it for the building's plumbing, thus reducing water consumption and the burden on municipal sources (Surya, n.d.).



Figure 61. Seoul Energy Dream Center. (WSP, n.d.)

Seoul City Hall

The seat of the Seoul Metropolitan Government is an all-round sustainable building, displaying a modern and unique appearance, and a complete switch from the previous colonial-era city hall (see Figure 74.). The building is branded as sustainable, and with about 28.3% of energy being supplied from eco-friendly sources, such as photovoltaic panels, solar, thermal, and geothermal. The building also features a green wall (see Figure 75.) that, with an area of 1,516 m² and spanning seven floors, is the world's biggest vertical garden. This green wall is home to over 70,000 plants from 14 species, and they help purify the air by removing contaminants and fine dust, as well as serve a thermoregulatory and humidity-control purpose (SMG, n.d.; Wang, 2014).

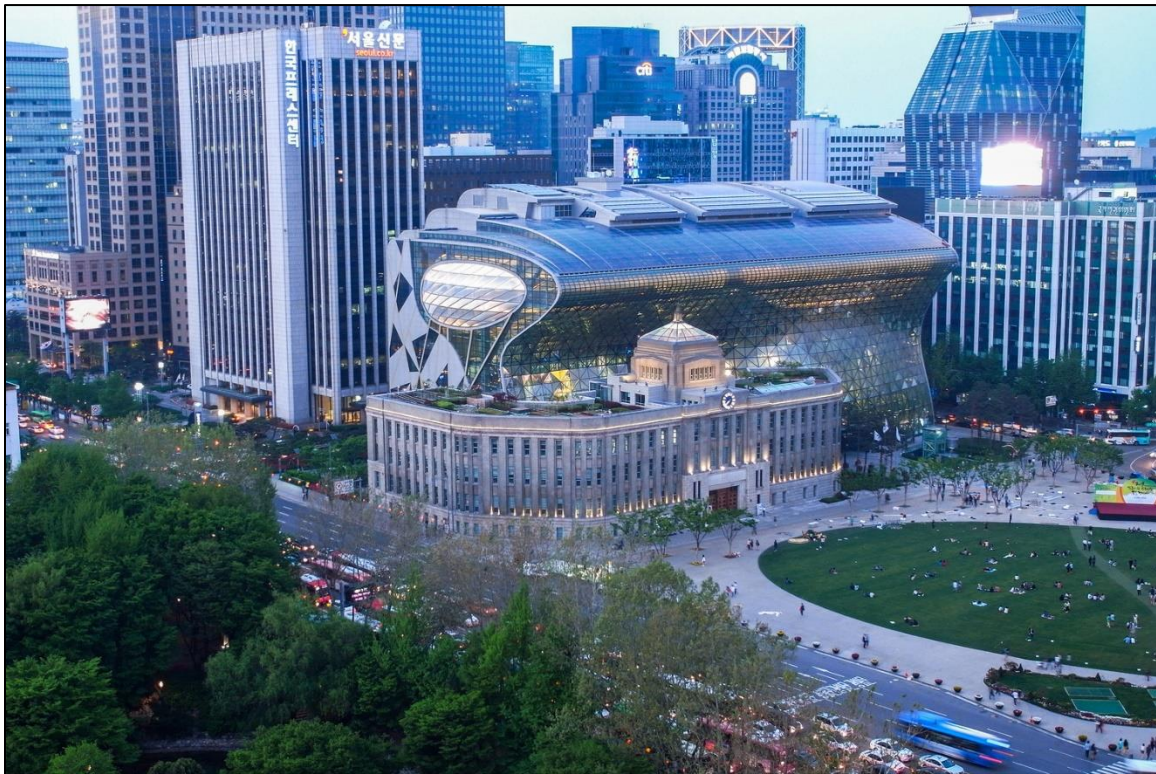


Figure 62. Seoul City Hall (Curved Glass Facade) Behind the Old Seoul City Hall. (Archframe, n.d.a)



Figure 63. Seoul City Hall Green Wall. (Archframe, n.d.b)

Seoul National University Green Roof

This green roof atop one of the buildings of Seoul National University covers 840 m² and it now covers a previously idle surface (see Figures 76. and 77.) Prior to implementation the roof actually posed a risk of flooding, due to piping connected to the sewer system being there. Moreover, during hotter months, the surface temperature of the roof could reach up to 50° C. Now, the site has become a gathering space and overall more pleasant, and it even hosts some events throughout the year (see Figure 78.) The 840 m² is divided between a flower garden covering 420 m², a vegetable garden covering 320 m², and a fish point taking up 100 m². The site is also a water-food-energy nexus (Restore Climate, n.d.).



Figure 64. Seoul National University's Rooftop Garden. (Anon., n.d.a)



Figure 65. Flower Garden on Seoul National University's Roof. (Anon., n.d.b)



Figure 66. Family-friendly Activities at Seoul National University's Rooftop Garden. (Anon., n.d.c)

Cape Town

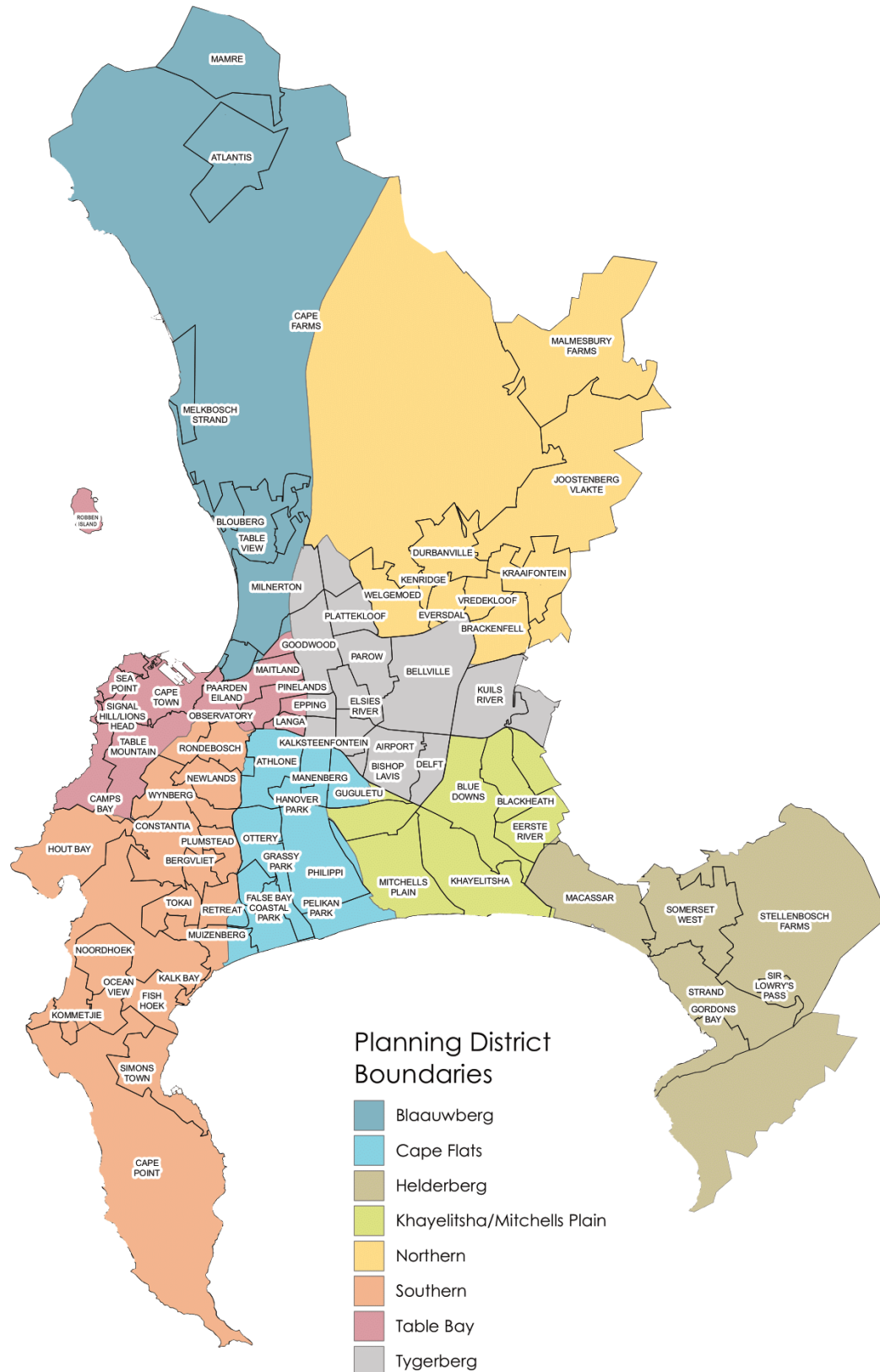
Cape Town is South Africa's legislative capital and the capital of Western Cape province, located in the west of the country on the Atlantic Ocean. The city's population in 2021 was 4,758,433 people (City of Cape Town, 2021c), making it the country's second largest city. According to City of Town's Urban Planning and Design Department (2019), the city is divided into the following eight districts (see Figure 79.): Blaauwberg, Cape Flats, Helderberg, Khayelitsha/Mitchells Plain, Northern, Southern, Table Bay, and Tygerberg. The reasons why Cape Town is selected to be featured in this study is because of it being a coastal city, and therefore especially vulnerable to climate change, then because of it being one of South Africa's capitals, and then because of its biodiversity.

Cape Town spreads around the Table Mountain, which is on the city's western coast, while the city's southern coast provides access to False Bay. The Table Mountain does not only affect the city's layout but it also locally modifies the city's climate, together with the cold Benguela Current of the South Atlantic (Axelson, 2023). The climate of the city is Mediterranean, with daily averages in winter being between 18° C (64° F) and 8.5° C (47° F), and 26° C (79° F) and 16° C (61° F) during summer (WMO, 2023).

According to the City of Cape Town (n.d.) and Western Cape Government. (n.d.), the city is or will soon be experiencing the following effects of climate change:

- Decreased average annual rainfall and changed rainfall patterns, resulting in changes to water supply.
- Increased average and maximum temperatures, i.e. more hotter days, and heat waves becoming more frequent and intense.
- Stronger winds.
- Sea level rise coupled with coastal erosion.
- Increased risk of fires.
- Increased incidence of extreme weather events (floods, droughts, storm surges).

Find your District



Sustainability Regulations, Plans, Initiatives, and Legislation

The City of Cape Town's commitment to sustainability is reflected primarily in the way it approaches the environment and its wealth of biodiversity, though the social and economic dimension are not disregarded either. There are several strategies and regulations that govern the city's approach to sustainability (City of Cape Town, n.d.), most notably the following as the facets of sustainability:

The 2017 Environmental Strategy

The 2012 Social Development Strategy

The 2021 Inclusive Economic Growth Strategy

The Environmental Strategy from 2017 is a comprehensive document created to address the environmental challenges, and the environmental dimension of sustainability in a broader sense, the city is facing in a structured and transparent manner. Some of the long-term aims of the city, and by extension the strategy, are: excellent air quality; properly managed wetlands and rivers; restored ecological quality of the marine environment; sustainable use of the city's municipal green infrastructure; improved public transport; easy access to green and ecologically diverse spaces; proper water management through modern technology; reduced urban sprawl; usage of renewable energy; and reduced waste production.

Nature-based infrastructure is mentioned in the document only once indirectly and the city equates it with ecological infrastructure. For instance, the city will promote the use of ecological infrastructure where that is cost-effective, then work to rehabilitate ecosystems and ecological infrastructure, and promote low impact urban design in favor of ecological infrastructure. Further, the strategy's Implementation Plan outlines that an Ecological Infrastructure Plan is proposed, though there is no indication that plan has already been accepted or passed. This plan would identify ecosystems and areas of importance and would work in synchronization with the City's Spatial Development Framework and Development Management scheme. The Under Coastal Program, the city also notes that there is a focus on ecological infrastructure along the shore.

Now, it is important to note that the entire Environmental strategy does not refer to nature-based solutions or nature-based infrastructure in the sense it was defined in this thesis. The focus seems to be put natural infrastructure, or rather ecosystem services provided by the natural environment. Nature-based solutions are somewhat touched upon when referring to wetland and coastal management.

The Social Development Strategy from 2012 is a document that attempts to contextualize solutions to societal challenges present in Cape Town. The strategy identifies poverty, inequality, substance abuse and crime, and further states that these issues hinder the city's development and the residents' ability to access opportunities for self-development. The city aims at several outcomes with this strategy, and among them: maximizing earning opportunities for residents at risk of exclusion, developing safe households and communities, and supporting vulnerable societal strata by enabling them to access infrastructure and services (City of Cape Town, 2012).

This strategy makes no mention of nature-based solutions, infrastructure, or biomimicry in any context. Despite that potentially being sensible, due to the strategy addressing social challenges, I believe that integrating these concepts would make it more comprehensive. This document is also fairly dated, and expanding the approaches to addressing social issues might be prudent in its revision. For instance, nature-based solutions such as urban green spaces promote social intermingling, which, if properly planned can address access to infrastructure for socially vulnerable populations. Further, all nature-based projects, similar

to any infrastructural projects, create new job opportunities. Developing areas by providing them with better infrastructure not only creates new jobs on the planning and execution of projects, but also drives economic activity. This improves chances of those areas' economic growth by bringing in new commercial and hospitality establishments, furthering earning potential. Considering the fact that a lot of the jobs that rely on such developments do not necessarily require higher education, this could definitely be a way to address this set of issues. Additionally, as neighborhoods become more developed, they also become more attractive to visitors, which also contributes to social integration, but also could add to the city's efforts to create safer communities.

Cape Town's Inclusive Economic Growth Strategy from 2021 identifies several problem areas the city is facing, and among them also climate change and unsustainable resource access, inadequate levels of economic collaboration, inadequate private and public investment, and skills deficiencies. The city finds itself playing several roles in the economy and addressing these challenges, namely as a service provider and facilitator, among others (City of Cape Town, 2021).

This strategy makes no mention of nature-based solutions or biomimicry, but it does note the proximity of a natural area – the Table Mountain, and underlines its role in economic processes. Now, the role nature-based solutions can play in addressing certain economic problems was already mentioned in relation to the Social Development Strategy, and they are equally as applicable to the Inclusive Economic Growth Strategy. Climate change and unsustainable resource access can be addressed with nature-based solutions, and this would lie on both the governance and infrastructural sides. Unsustainable resource access is purely related to governance, and as nature-based solutions are also actions to manage natural systems, mindful planning that would prevent resource depletion in short periods of time could be considered a nature-based solution. Climate change will be discussed more subsequently. Economic collaboration can be driven by nature-based and biomimicry projects, as they require interaction between various stakeholders and often sectors, as evidenced in several projects presented above, similarly how these developments can drive economic activity and investment.

When it comes to climate change, Cape Town acknowledges its threats as well as the impacts that it has had on the city already and logically connects them to all three aspects of sustainability. The city has two main documents that guide its responses and reactions to climate change, as well as a commitment to achieve carbon neutrality by 2050 and several reports on climate change hazards (City of Cape Town, n.d.):

- The 2021 Climate Change Strategy
- The 2021 Change Action Plan (and the Strategy's Summary)

The City of Cape Town's (2021a) Climate Change Strategy offers guidance to various actors and stakeholders in relation to climate change and its effects, and it is designed to be used together with the Action Plan. The strategy covers both mitigation (which the city understands as reduction in greenhouse gas emissions) and adaptation. The strategy possesses all the elements outlined in Major and Juhola's (2021) framework, and what is perhaps most admirable about the strategy is that it provides a comprehensive list of linkages to various policies on different administrative levels that already exist and address some aspect of climate change. The strategy mentions nature-based solutions in two places, and contextualizes them in the following ways:

- Their ability to ensure sustainable and resilient ecosystems;
- Their ability to support the delivery of ecosystem services as a response to climate change;

- Their integration in infrastructure development as a balancing factor;
- Their ability to prevent stranded economic assets; and
- Their ability to reduce the impact of extreme climate events.

The strategy does not mention nature-based solutions when discussing strategic focus areas and goals (SFA), although they could be included in some of them. In discussing SFA 1 – urban cooling, nature-based solutions can be used for thermoregulation. The strategy notes that a mitigation aspect is urban greening, which can absorb some of the additional heat gain, and it is a form of nature-based solution. Under SFA 4 – coastal management, for instance, the strategy understands the risks of sea level rise and coast erosion, both of which can be addressed with nature-based solutions, such as different types of vegetation in coastal areas that can prove helpful with sea level rise.

Biomimicry is not mentioned at all, which is perhaps even more surprising, as it can be used in architecture to address several of the SFAs. Under SFA 8 – zero-emission buildings, biomimicry can be used, often in conjunction with nature-based solutions, to reduce reliance on artificial cooling and ventilation systems. This is also partially related to SFA 1 – urban cooling, as specific types of concrete and paint can help with cooling urban areas, which is a combination of nature-inspired design and technology.

The City of Cape Town's (2021b) Climate Action Plan is an accompanying document to the Climate Change Strategy, and provides additional details and clarifications on how the strategy should be implemented. The plan includes a set of detailed actions, with information on stakeholders responsible for their implementation, making for a document that is user-friendly and easy to navigate. The document also shows the status of all the actions, though depending on when its next revision will be, these will eventually become obsolete. Nature-based solutions are mentioned in the strategy six-fold compared to the strategy, that is in 12 places, but they are mentioned only in relation to one action.

The plan proposed developing a database of nature-based solutions and climate change response best practices, under Action 35.3. Here, nature-based solutions are mentioned in the context that their co-benefits can also increase ecosystem's capacity to sequester carbon, but it also notes that they cannot be generalized due to their local nature. The action also entails developing new nature-based solutions and monitoring their outcomes and decide on their successfulness. This action's is still just a concept.

Nature-based and biomimicry projects

Table 5. below presents the basic information about the reviewed biomimetic and nature-based solutions analyzed in Seoul. Each solution's description is paired with features that make it nature-based and/or biomimetic, as well as the cost of the projected. Information about the costs could not be found for any of the reviewed projects.

Table 5. Solutions from Cape Town. (Prepared by author)

Cape Town	Project Name	Brief Description	Relevant Nature-based and Biomimetic Features	Cost
1	Princess Vlei Restoration Project for the Greater Conservation Area	Revitalization and greening project in a low-income area that aims at improving local biodiversity and providing a traditionally disadvantaged population with easy access to green spaces	Natural park	No data
2	Cape Town Civic Center Roof Garden	Garden on top of a building that aims at improving local biodiversity and greening otherwise grey infrastructure	Roof garden	No data
3	The Arderne Gardens	Park that helps improve local biodiversity and provides easy access to a green space	Park, ponds	No data
4	Maynardville Park	Park and open-air theater providing visitors with easy access to a green area and social and cultural activities, as well as protection from heat and supports local biodiversity	Park, pond	No data
5	The Ridge	Commercial building built using biomimetic principles and sustainable materials, aiming primarily at energy and water conservation	Natural ventilation, rainwater collection, greywater treatment, sustainable materials – cross-laminated and modified timber and ecobricks	No data

Most of the solutions analyzed in Cape Town are related to the provision or restoration of outside green spaces. One of them is significantly larger in scale and is a restoration of what is a natural park. Moreover, one solution stands out as an infrastructural feat, and it utilizes nature-based and biomimetic principles to be more sustainable and energy efficient.

After reviewing the policies the city has in place both to address sustainability challenges and climate change, the case will present some of the nature-based solutions and biomimicry projects in the city.

The Ridge

The Ridge is a commercial building in Cape Town's V&A Waterfront created by StudioMAS and Arup, where sustainability and technology come together (see Figure 80.) The building's main sustainability features revolve around air and temperature. It possesses a natural ventilation system and it is constructed with two wings, which are both there to distribute its weight and act as the building's lungs. This allows for minimal energy consumption on air conditioning, and it is projected that the building saves 64% in energy in comparison to similar buildings. The building also uses cross-laminated timber and modified timber cladding on its façade, relying on the material's robust and light nature. In fact, it is through the façade that the building achieves natural ventilation. Moreover, the building also collects rainwater and treats greywater, and then uses water from both these sources for its toilets and greenery irrigation, allowing it to halve its potable water consumption. There is also the fact that the designers were sustainably-minded in materials selection, and opted to use around 14,000 ecobricks, thus reducing the use of concrete (Arup, n.d.).



Figure 68. The Ridge. (Gareth Griffiths Imaging, n.d.)

Maynardville Park

The Wynberg residential area in Cape Town's southern suburbs boasts Maynardville Park and open-air theater. The park can be found in the bustle of the city and it provides a refuge from the noise and heat, as well as activities related to the arts, drama, and music. The park is given structure with many walking paths that divide the grassy areas covered by flower and tree species, including the Mediterranean cork oak, swamp cypress, and lemon-scented gum trees. There is also the Krakeelwater pond in the park (see Figure 81.), which contributes to the area both aesthetically and in terms of heat absorption (City of Cape Town, n.d.).



Figure 69. Maynardville Park. (Abu Shawka, 2014)

The Arderne Gardens

The Arderne Gardens is another park in Cape Town (see Figure 82.) managed by the city and it features shady glades and Japanese-style ponds. The garden is also a biodiversity reserve of sorts and it contains the most diverse collection of exotic trees in the country, as well as hosts one of the country's largest trees, the vast Moreton Bay Fig and potentially the biggest Aleppo Pine in the world (The Arderne Gardens, n.d.).



Figure 70. The Arderne Gardens. (The Arderne Gardens, n.d.)

Cape Town Civic Center Roof Garden (McKellar-Basset, 2010)

The roof garden (see Figures 83. and 84.) on the Cape Town Civic Center is a joint endeavor by architects Kruger and Roos and several other consultants is a way to breathe life into the glass and aluminum construction that would be the center's expansion. One of the guiding ideas in designing the garden was that it must not be too reliant on water, both for the objective need and in order for the city to set an example in adhering to water policies. The garden, utilizing decorative gravel circulation areas for water conservation, includes a variety of plants divided into seven landscapes:

- 1 is the Succulent Karoo, displaying succulents, shrubs, trees, and bulbs
2. and 3. are for the Cape fynbos, with restios, aromatic shrubs, and seasonal bulbs
4. through 7. showcase various Eastern Cape thicket and bushveld plants

Different elements were equally considered in the design process, in order for the garden to provide suitable shade, seating, greenery, and paving, and maximize the overall utility of the project. The project was also a learning experience, as certain challenges had to be overcome. The selection of soil was one of them, and the designers had to come up with special soil mixes with appropriate drainage and water-retention, and the

ability to support the various plants. Waterproofing was another, and different approaches were used to avoid water waste, such as drip irrigation.



*Figure 71. Cape Town Civic Center Roof Garden.
(Marijke Honig, n.d.a)*



*Figure 72. Cape Town Civic Center Roof Garden
Planting. (Marijke Honig, n.d.a)*

Princess Vlei Restoration Project for the Greater Conservation Area

The Princess Vlei Restoration Project is still ongoing in the Greater Princess Vlei Conservation Area (GPVCA), a wetland in one of Cape Town's low income areas, that has historically been disadvantaged by apartheid policies, which has also led to its ecological decline (see Figure 86. in Appendix 5.) The area is currently undergoing a rehabilitation and revitalization process (see Figure 85.) for three main types of flora: Cape Lowlands Freshwater Wetlands (endangered), Cape Flats Dune Strandveld (endangered) and Cape Flats Sand Fynbos (critically endangered). The project has environmental and socio-economic aims, such as eliminating existing threats to the ecosystem; restoring a desirable species composition and structural diversity; economic vitality and local livelihoods; and regulation of climate, floods, disease, erosion, and water quality. The project has been in progress for several years already, and it is planned to be completed in 2024 (Society for Ecological Restoration, n.d.).



Figure 73. Princess Vlei Area Before and After Restoration. (Alex Landsowne, n.d.)

DISCUSSION

After developing the cases for the five cities, the thesis can now refer back to its aims and answer the initially posed research questions. As the main aim of this thesis is to summarize the findings about the effectiveness of nature-based solutions and biomimicry projects on environmental sustainability and their role in climate change mitigation and adaptation plans in the selected cities, the thesis presented 39 different infrastructural solutions across the five cities, as well as their relation to sustainability and role in climate change mitigation and adaptation. All the solutions are summarized and accompanied by visual materials. Through the cases, I have found that the implemented nature-based and biomimicry, either in design or outcomes, affect different aspects of sustainability as well as climate change mitigation and adaptation, as presented in the Table 6. in Appendix 1.

A total of 39 projects were presented, and all of them seem to have impacted sustainability and climate change mitigation and adaptation in different ways and on different level. Out of the analyzed nature-based and biomimicry solutions, **97% addressed environmental sustainability** in some way. Mostly, the solutions were greening the areas in which they were present or were applied to already existing infrastructure to make it greener. In addition to this, energy efficiency or water conservation in some form were also fairly common, and the solutions included some form of decreased grid-energy reliance (e.g. solar panels or geothermal rods) or water grid reliance (e.g. by harvesting rainwater). In cases where rainwater was harvested by the solution, it is never intended for human consumption through ingestion, but for technical purposes such as irrigation or toilet systems.

Social sustainability was addressed through the projects less than environmental sustainability, that is by **79%** of the projects. Social sustainability was addressed in different ways as well, and most commonly it was through improving access to green areas, including for a low-income area in one case. Additionally, the social sphere was affected by the provision of social amenities, or spaces for cultural, educational, and art purposes. Aesthetics were also a common theme, improving the overall appearance of either a building or the area.

Economic sustainability was, according to the analysis, least affected of the sustainability dimensions – by **51%** of the projects. From an economic standpoint, the projects mostly contributed to the infrastructure's energy and water efficiency and decreasing reliance on the grid. Moreover, in some cases, when grey infrastructure was being repurposed, that was also a form of economic benefit, as no demolition or new construction needed to be undertaken.

The effects of the analyzed solutions were mostly multi-dimensional, and all the projects affected at least two dimensions. What is perhaps more interesting is that the effects of the solutions were present on multiple levels. With projects implemented or initiated by the city, effects on overall urban sustainability or climate change impacts can be observed. The parks, recreational areas, or coastal restoration projects, all can be observed to have effects that are not necessarily localized or related to the infrastructure itself.

With private projects, such as buildings that incorporate some kind of nature-based or biomimetic solution, effects are project-specific. This means that the benefits are reaped by the users of that infrastructure, such as indoor green walls or energy-efficient systems, as only the building's occupants will enjoy increased air quality or lower costs for energy, for example.

In some cases, I would note, these local or infrastructure-specific benefits can be transposed onto the system at large. This is primarily noticeable with energy and water efficiency, for example. In future scenarios where cities are at risk of experiencing water scarcity, harvesting rainwater or processing greywater for a building's green wall irrigation or toilet flushing system means lower reliance on water from the grid. By extension, the building in question would not be an additional burden to the grid.

The same applies to energy efficiency if achieved through photovoltaic panels, as the building does not burden the grid, which automatically contributes to overall energy supply not being depleted unnecessarily. An added benefit in this case is also that there are no emissions from energy generation, therefore adding on to overall contributions to climate mitigation efforts, for instance. These effects, however, have not been included in the table in Appendix I, as they are implied and not specifically mentioned anywhere, as well as that the impacts of equipment production would also need to be considered to assess whether something like the postulated situation is entirely sustainable and climate-friendly.

In relation to climate change mitigation, the included projects' effects on **climate change mitigation** were the overall lowest, at **41%**. These effects are that the solutions can act as carbon sinks, or to absorb more carbon than they produce. **Climate change adaptation** efforts were affected more, by **69%** of the projects. These effects were noted mostly on the solutions' capacity to offset the effects of increased temperatures, but then also through water conservation, thermoregulation (building-level), and infrequently in flood risk mitigation. Frankly, it was somewhat surprising to see that there were not a lot of solutions implemented with the main purpose being addressing climate change challenges. Nonetheless, there are climate co-benefits in a fair share of the analyzed solutions.

The first research question considered what influences the effectiveness of nature-based and biomimicry solutions in relation to urban sustainability in the selected cities. Based on the case studies, I would argue that, from a scale perspective, one should look at the infrastructural project's use type and owner or initiator. Mixed-use developments and infrastructure, that is any projects that were from their inception intended to serve various purposes and audiences, seem to have an overall better impact on sustainability. All parks can be relevant for both the environmental and social dimensions, and therefore have an overall sizeable potential to affect urban sustainability. However, some parks and walkways that were analyzed affected all three facets of sustainability, if they managed to include hospitality or commercial establishments.

Solutions that were developed for a specific purpose, and this seems more relevant for projects that are privately developed, have a lesser impact on urban sustainability. The specific purpose limits the ability of what the project can include, and therefore the scope of its impacts. If the owner or initiator is also considered, it can be observed that private projects can have sustainable features, though their effects on sustainability overall are less than those initiated by the cities. Several of the commercial or aesthetic projects and buildings testify to this, as they can affect only some facets of sustainability, or if they affect all facets of the city's sustainability, the scale at which this happens is much lower.

In this sense, the city should take the lead on planning and initiating the development of multi-purpose and mixed use nature-based and biomimetic projects, in order to maximize their effectiveness in the context of urban sustainability. Concerning private projects, they should be developed with several purposes in mind, in order for their sustainability impacts to be maximized. This, of course, is not the primary responsibility of private investors, and therefore it is not surprising to see these trends among the analyzed cases.

Something that was also observed is that the project's locality can also be related to how effective it is from a sustainability perspective. Excluding, of course, coastal infrastructure, certain the ability of a development to possess aesthetic value and thus contribute to social sustainability is only possible if the appropriate location for the implementation is selected. Certain building embodying biomimetic principles, if placed in areas where they do not fit in well, will not be able to contribute to this dimension. The most plastic example would be placing a high-rise among three-story buildings; it would disrupt the natural flow of the skyline and feel out of place.

The most important considerations related to biomimetic and nature-based projects have been already touched upon in answering the first question, but they can be further elaborated (see Figure 87.).

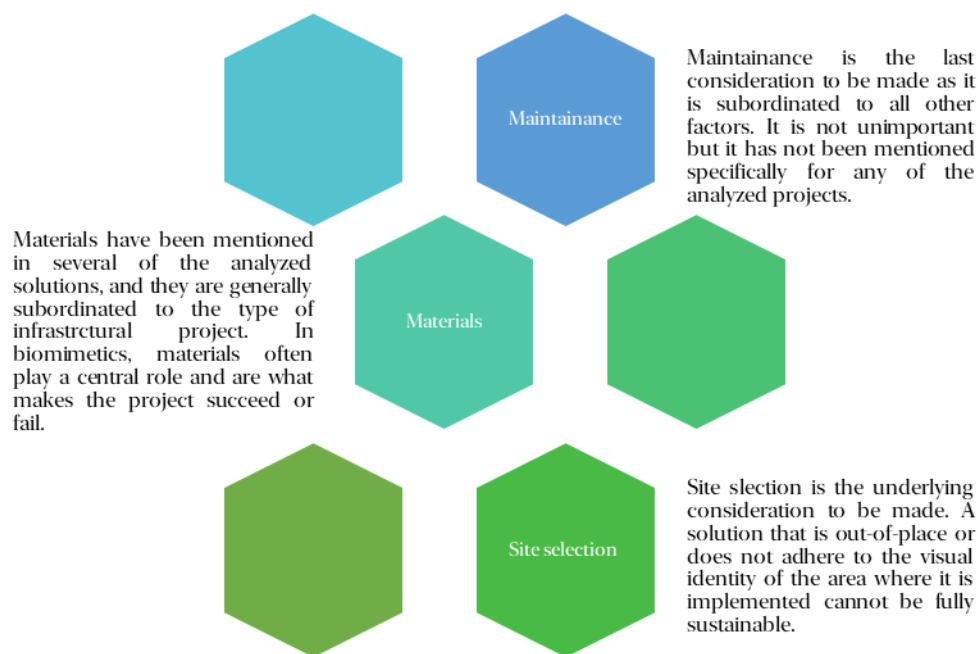


Figure 74. Considerations for Implementation of nature-based and biomimetic projects. (Developed and prepared by the author.)

Site selection for the implementation of the solutions, aside from what was already mentioned, also includes proprietary rights. That is, if a solution finds itself on or in close proximity to the property of an entity other than the owner, extra care needs to be taken as to avoid damage. This is perhaps most prevalent when it comes to water damage, in the cases of roof gardens and green walls.

Some of the most common materials the analyzed solutions utilized, which are related to sustainability and the solutions' effectiveness are:

- Wood – light-weight and breathable, contributes to natural ventilation
- Glass – enhances solar gain
- Recycled materials – can reduce the reliance on concrete (e.g. ecobricks)
- 3D printed materials – easily recyclable during the production process

These are, of course only some of the materials the projects have utilized and that can be connected to sustainability or the buildings' sustainable properties. Most interesting here were 3D printed materials, and their sustainability is reflected in the fact that they can be recycled during construction. That is all parts can be reused and remade in case there are changes in plans.

Maintenance is, though it might seem counter-intuitive, the last link in the chain, as all the analyzed solutions' maintenance does not seem to differ than regular infrastructural maintenance, except in the case of vertical gardens which are somewhat specific in design. Overall, maintenance can be learned, and it therefore can come as the last consideration to be made in the process.

Progressing naturally to next question related to the opportunities and challenges with respect to the solutions' inclusion in sustainability initiative. The first observation is that the selected cities have surprisingly underdeveloped sustainability strategies, or rather strategies that are not well integrated. Only one of the analyzed cities had a fully integrated sustainability strategy that addressed all dimensions of sustainability, as well as the city's climate change responses. The other cities either had separate strategies for different aspects of sustainability, strategies that build upon the 2030 Agenda, or only environmental sustainability strategies.

When this is considered, it is clear that nature-based and biomimetic solutions' inclusion in these strategies was not on a high level. Each strategy mentioned nature-based solutions or infrastructure in several instances only or not at all, and when the term was mentioned the claims were mostly generic. Biomimicry was not mentioned in any of the analyzed strategies. This was the ultimate challenge, as there was no baseline to analyze. On the other hand, this also doubles as an opportunity, as the potential for the inclusion of the concepts is great.

I also found that, despite the lack of nomenclature, the strategies and plans included generally did include nature-based or biomimetic solutions in everything but name. Principles related to the solutions or some solutions themselves were nature-inspired. This could imply a competence gap, or the fact that the cities have not made it a priority for themselves to promulgate the use of nature-based solutions or biomimetic principles. Either way, I believe this is an opportunity for future plans to utilize the concept when conducting sustainability planning. The outcomes of the implemented nature-based and biomimetic projects are the theme of the next question. This summarized as the main aim of the thesis, but there are more details related to this (see Figures 88.-90.)



Figure 75. Main outcomes of nature-based and biomimetic projects in Cape Town. (Developed and prepared by the author.)

From Figure 88 above, one can see that there are some commonalities between projects' outcomes in relation to urban sustainability, despite them not being related. All the cities have solutions that bare benefits for biodiversity and ecosystem restoration, as well as a consequent increase in green areas. Interestingly enough, only Cape Town has a solution that is located in a low-income area, which I believe can have additional co-benefits in both social and economic sustainability. Once the project is completed, the area could draw in more people who do not reside in the area and who belong to different societal strata, thus helping with the residents' social mobility. This could also contribute with some economic co-benefits, though this would require targeted action by the city as well.

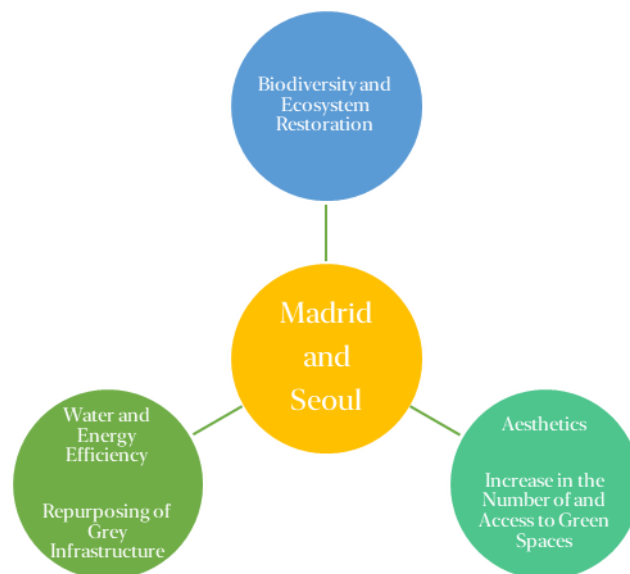


Figure 76. Main outcomes of nature-based and biomimetic projects in Madrid and Seoul. (Developed and prepared by the author.)

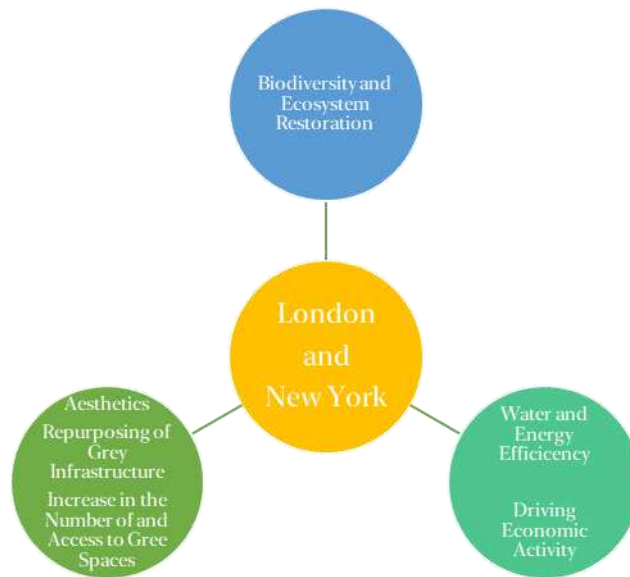


Figure 77. Main outcomes of nature-based and biomimetic projects in London and New York. (Developed and prepared by the author.)

The remaining four cities, all have the outcome of repurposed grey infrastructure. I would argue that this is related to overall levels of development and the land availability. Urban sprawl, being a negative phenomenon as is, is also not an option for some cities, due to their layout and surrounding areas. Also, I am also ascribing this to the sheer built infrastructure in these cities, and the fact that it is more practical to repurpose what already exists, rather than to demolish and then rebuild or recreate. This is how I would explain these connections.

Only London and New York have solutions that also drive economic activity. I would argue that these two cities are the most advanced out of the analyzed cities, which is also supported by the presented indicators, and therefore also consider economic benefits more. However, it is also important to consider that this can be attributed to some of these two cities' projects being implemented in already fairly developed areas with existing propensity toward commercial and entertainment activities or the area's historic heritage, and therefore this potential could be achieved. Overall, it is undeniable that all the combined solutions in all of the cities positively contribute to urban sustainability, with the only difference being the type of outcome and scale.

All the findings already make up a modest body of knowledge on nature-based and biomimetic effectiveness and implementation in urban context that can fairly easily be transposed onto other urban areas. Below are some of the main takaways that I consider most easily applicable, regardless of the size of the city (see Figure 91.)

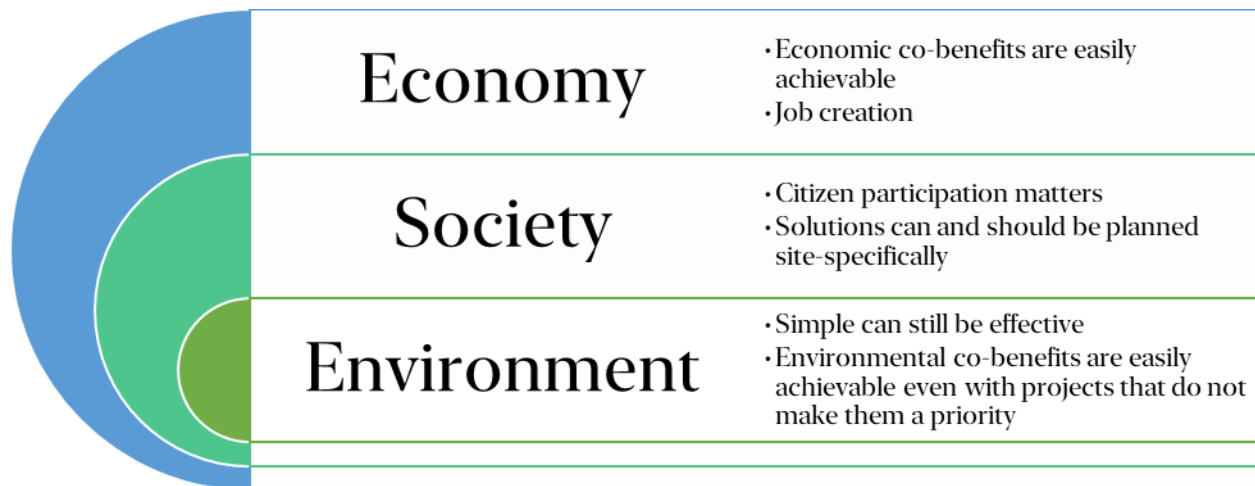


Figure 78. What other cities can learn about nature-based and biomimetic projects' impact on urban sustainability? (Developed and prepared by the author.)

Regardless of the implementing city's size or scale, nature-based and biomimetic solutions can be utilized in improving urban sustainability. Starting with the environmental sphere, I would first underline that a nature-based solution can be as small as greening an area by planting trees which will provide heat or sound pollution protection. Similarly, biomimetics can be applied to infrastructural projects, with some research, which need not necessarily be conducted by the city itself. Simplicity can therefore be the basis for all these projects, and these solutions are related mostly to the way of thinking.

Some of the reviewed projects included citizen participation at some stage of the process. This can be as small as naming the final project, over questions on the design or site of implementation, or the type of amenities that the solution will include. Citizen participation can also benefit the projects themselves, rather than just be positive practice and policy mechanism for its own sake, in the sense that new ideas can be generated through co-creation. Planning projects should begin with the site in mind. I believe this is crucial for the projects' success, as if it is created in an area that it does not feel organic to and whose residents dislike it, then the purpose of social sustainability is defeated. When considering what to do, it is necessary to also consider if there is a specific site where it can and should be done.

Although economic (co-)benefits might not be the first thought that comes to mind when thinking about these solutions, they are definitely beneficial to have, and are easily achievable. Designating a site within the selected area for commercial or hospitality establishments will easily impact economic sustainability. Not only that, but this can also increase the attractiveness of the project itself, and enable it to offer its other benefits to more people. Furthermore, nature-based and biomimetic solutions' potential to contribute to job creation should not be disregarded. This helps not only those who are to benefit from these job opportunities, but also to the city itself, as more jobs contribute to bolstering the economy.

Finally, the last considerations are on how to quantify the benefits of nature-based and biomimetic solutions and communicate them adequately to policy makers, residents, and other various stakeholders. This question is interesting because it is, in a way, how this thesis gains applicability and practical value. Unfortunately, there was very little verifiable quantitative data that I could use to make any inferences from. However, this is also something that I see as an opportunity as it gives me some creative freedom in how I think benefits could be quantified and communicated, and I propose the model below (see Figure 92.)

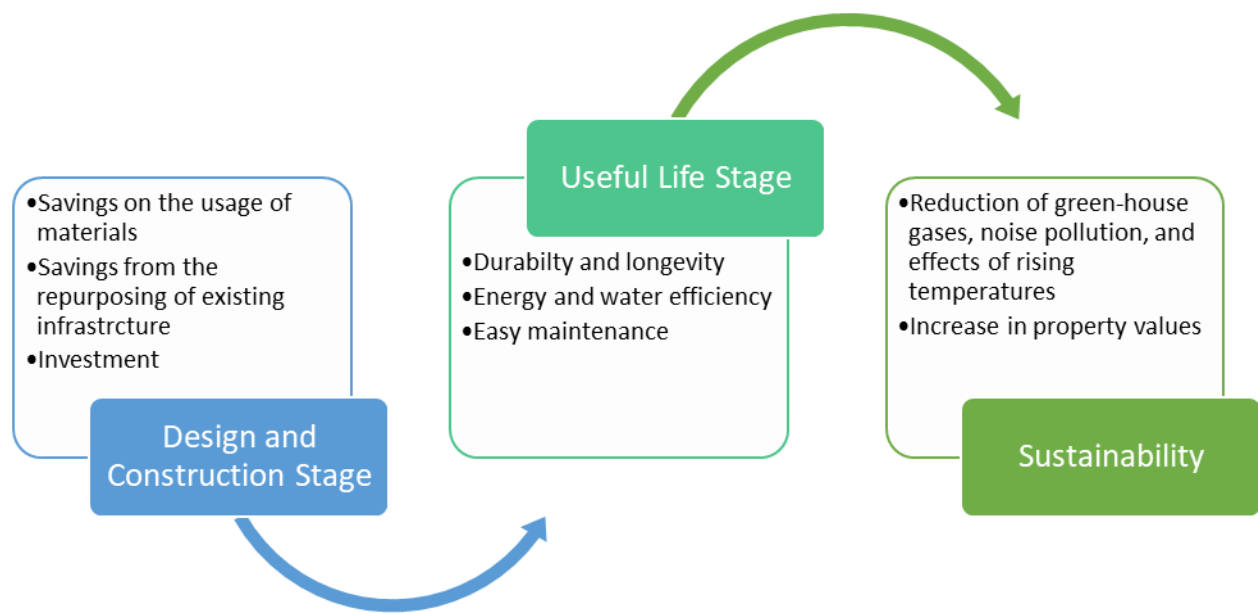


Figure 79. How to communicate the (co-)benefits of nature-based and biomimetic solutions with stakeholders? (Prepared and designed by the author,)

I think that these are the main points that can and should be quantified with nature-based and biomimetic solutions in order to effectively communicate the solutions' (co-)benefits to a variety of stakeholders. The idea is rooted in the fact that the majority of decisions are made with economic legitimacy in mind, as was also mentioned in the definition of nature-based solutions. Despite the pressing urgency of changing human behavioral patterns if the climate crisis is to be thwarted, I do not consider such a shift to be highly realistic, and therefore think communicating about these solutions should be put in terms that are easily understood.

From the analyzed solutions, there is an indication that sustainable construction processes could cost less than traditional ones, but what needs to be done, and this could be the topic of another scientific or other paper, is a cost analysis of different nature-based and biomimetic projects, to ascertain that their implementation would not pose a financial challenge for the urban area in question.

Further, and this is evidenced by some of the decisions in Seoul and New York with the analyzed solutions, repurposing grey infrastructure, instead of tearing it down is a more cost-effective way of implementing nature-based solutions. Depending on what a city's plan is for an area that is to be redeveloped, the city itself should conduct a financial analysis to decide what the best approach is. In Seoul, for instance, a successful revitalization and repurposing project has already influenced another project to take the same approach, and it is currently underway.

Given that these solutions are gaining momentum in popular media and academia, and that there is already a number of solutions being implemented worldwide, there is also investment that can be expected if a city opts for one such solution, instead of a traditional grey infrastructural project. An academic study could be conducted on the availability of funding from international institutions, such as the European Union, or the availability of green bonds and loans, that could be considered investments in nature-based and biomimetic solutions. Furthermore, it would be worth conducting an analysis of investments these solutions have already contributed to bringing in cities where they are implemented.

I have not found any indication that nature-based solutions are not durable or that they cannot last long. A study into the useful life of specific solutions, perhaps by category, could be conducted in order to understand how long solutions could be expected to provide benefits for, on average. This could be integrated with a maintenance costs study that would research whether the solutions' upkeep costs more than that of different infrastructural solutions, or what these expenses are in general.

Energy and water efficacy has actually already been mentioned in the analysis in several places, and I believe making facts and figures such as these are beneficial in communicating with policy makers on the benefits of nature-based and biomimetic solutions. Different solutions analyzed in this thesis self-report energy saving between 64 and 70% because of their nature-based or biomimetic properties, as well as decreased dependance on the water grid, as one is 90% water self-sufficient. These numbers are a good indicator that these solutions can financially benefit from energy and water efficiency.

Further, the co-benefits of nature-based and biomimetic solutions in dealing with the effects of climate change can also be quantified, though there was not sufficient data for me to make such projections. Another study could be conducted to see, for example, how much carbon can be stored in a park, or the ability of a green roof to lower the surface temperature during the summer or a heatwave. This information would, I think, contribute to better contextualizing the benefits of the solutions.

Ultimately, something that I consider valuable is the information that difference in property prices in downtown Seoul with a difference being the property's proximity to the redevelopment project have doubled. With this in mind, cities could expect increased property prices, and with this knowledge, they would have the ability to improve the affluence of a neighborhood, whilst simultaneously providing the residents with access to more green areas. All of these are points that I think would make for good quantitative and economic studies, that could help make a case for nature-based and biomimetic solutions.

CONCLUSIONS AND RECOMMENDATIONS

As cities grow, so does their importance in the international system, but also the challenges they face. These challenges are both within the realm of sustainability, affecting their environmental, social, and economic spheres, as well as those exacerbated by climate change. Despite geographic, cultural, and other differences, all cities have the common denominator that they cannot avoid challenges, but, all cities have a way of addressing the challenges and problems they face. The response the cities provide will shape their futures, and sometimes it is not necessary to reinvent the wheel to be successful. In fact, some solutions to some problems man and cities face are timeless. They can be found in nature and its principles, and this is precisely what nature-based and biomimetic solutions are.

This thesis aimed at contributing to developing a better understanding of how nature-based and biomimetic solutions can be used to improve sustainability and climate change responses in urban settings. I reviewed selected urban development projects and most notable policy documents addressing sustainability and climate change in five global cities across four continents in order to achieve this aim and answer seven research questions. I was interested in the outcomes of the implemented solutions, what are some factors of success, what can be learned, and what of these cities' experience can be transposed onto other urban areas in the world.

Through this research, I found that there are several factors that contribute to the success of these solutions, especially during the planning stage, as well as that the environmental facet of sustainability is most commonly addressed by the analyzed solutions, at least to some extent. This is congruent with my findings that sustainability in urban areas is mostly understood through the prism of environmental questions, as all the cities I analyzed had some sort of environmental sustainability strategy, while integrated sustainability strategies or strategies focused on social and economic sustainability were rare. Something else that I found was rare in the case cities is their utilization of nature-based and biomimetic solutions primarily and specifically for climate change mitigation and adaptation, which was surprising considering their potential in these two fields.

In conceptualizing the contributions of nature-based and biomimetic solutions to both sustainability and the fight against climate change, I also considered how can the benefits of the solutions in relation to these two foci can best communicated to stakeholders. Considering the current paradigms in policy, I proposed a way to communicate these benefits, and especially toward policy makers.

A prominent challenge was finding statistical data on city levels, as well as finding the financial details about specific projects. This, however, has birthed something positive – several ideas for research projects in the field that could serve as communication points for the practical and financial benefits of nature-based and biomimetic solutions. This potential research would, hopefully, contribute not only to an increased usage of the solutions in practice, but also their inclusion in policies, as the research also found that they are very infrequently mentioned explicitly or implicitly.

The value of this thesis is in it bridging understandings about nature-based and biomimetic solutions across four continents, linking the similarities in outcomes, and finding some differences, which are equally as relevant. As a contribution toward the practice of nature-based and biomimetic solutions, there are also certain policy recommendations that, if enacted, could contribute to urban sustainability and thwarting climate change.

First, I recommend the inclusion of nature-based solution and referring to biomimetic concepts in conceptualizing sustainability strategies and climate change action plans.

Second, with the understanding of the breadth of impacts that nature-based solutions and biomimicry can have on improving urban sustainability, I recommend purposefully planning these solutions in an integrated manner. This can help maximize their benefits, and simultaneously provide the optimal return on the investment in such infrastructure.

Third, considering the above and the growing popularity of nature-based and biomimetic infrastructure, I recommend utilizing the investment such infrastructure can bring to urban areas. This pre-implementation incentive, coupled with the post-implementation benefits should inspire their usage.

Fourth, and last, I recommend considering the utilization of these solutions in climate change adaptation and mitigation. It is evident that both these sets of measures must be fully implemented if cities are to be resilient to climate change. Even if there were not many solutions used specifically with climate change in mind, there was a considerable number of those that had co-benefits in action against climate change. Planning them specifically can only bring about better results.

Ultimately, this thesis was about bridging theory and practice about a field I personally important and academic stimulative. More research is definitely necessary to understand the full extent of what nature-based solutions can do and how biomimetic principles can be applied, but what is understood at the moment is that they could be one of humanity's best chances for a livable future.

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Appendix 1. – Impacts of Reviewed Nature-based and Biomimetic Solutions

Table 6. Impact of Reviewed Nature-based and Biomimetic Solutions. (Prepared by author)

City	Nature-based / biomimicry project	Dimension					Challenges
		Environmental sustainability	Social sustainability	Economic sustainability	CC Mitigation	CC adaptation	
Cape Town	Princess Vlei Restoration Project for the Greater Conservation Area	Ecosystem restoration Biodiversity	Improving access to green zones in a low-income area			Erosion prevention Flood prevention	
Cape Town	Cape Town Civic Center Roof Garden	Water conservation Biodiversity				Water conservation	Soil selection Irrigation systems
Cape Town	The Arderne Gardens	Biodiversity	Aesthetics Pleasant environment with two of the country's largest trees		Carbon sink		
Cape Town	Maynardville Park	Biodiversity Green area among the grey infrastructure Refuge from noise pollution	Aesthetics Open-air theater and activities withing the arts		Carbon sink	Offsetting effects of increased temperatures	
Cape Town	The Ridge	Natural ventilation system Lower energy consumption Rainwater collection Greywater treatment Sustainable building materials	Aesthetics	Water efficiency		Natural ventilation Water efficiency	

Seoul	Seoul National University Green Roof	Biodiversity	Utilizing an idle grey area Serves as a social point in the building			Reduces the risk of flooding Innovation in food production Thermoregulation	
Seoul	Seoul City Hall	Biodiversity Air purification Energy efficiency		Energy efficiency		Thermoregulation Humidity control	
Seoul	Seoul Energy Dream Center	Geothermal heating Rainwater collection Natural ventilation		Energy efficiency (net-zero building) Water efficiency		Thermoregulation Natural ventilation	
Seoul	Garden of the Sun in Jonggak Subway Station	Biodiversity	Aesthetics Utilizing an idle grey area Citizen engagement in decision-making	Energy efficiency			
Seoul	The Galleria in Gwanggyo	Biodiversity	Aesthetics				
Seoul	The Weaves	Biodiversity Green area among grey infrastructure Ecosystem restoration	Repurposing grey infrastructure Provision of new social services	Repurposing grey infrastructure	Creating a green-blue area	Waterway management Offsetting effects of increased temperatures	
Seoul	Seoullo 7017	Biodiversity and "tree nursery" Green area among grey infrastructure	Aesthetics Repurposing grey infrastructure Improving access to green zones		Carbon sink	Offsetting effects of increased temperatures	

Seoul	Dongdaemun Design Plaza and Park	Biodiversity Sustainable building process	Social green point among grey infrastructure Provision of social amenities				
Seoul	Cheonggyecheon Stream Restoration	Biodiversity Green area among grey infrastructure Ecosystem restoration and reconnection Waterflow management	Aesthetics Repurposing grey infrastructure Improving access to green zones Citizen engagement in decision-making Opportunities for cultural, social, and recreational activities		Carbon sink	Offsetting effects of increased temperatures Flood protection	Gentrification
Madrid	3D printed bridge in Alcobendas	Sustainable building process	Aesthetics				
Madrid	The Vertical Garden of Torre de Cristal	Biodiversity Air purification					
Madrid	The Vertical Garden of Caixa Forum Museum	Biodiversity	Aesthetics		Carbon sink		Moisture damage prevention
Madrid	Avenida de Portugal	Biodiversity	Aesthetics Improving access to green zones		Carbon sink		

Madrid	Salon de Pinos	Biodiversity	Repurposing grey infrastructure Provision of social amenities Improving access to green spaces		Carbon sink	Offsetting effects of increased temperatures	
Madrid	The Madrid Rio Park Redevelopment	Biodiversity Green-blue area among grey infrastructure	Aesthetics Repurposing grey infrastructure Provision of social amenities Improving access to green spaces		Carbon sink	Offsetting effects of increased temperatures	
London	The Bloomberg European Headquarters	Natural ventilation system Lower energy consumption	Aesthetics	Energy efficiency Driving economic activity			
London	30 St Mary Axe (The Gherkin)	Natural ventilation system Lower energy consumption	Aesthetics	Energy efficiency Driving economic activity			
London	Camley Street Natural Park	Biodiversity Green area among grey infrastructure	Aesthetics Social and educational activities		Carbon sink	Offsetting effects of increased temperatures	
London	Bagley Walk	Biodiversity Green area among grey infrastructure	Aesthetics Repurposing grey infrastructure Provision of social amenities Improving access to green spaces	Driving economic activity Repurposing grey infrastructure	Carbon sink	Offsetting effects of increased temperatures	

London	Gasholder Park	Green area among grey infrastructure	Aesthetics Repurposing grey infrastructure Provision of social amenities Improving access to green spaces		Carbon sink	Offsetting effects of increased temperatures	
London	Regent's Canal	Biodiversity Green-blue area among grey infrastructure Ecosystem restoration	Aesthetics Opportunities for cultural, social, and recreational activities	Repurposing grey infrastructure	Carbon sink	Waterway management Offsetting effects of increased temperatures	
London	Jellicoe Gardens	Biodiversity Green area among grey infrastructure	Aesthetics Intercultural ties Improving access to green spaces		Carbon sink	Offsetting effects of increased temperatures	
London	The King's Cross Central Development	Biodiversity Green area among grey infrastructure Ecosystem restoration	Aesthetics Opportunities for cultural, social, and recreational activities Provision of social amenities Improving access to green spaces Repurposing grey infrastructure Landscape revitalization	Driving economic activity Repurposing grey infrastructure	Carbon sink	Waterway management Offsetting effects of increased temperatures	
London	The Sky Garden	Green area among and in grey infrastructure Biodiversity	Aesthetics Social and educational activities	Driving economic activity			
London	The Crystal	Water conservation	Aesthetics Opportunities for cultural, social, and recreational activities	Energy efficiency Water efficiency Driving economic activity		Thermoregulation Water efficiency	

New York	The Living Breakwaters Project	Ecosystem restoration Biodiversity	Infrastructure protection	Infrastructure protection		Storm surges adaptation Coastline erosion adaptation Flood management Infrastructural hazards prevention	
New York	The Vessel		Aesthetics Opportunities for cultural, social, and recreational activities	Driving economic activity			
New York	The Bank of America Tower	Biodiversity Water conservation Energy efficiency		Energy efficiency		Thermoregulation	
New York	The Brooklyn Grange Rooftop Farm	Green area among grey infrastructure Energy efficiency Water conservation	Utilizing an idle grey area Serves a social and educational purpose	Job creation		Innovation in food production Thermoregulation Water conservation Energy efficiency	
New York	The Newtown Creek Wastewater Treatment Plant	Biodiversity Green area among grey infrastructure				Thermoregulation Stormwater management Offsetting effects of increased temperatures	
New York	The Cooper Union Building	Energy efficiency Water conservation		Energy efficiency Water efficiency		Thermoregulation Energy efficiency Water efficiency	

New York	The Barclays Center	Biodiversity Green area among grey infrastructure Water conservation	Aesthetics Opportunities for cultural, social, and recreational activities Improving access to green spaces	Energy efficiency Water efficiency	Carbon sink	Stormwater management	
New York	The Solaire	Biodiversity Energy efficiency		Energy efficiency		Thermoregulation Energy efficiency Offsetting effects of increased temperatures	
New York	The High Line	Biodiversity Green area among grey infrastructure	Aesthetics Utilizing an idle grey area Serves a social and educational purpose Opportunities for cultural, social, and recreational activities Improving access to green spaces Repurposing grey infrastructure	Repurposing grey infrastructure	Carbon sink	Offsetting effects of increased temperatures	

Appendix 2. – Solutions from New York City

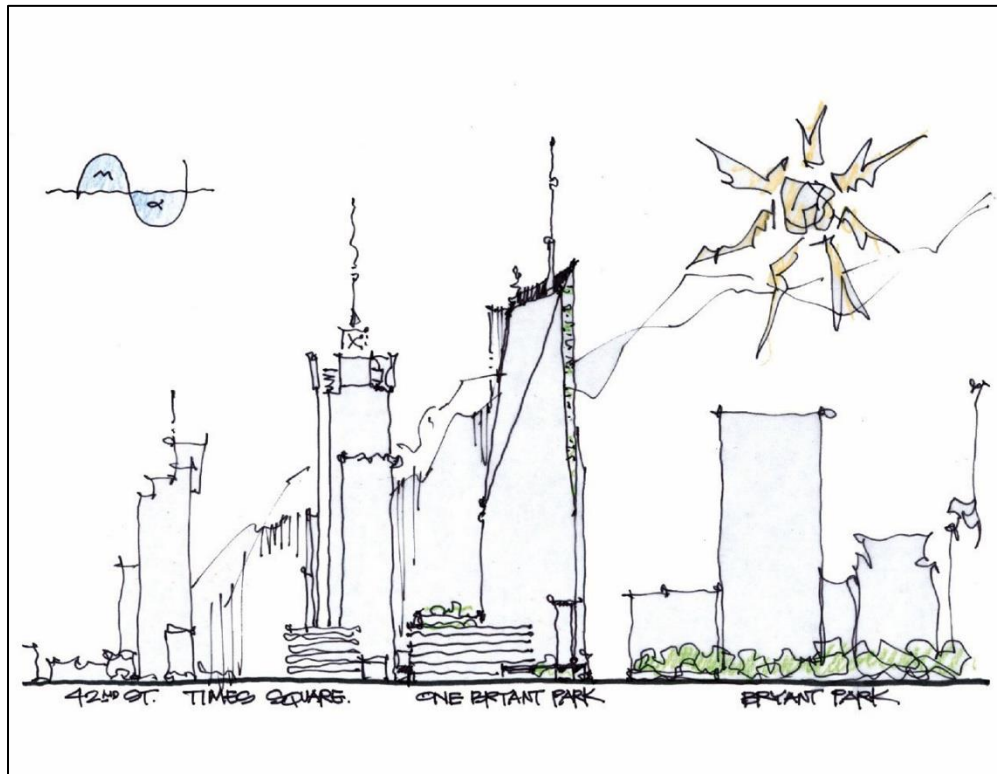


Figure 80. Conceptual Sketch of One Bryant Park in its Ecological and Urban context. (Cook Fox Architects, n.d.)

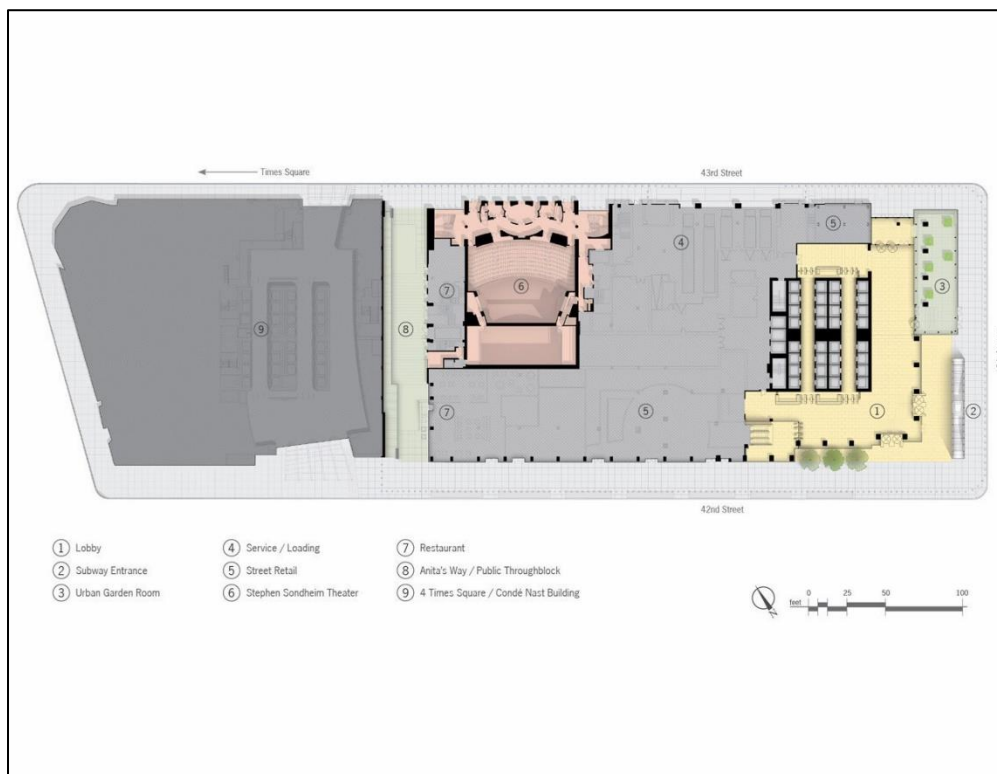


Figure 81. One Bryant Park Street Level Floor Plan. (Cook Fox Architects, n.d.)

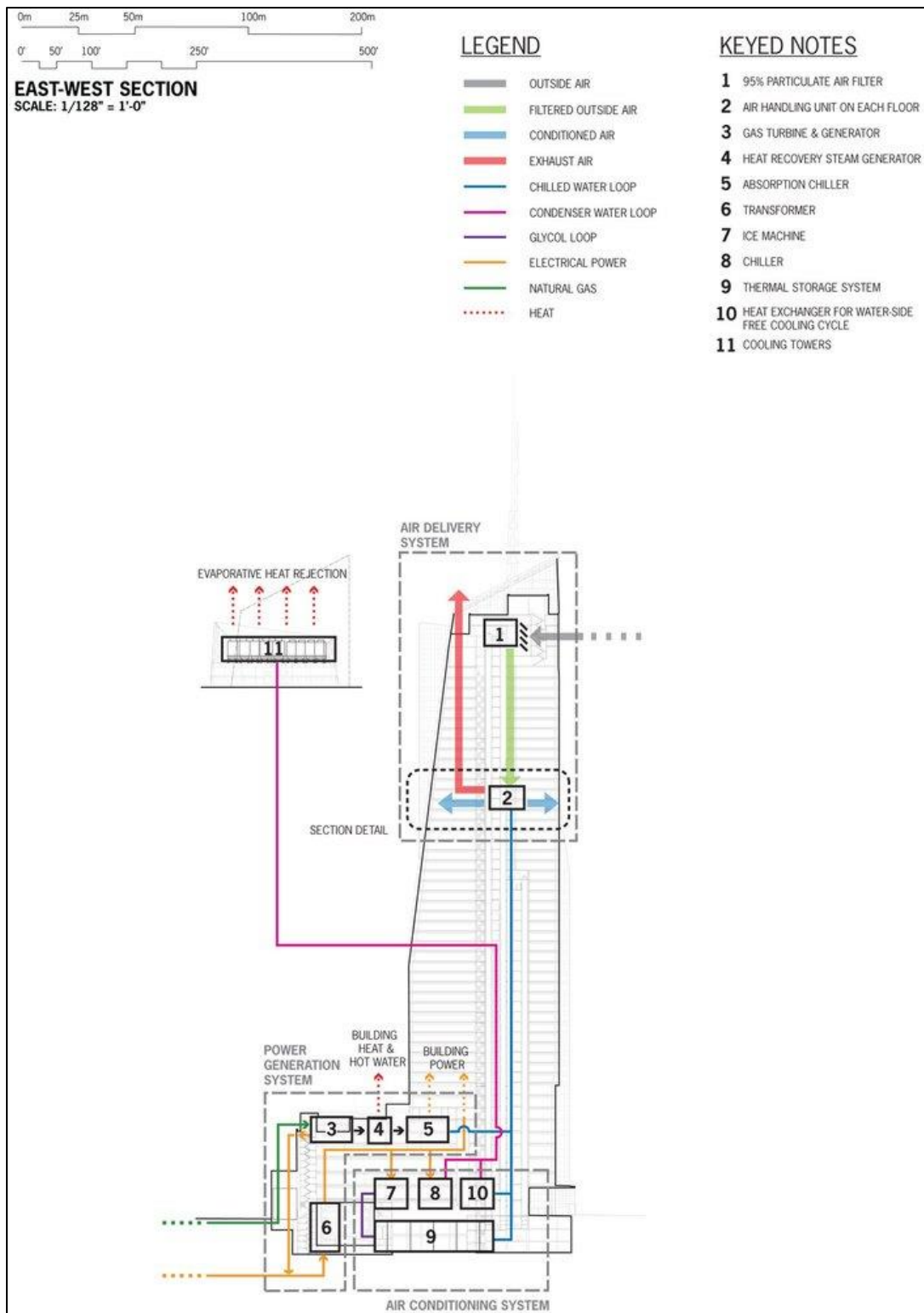


Figure 82. One Bryant Park Air and Thermal Systems Scheme. (Cook Fox Architects, n.d.)

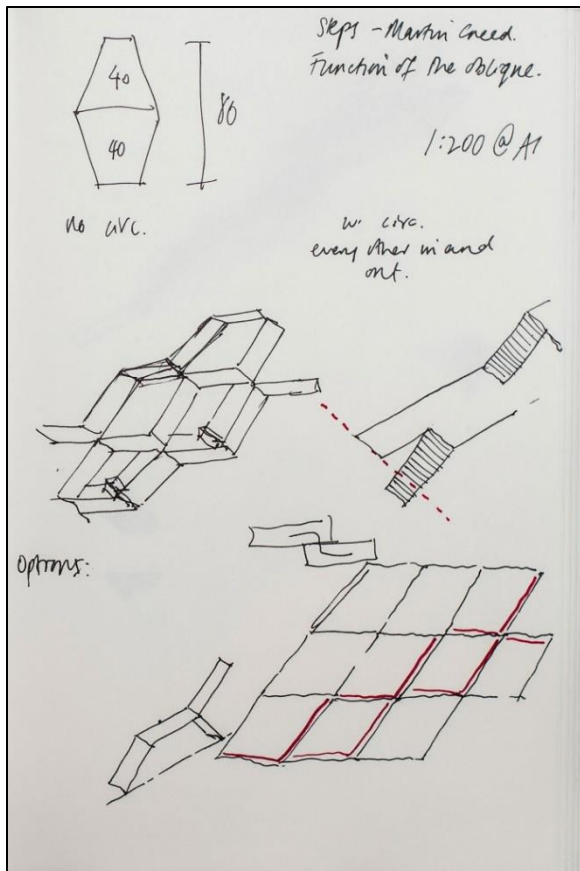


Figure 83. The Vessel's Conceptual Sketches - Cells.
(Heatherwick Studio, n.d.)

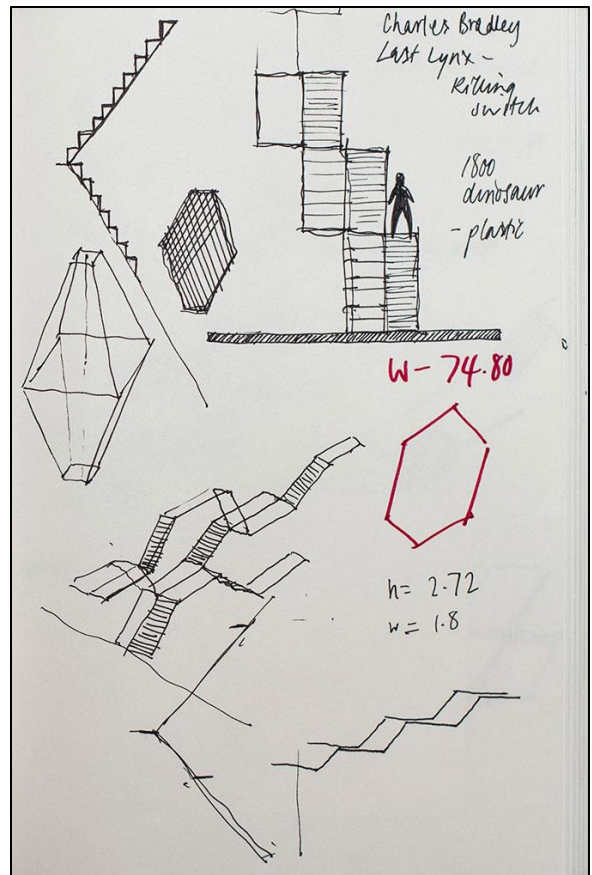


Figure 84. The Vessel's Conceptual Sketches - Stairs.
(Heatherwick Studio, n.d.)



Figure 85. The Living Breakwaters Plan and Map. (SCAPE/NYS Governor's Office of Storm Recovery (GOSR), n.d.)

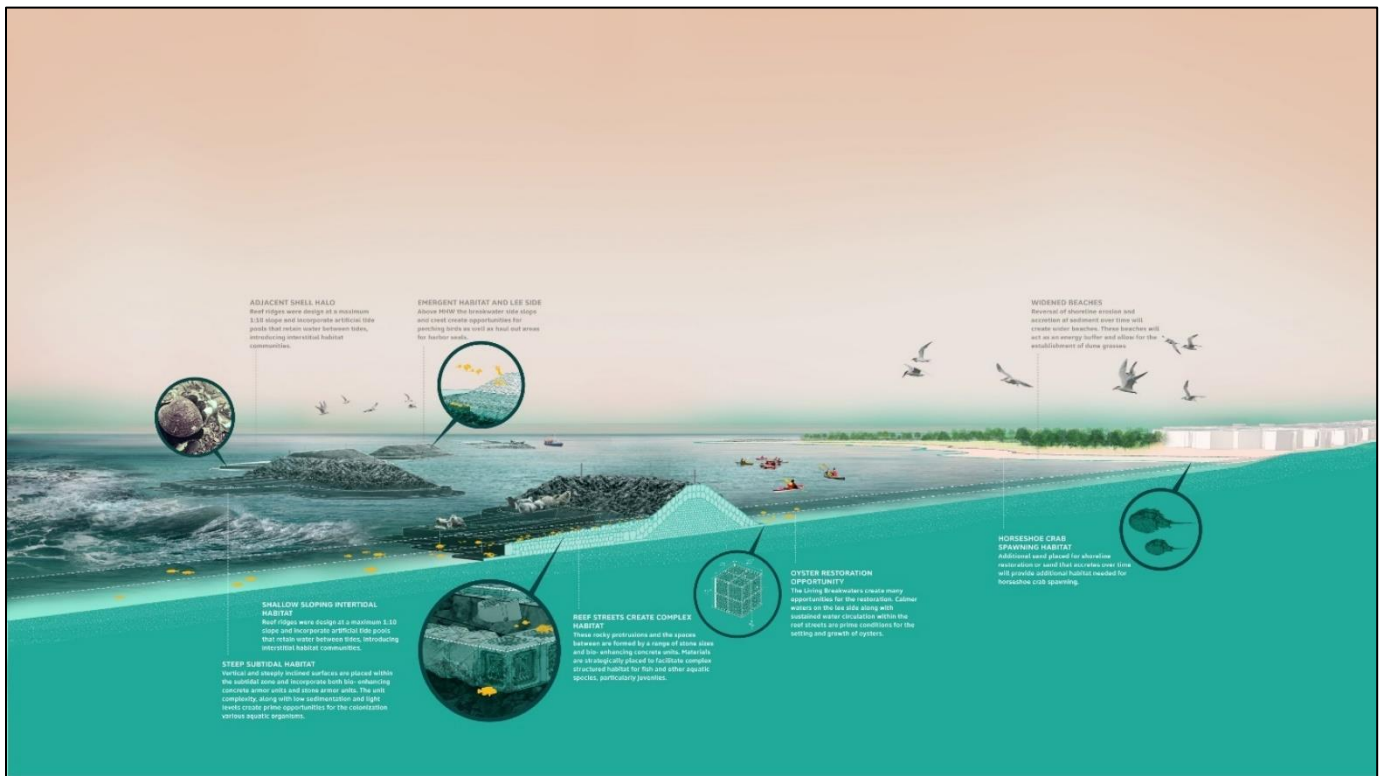


Figure 86. Ecological Benefits of the Living Breakwaters Project. (SCAPE/NYS Governor's Office of Storm Recovery (GOSR), n.d.)

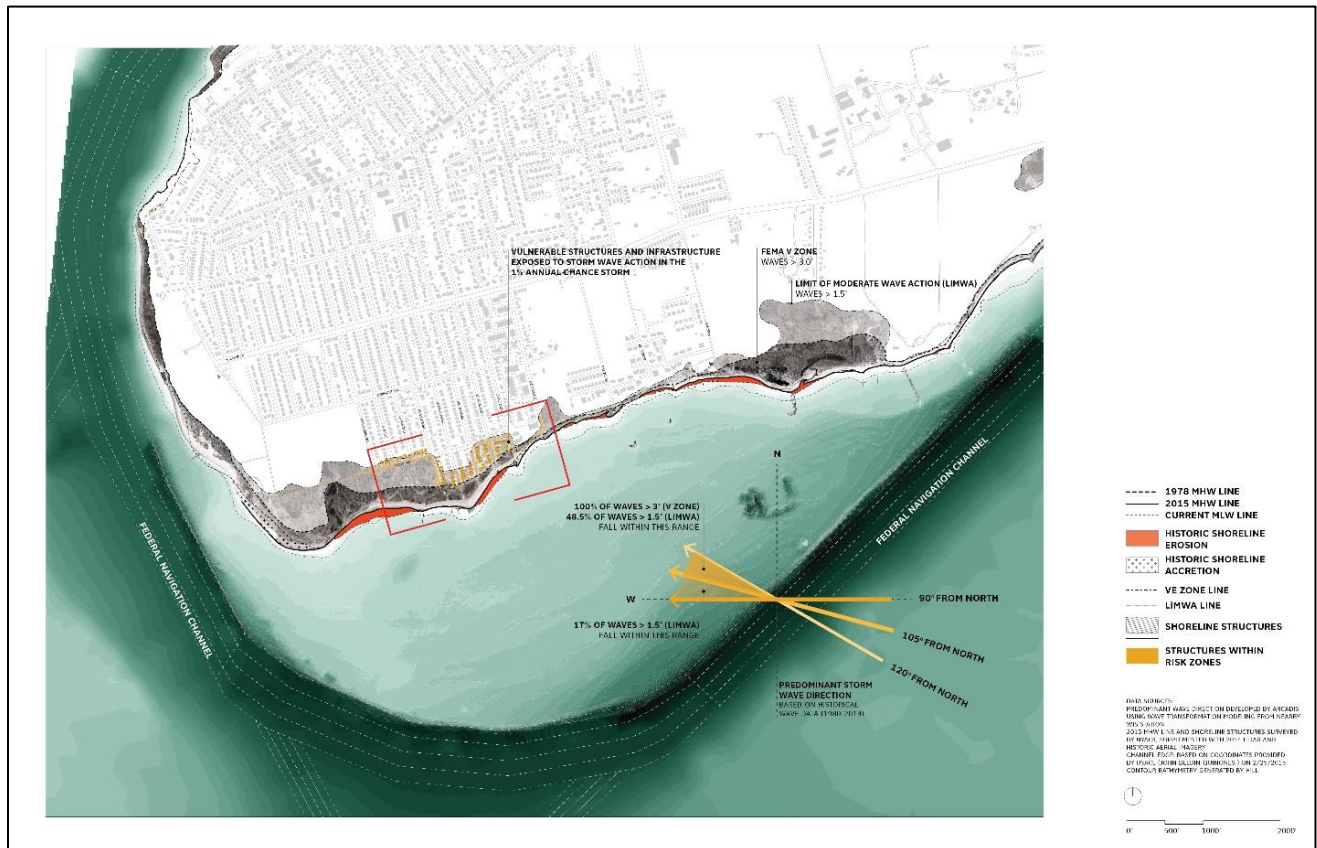


Figure 87. The Living Breakwaters Project's Location in Relation to the Vulnerable Area. (SCAPE/NYS Governor's Office of Storm Recovery (GOSR), n.d.)

Appendix 3. – Solutions from London

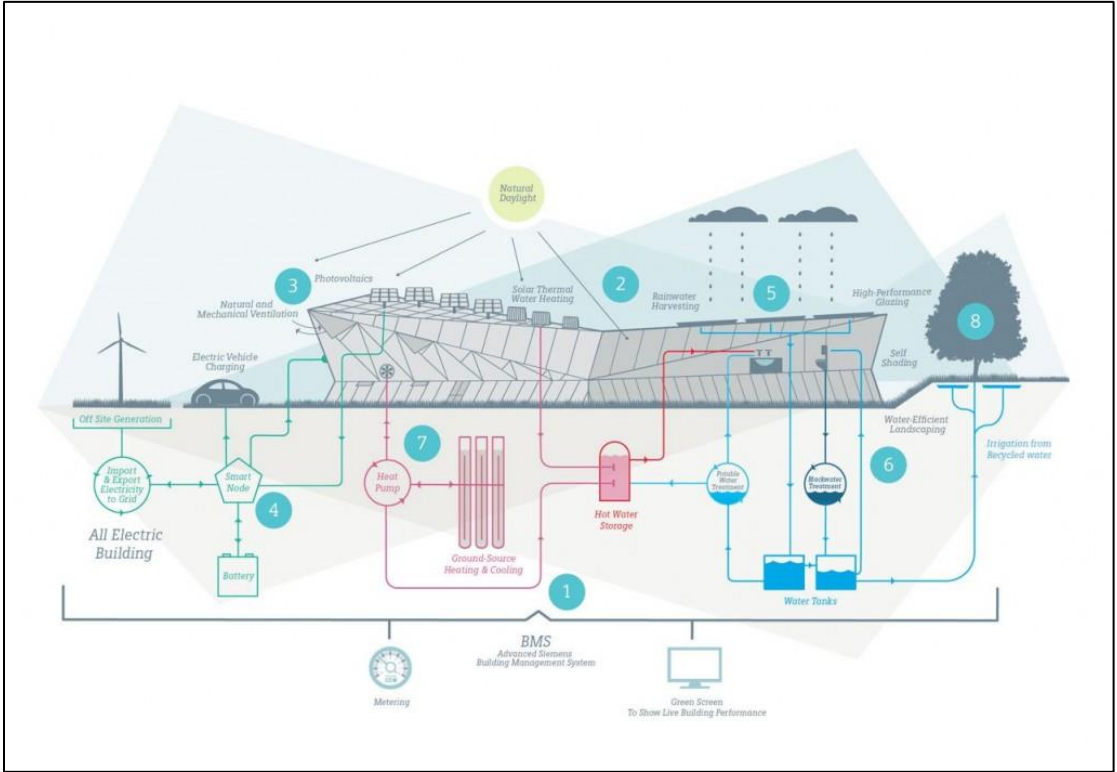


Figure 88. The Crystal Sustainable Features. (WilkinsonEyre, n.d.)

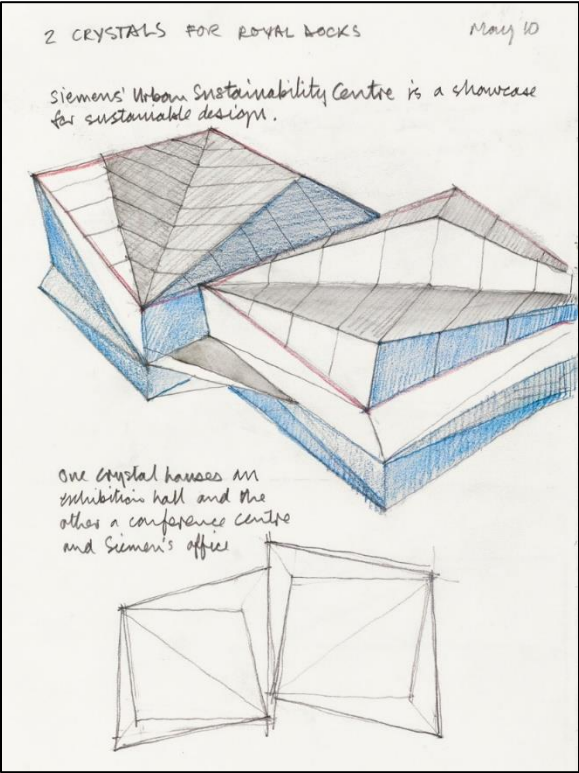


Figure 90. The Crystal Conceptual Sketch. (WilkinsonEyre, n.d.)

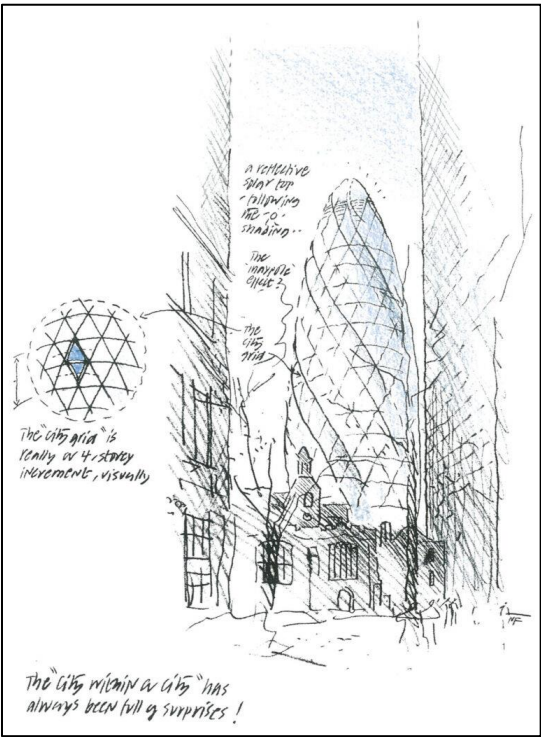


Figure 89. The Gherkin Conceptual Sketch Placing it in the Existing Infrastructure. (Foster + Partners, n.d.a)

Appendix 4. – Solutions from Madrid

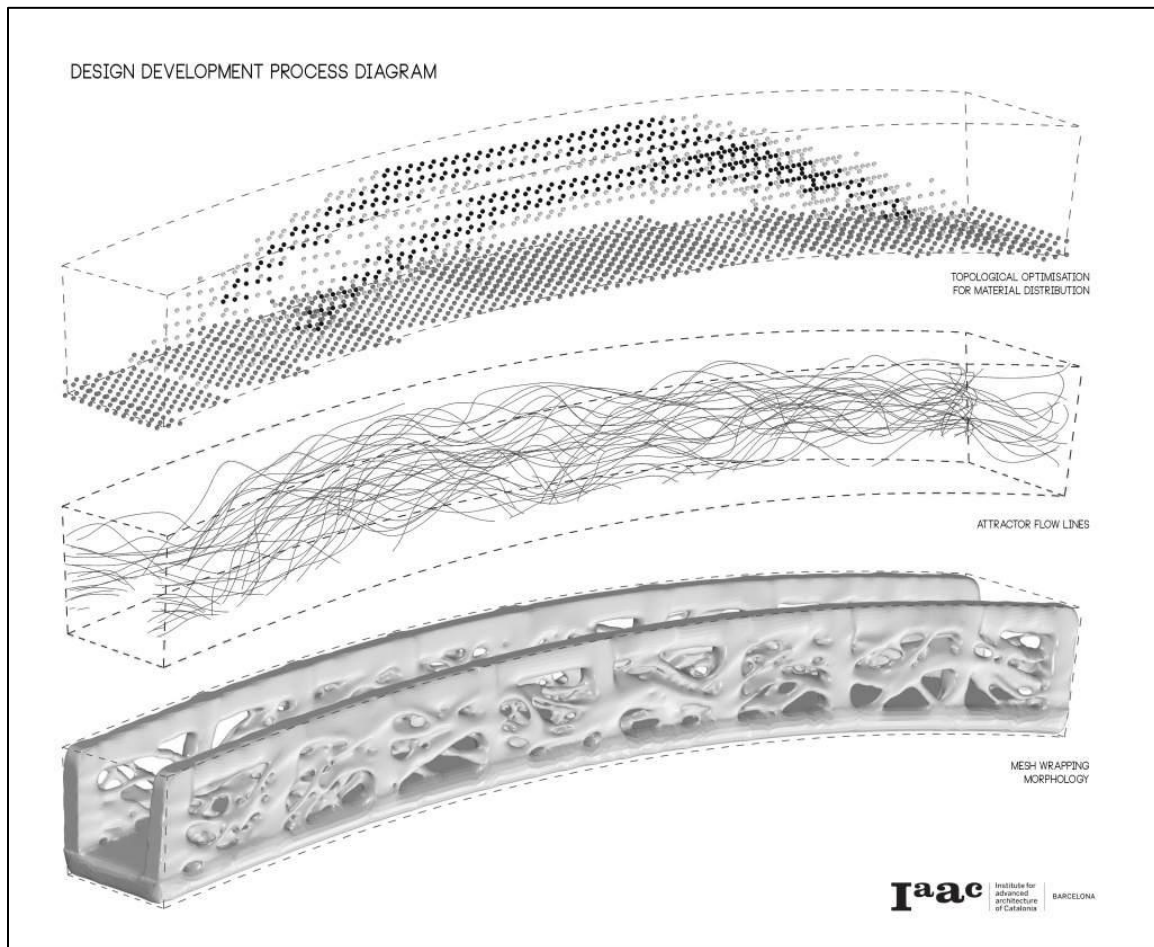


Figure 91. Design Process of the Bridge. (Enric Rodon, n.d.)

Appendix 5. – Solutions from Cape Town

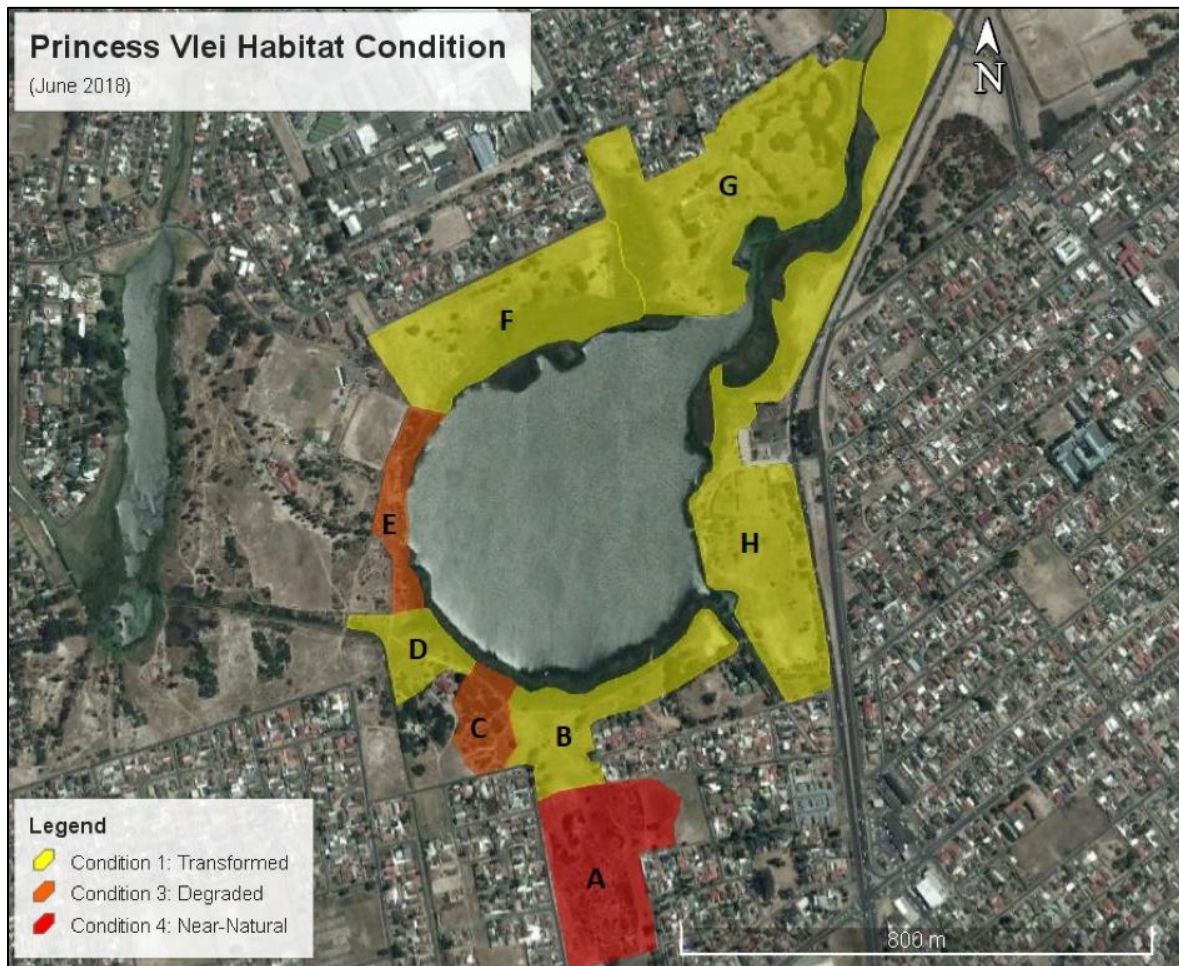


Figure 92. Map Assessing the Vegetation of the Princess Vlei Area. (Society for Ecological Restoration, n.d.)